

Canonical GHZ configurations - cluster P1 - helicity conserving

species: electron
polarization cluster: P1
source cluster configurations: 111
canonical configurations collected: 26

canonical display criterion: exactly 2 retained final-state components
retained-component cutoff: fraction $\geq 2.0\%$
normalized amplitude display: bar length = $|A_{\text{tilde}}|$, bar color/text = phase

helicity-mechanism subset: helicity conserving
classification: complete-chain R_flip/conserves ≤ 1
per-page labels: isolated flip/conserves and T/L powers; coherent cross terms are reported separately

cluster retained-component count distribution:

- 2 components: 72 configurations
- 3 components: 1 configurations
- 4 components: 9 configurations
- 5 components: 7 configurations
- 6 components: 18 configurations
- 7 components: 4 configurations

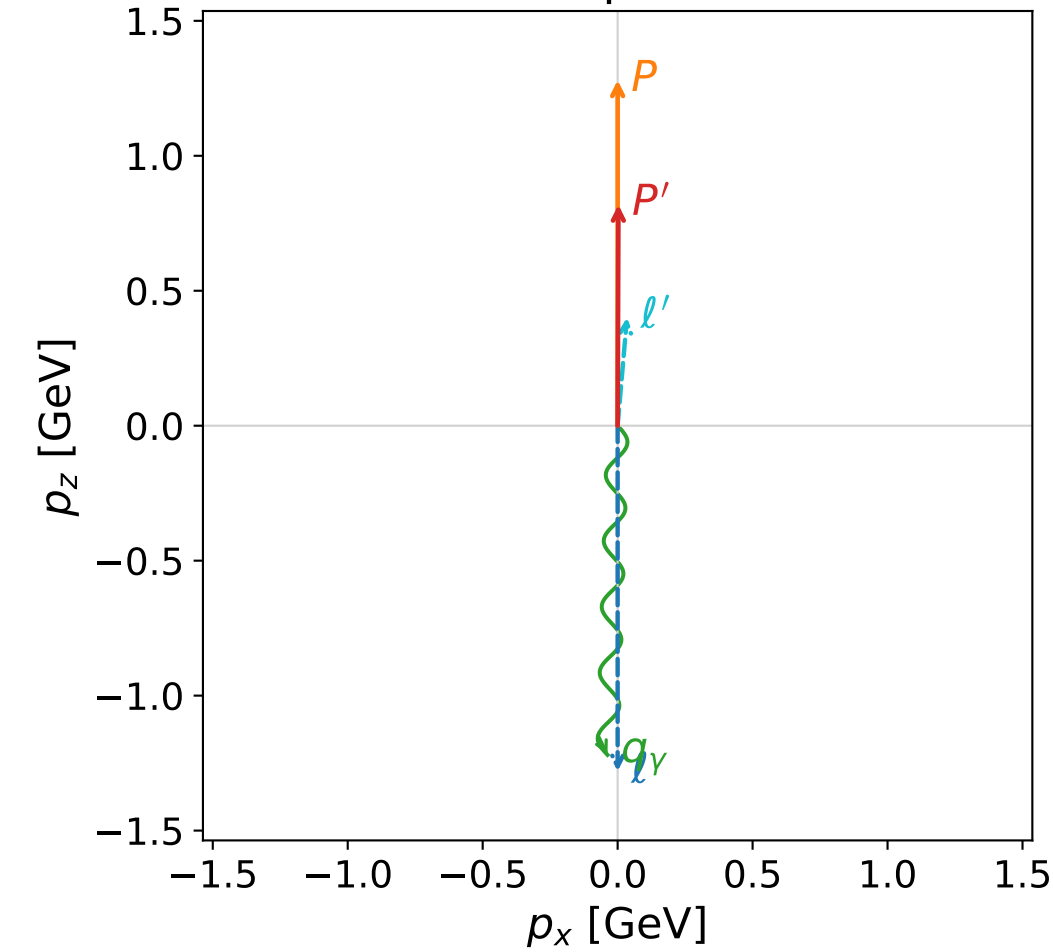
CSV index:

Output\GradientPhaseSpaceScan\electron\Data\GHZ\dghz\polarization_cluster_01\combined\
helicity_conserving\canonical_dghz_polarization_cluster_01_helicity_conserving_configu
rations_electron.csv

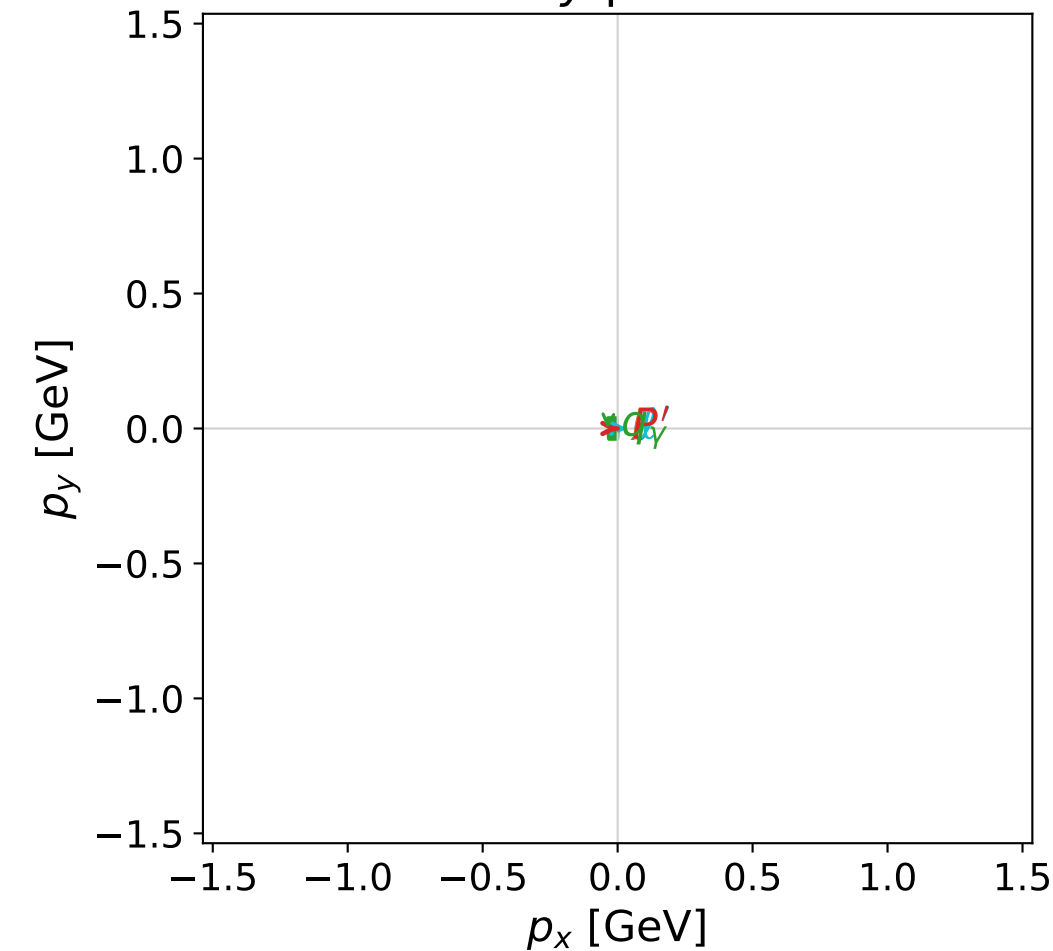
Each following page is one complete momentum, kinematic-summary, and normalized-amplitude configuration.

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

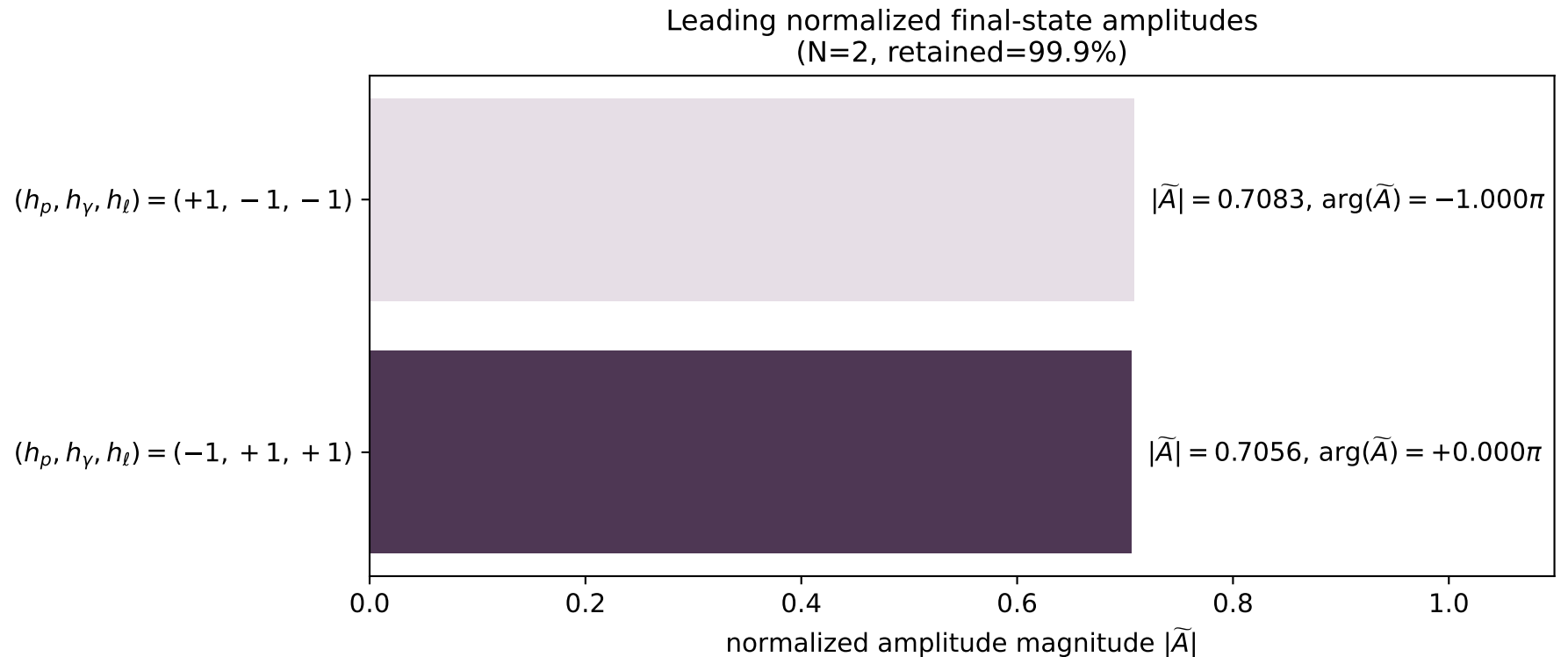


dghz_mixing_angles_polarization_01_local_minimum_193
 $d_{\text{GHZ}}=1.77496\text{e-}05$
 region: polarization_cluster_01_local_minimum_193
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0193
 lepton mixing angle: $\alpha_e=0.24273564$ rad
 incoming lepton: $0.970684|+\rangle +0.240359|-\rangle$
 proton mixing angle: $\alpha_p=1.3360225$ rad
 incoming proton: $0.232623|+\rangle +0.972567|-\rangle$

$s=8.22294$, $\sqrt{s}=2.86757$, $|\vec{P}|=1.28037$, $|\vec{P}'|=0.818827$
 $\theta_{p'}=0.00352053$, $\theta_\gamma=3.11084$, $\phi_{p'}=6.28317$, $\phi_\gamma=3.14158$
 $E_\gamma=1.22018$, $\theta_{l'}=0.086202$, $\phi_{l'}=6.28317$
 $Q^2=2.0564$, $x_B=0.289943$, $t=-0.0960184$, $W^2=5.91586$, $y=0.965862$

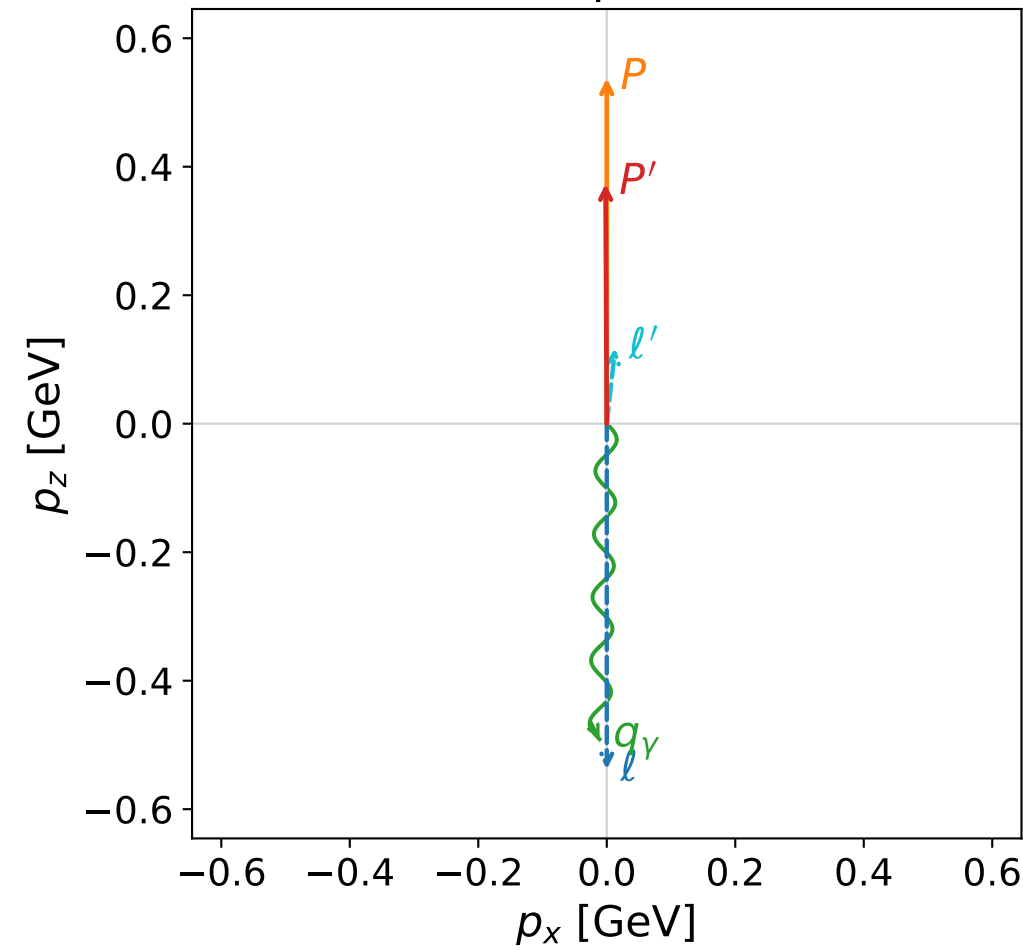
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.2804$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.2804$
 P $E= 1.5872$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2804$
 ℓ' $E= 0.4023$ $p_x= 0.0346$ $p_y= -0.0000$ $p_z= 0.4008$
 P' $E= 1.2451$ $p_x= 0.0029$ $p_y= -0.0000$ $p_z= 0.8188$
 q_γ $E= 1.2202$ $p_x= -0.0375$ $p_y= 0.0000$ $p_z= -1.2196$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.124%, conserve 99.876%
 $R_{\text{flip/conserve}}=1.242\text{e-}03$; $I_h=-2.094\text{e-}03$
 virtual-photon powers: T 99.859%, L 0.141%
 $R_L/T=1.414\text{e-}03$; $I_{TL}=+7.848\text{e-}03$

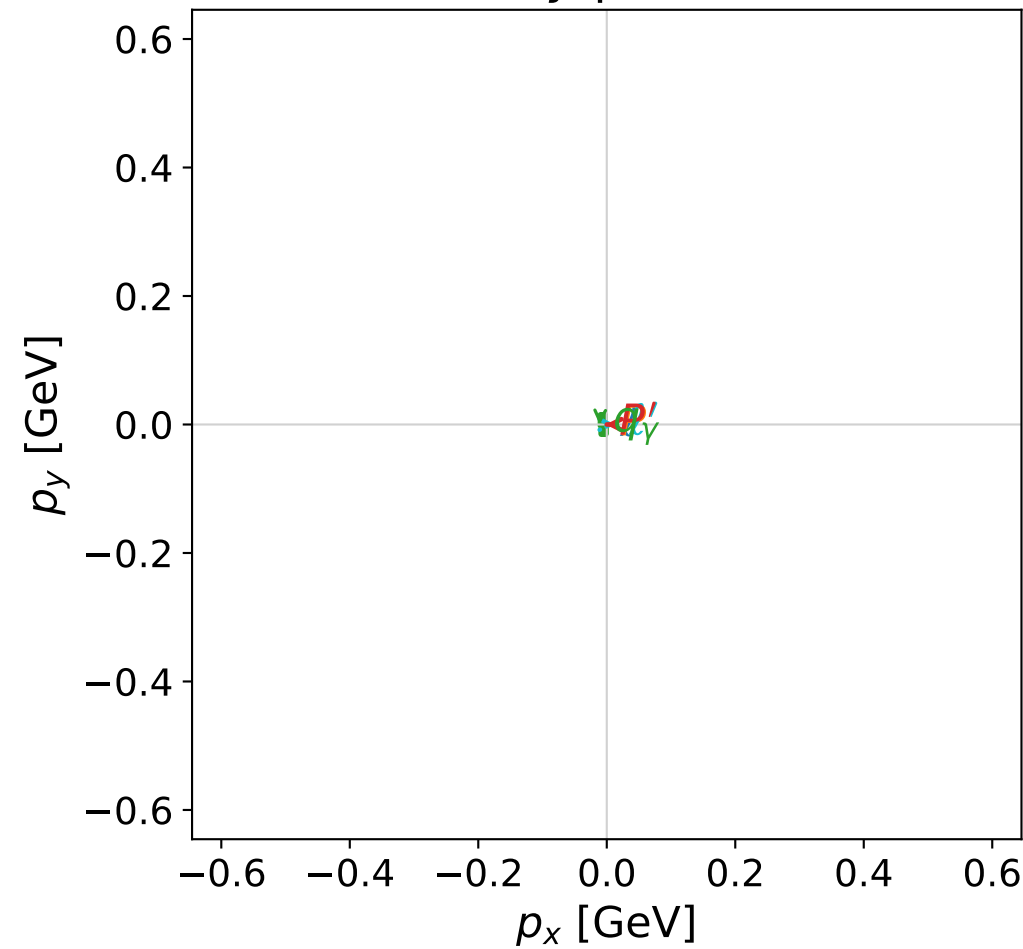


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



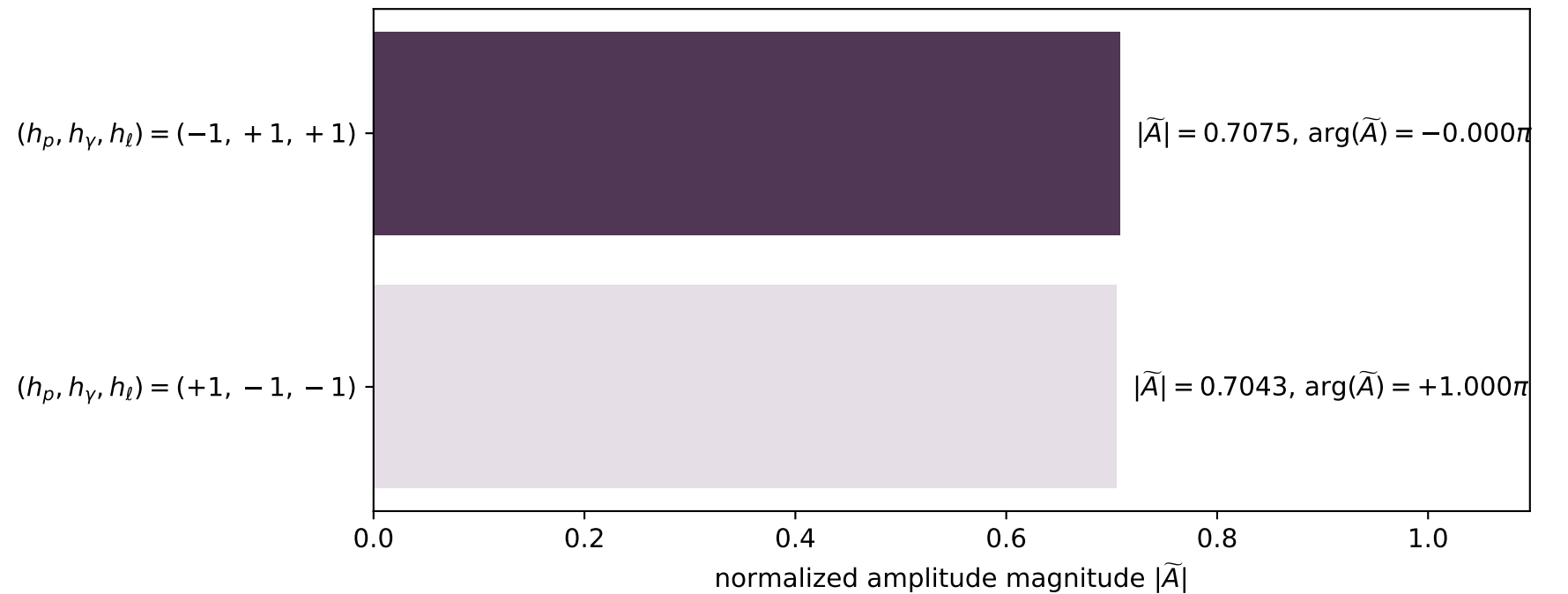
dghz_mixing_angles_polarization_01_local_minimum_198
 $d_{\{\rm GHZ\}}=2.02825\text{e-}05$
 region: polarization_cluster_01_local_minimum_198
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0198
 lepton mixing angle: $\alpha_e=0.31842078$ rad
 incoming lepton: $0.949731|+\rangle +0.313067|-\rangle$
 proton mixing angle: $\alpha_p=1.2260476$ rad
 incoming proton: $0.33796|+\rangle +0.94116|-\rangle$

$s=2.62187$, $\sqrt{s}=1.61922$, $|\vec{P}|=0.537922$, $|\vec{P}'|=0.373454$
 $\theta_{p'}=0.00509094$, $\theta_\gamma=3.11889$, $\phi_{p'}=3.14157$, $\phi_\gamma=3.14159$
 $E_\gamma=0.491231$, $\theta_{l'}=0.110485$, $\phi_{l'}=6.28318$
 $Q^2=0.253934$, $x_B=0.157469$, $t=-0.0219156$, $W^2=2.23851$, $y=0.925704$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.5379$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5379$
 P $E= 1.0813$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5379$
 l' $E= 0.1184$ $p_x= 0.0131$ $p_y= -0.0000$ $p_z= 0.1177$
 P' $E= 1.0096$ $p_x= -0.0019$ $p_y= 0.0000$ $p_z= 0.3734$
 q_γ $E= 0.4912$ $p_x= -0.0112$ $p_y= 0.0000$ $p_z= -0.4911$

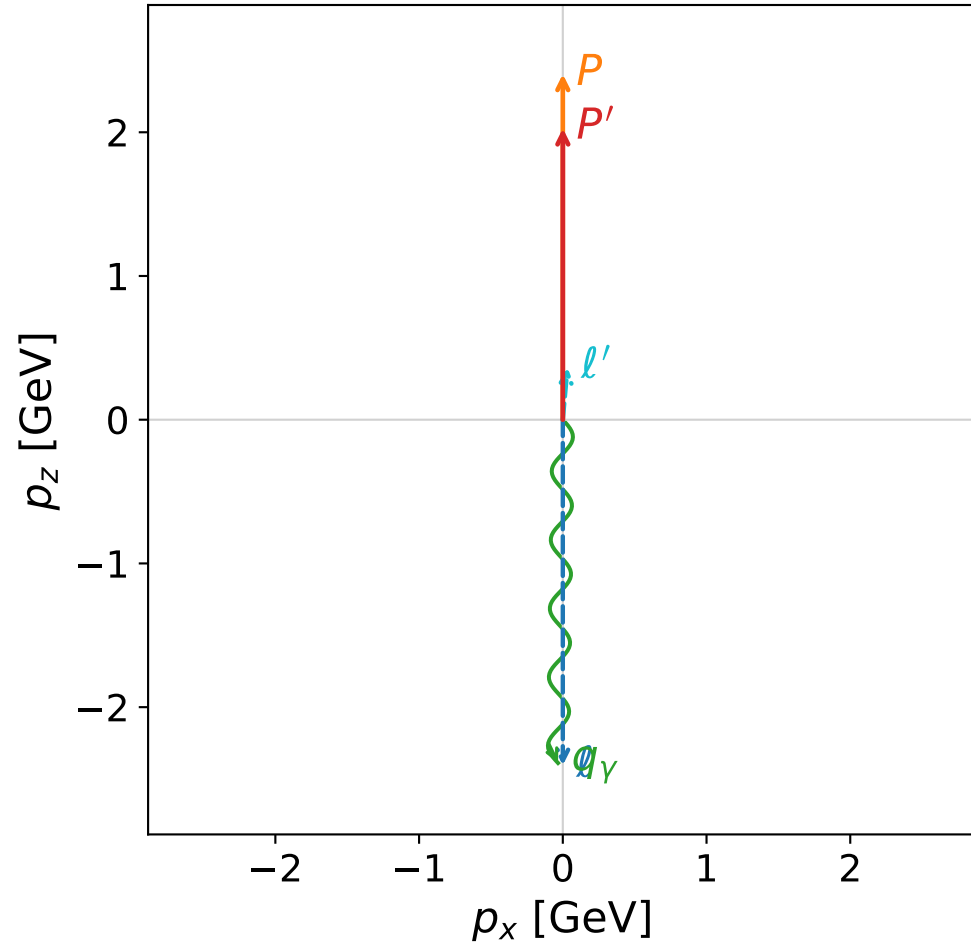
GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.641%, conserve 99.359%
 $R_{\text{flip/conserve}}=6.453\text{e-}03$; $I_h=-7.749\text{e-}03$
 virtual-photon powers: T 99.349%, L 0.651%
 $R_L/T=6.557\text{e-}03$; $I_{TL}=-4.905\text{e-}02$

Leading normalized final-state amplitudes
(N=2, retained=99.7%)

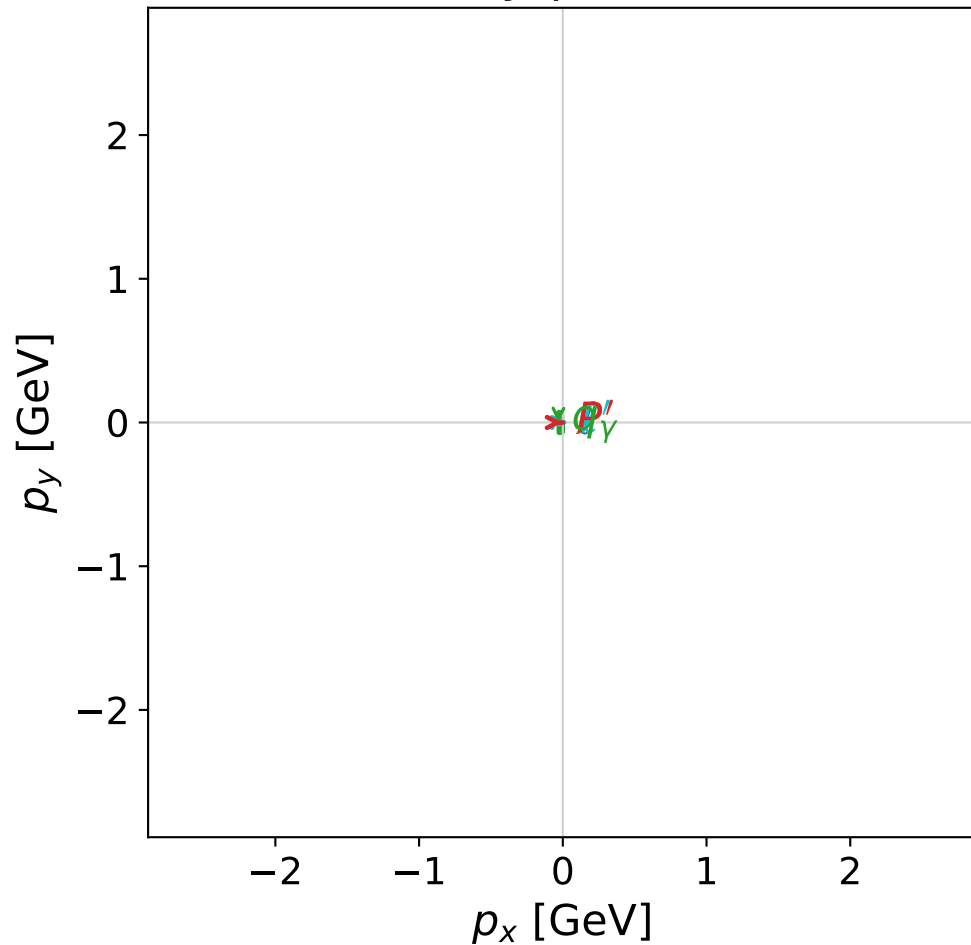


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

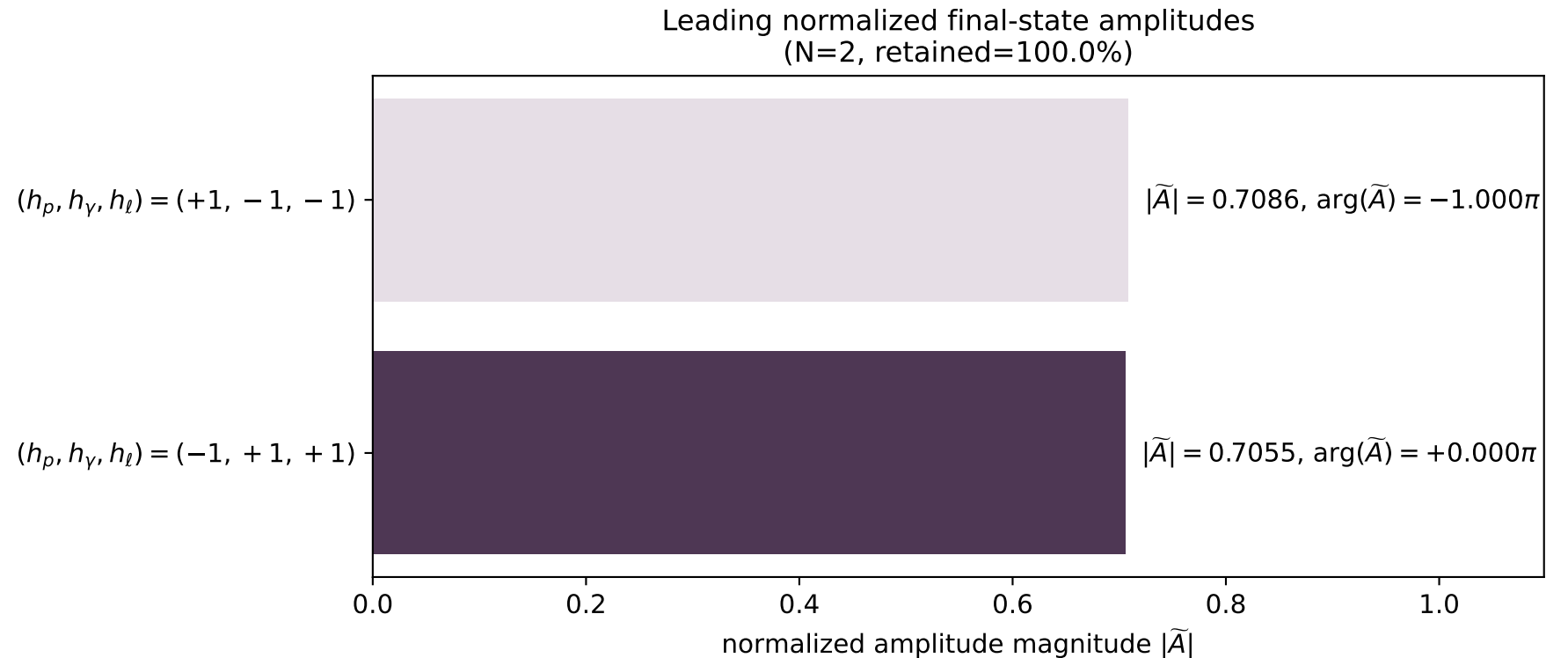


dghz_mixing_angles_polarization_01_local_minimum_203
 $d_{\text{GHZ}}=2.34128\text{e-}05$
 region: polarization_cluster_01_local_minimum_203
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0203
 lepton mixing angle: $\alpha_e=0.64267685$ rad
 incoming lepton: $0.800494|+\rangle +0.59934|-\rangle$
 proton mixing angle: $\alpha_p=0.93529282$ rad
 incoming proton: $0.593583|+\rangle +0.804773|-\rangle$

$s=24.8566$, $\sqrt{s}=4.98564$, $|\vec{P}|=2.40458$, $|\vec{P}'|=2.02645$
 $\theta_{p'}=0.000191071$, $\theta_\gamma=3.12775$, $\phi_{p'}=6.28238$, $\phi_\gamma=3.14161$
 $E_\gamma=2.38892$, $\theta_{l'}=0.0899866$, $\phi_{l'}=2.49864\text{e-}05$
 $Q^2=3.49121$, $x_B=0.146436$, $t=-0.0218485$, $W^2=21.2299$, $y=0.994351$

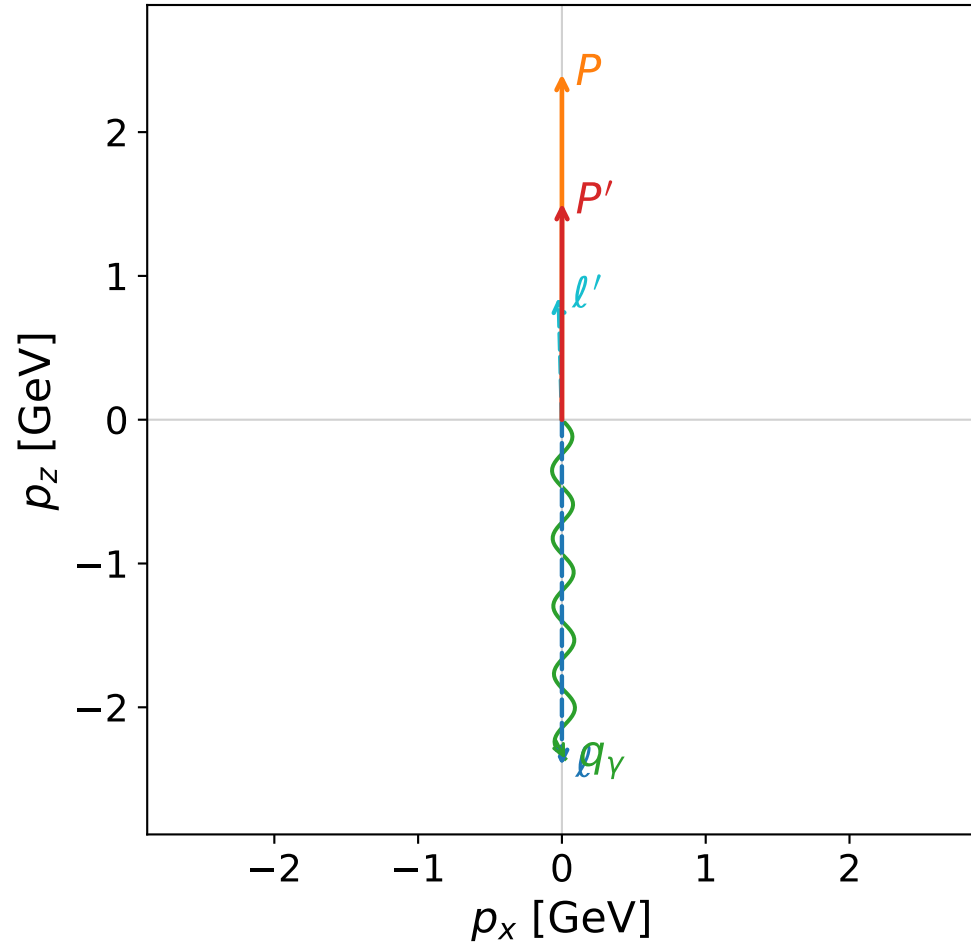
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4046$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4046$
 P $E= 2.5811$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4046$
 l' $E= 0.3637$ $p_x= 0.0327$ $p_y= 0.0000$ $p_z= 0.3622$
 P' $E= 2.2330$ $p_x= 0.0004$ $p_y= -0.0000$ $p_z= 2.0264$
 q_γ $E= 2.3889$ $p_x= -0.0331$ $p_y= -0.0000$ $p_z= -2.3887$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.027%, conserve 99.973%
 $R_{\text{flip/conserve}}=2.716\text{e-}04$; $I_h=-4.915\text{e-}05$
 virtual-photon powers: T 99.991%, L 0.009%
 $R_{L/T}=8.798\text{e-}05$; $I_{TL}=+2.688\text{e-}04$

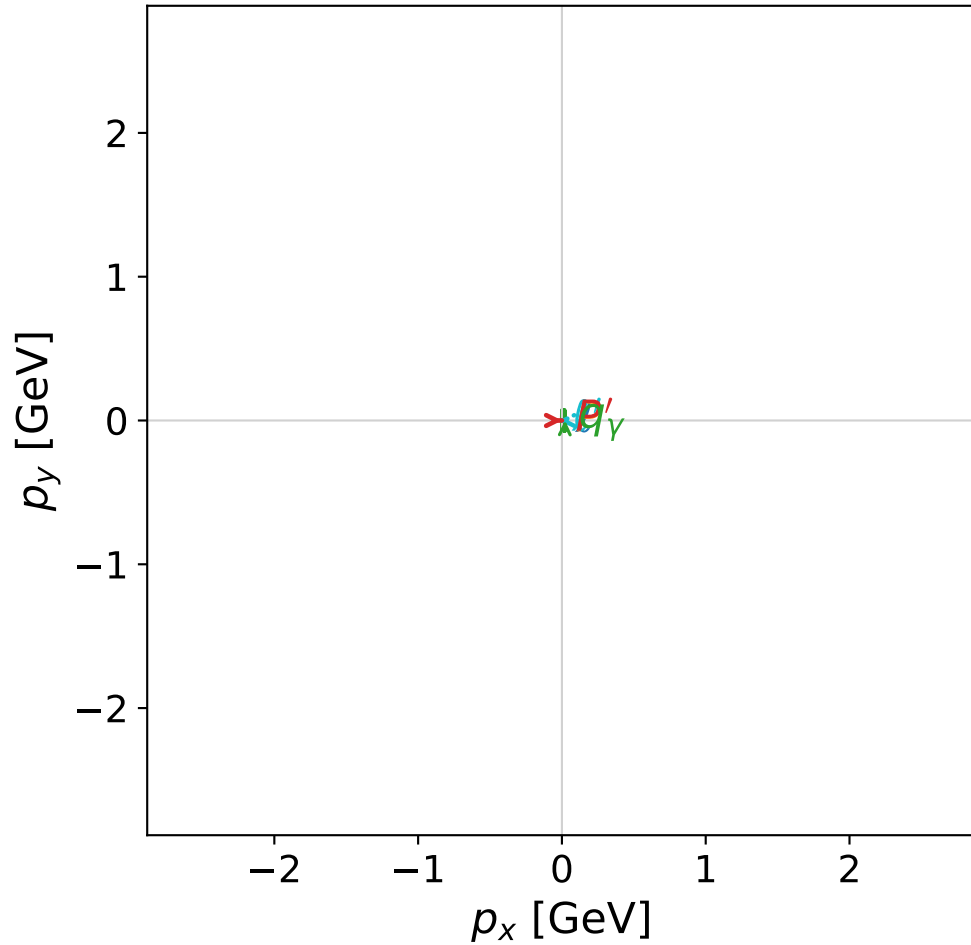


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

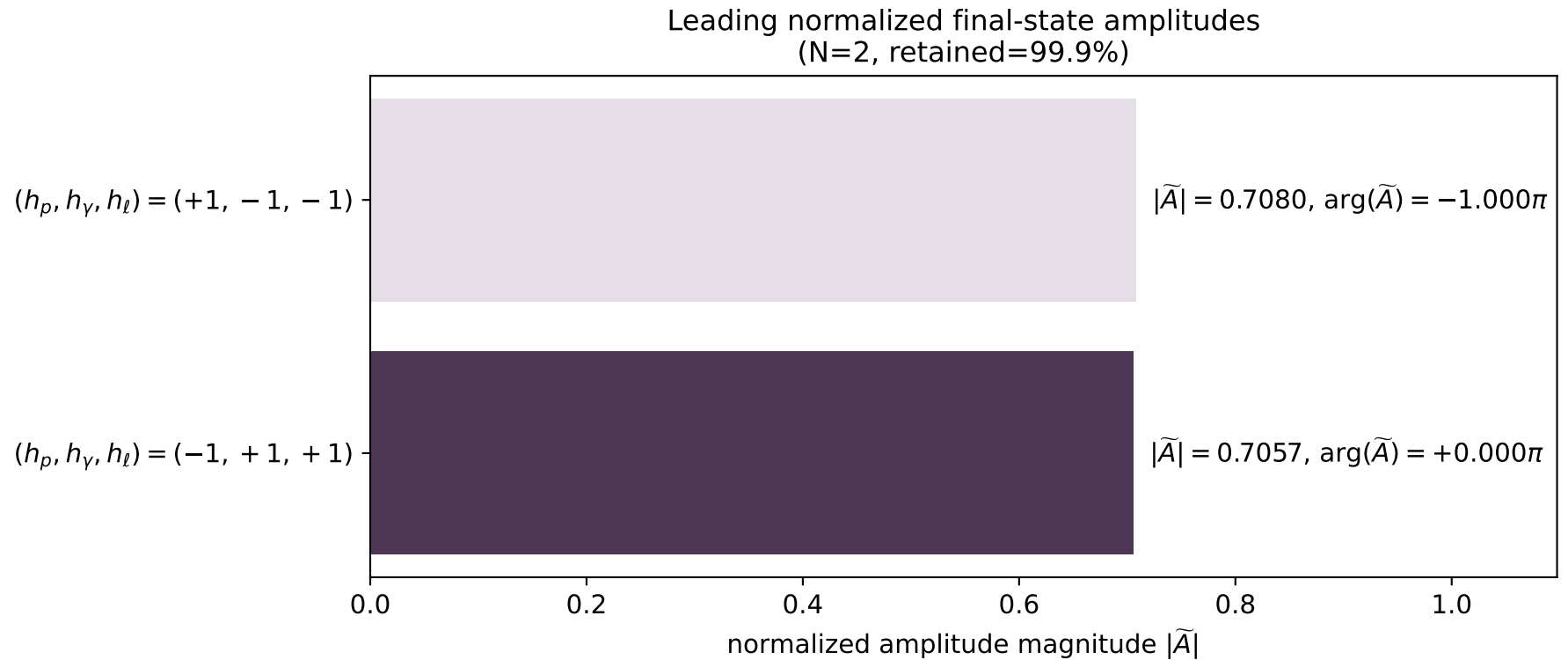


dghz_mixing_angles_polarization_01_local_minimum_206
 $d_{\{\rm GHZ\}}=2.59585\text{e-}05$
 region: polarization_cluster_01_local_minimum_206
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0206
 lepton mixing angle: $\alpha_e=1.1210285$ rad
 incoming lepton: $0.434756|+\rangle +0.900548|-\rangle$
 proton mixing angle: $\alpha_p=0.45230063$ rad
 incoming proton: $0.899444|+\rangle +0.437036|-\rangle$

$s=24.8343$, $\sqrt{s}=4.9834$, $|\vec{P}|=2.40342$, $|\vec{P}'|=1.50557$
 $\theta_{P'}=0.000624821$, $\theta_\gamma=3.13227$, $\phi_{P'}=6.28282$, $\phi_\gamma=0$
 $E_\gamma=2.35745$, $\theta_{\ell'}=0.0268985$, $\phi_{\ell'}=3.14158$
 $Q^2=8.19023$, $x_B=0.346279$, $t=-0.156318$, $W^2=16.3417$, $y=0.987378$

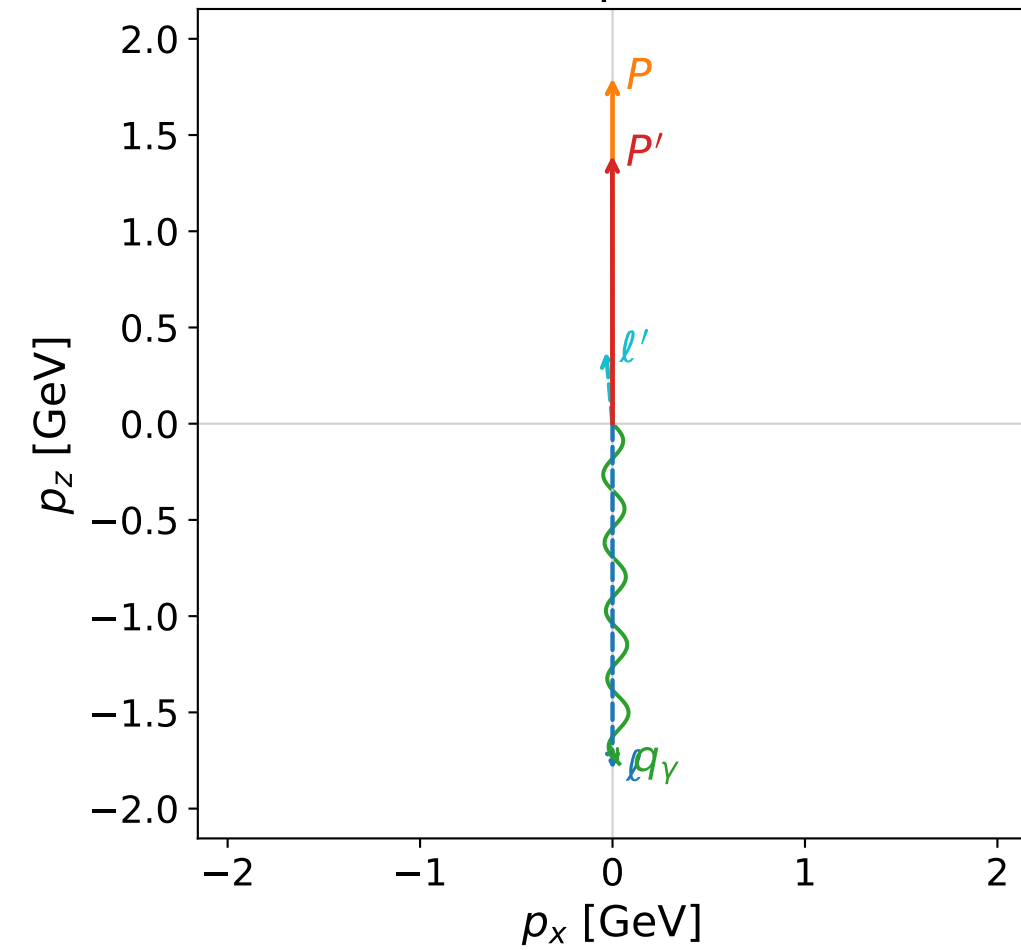
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4034$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4034$
 P $E= 2.5800$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4034$
 ℓ' $E= 0.8521$ $p_x= -0.0229$ $p_y= 0.0000$ $p_z= 0.8518$
 P' $E= 1.7739$ $p_x= 0.0009$ $p_y= -0.0000$ $p_z= 1.5056$
 q_γ $E= 2.3575$ $p_x= 0.0220$ $p_y= 0.0000$ $p_z= -2.3574$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.112%, conserve 99.888%
 $R_{\text{flip/conserve}}=1.125\text{e-}03$; $I_h=-6.653\text{e-}04$
 virtual-photon powers: T 99.999%, L 0.001%
 $R_{L/T}=1.055\text{e-}05$; $I_{TL}=-3.723\text{e-}04$

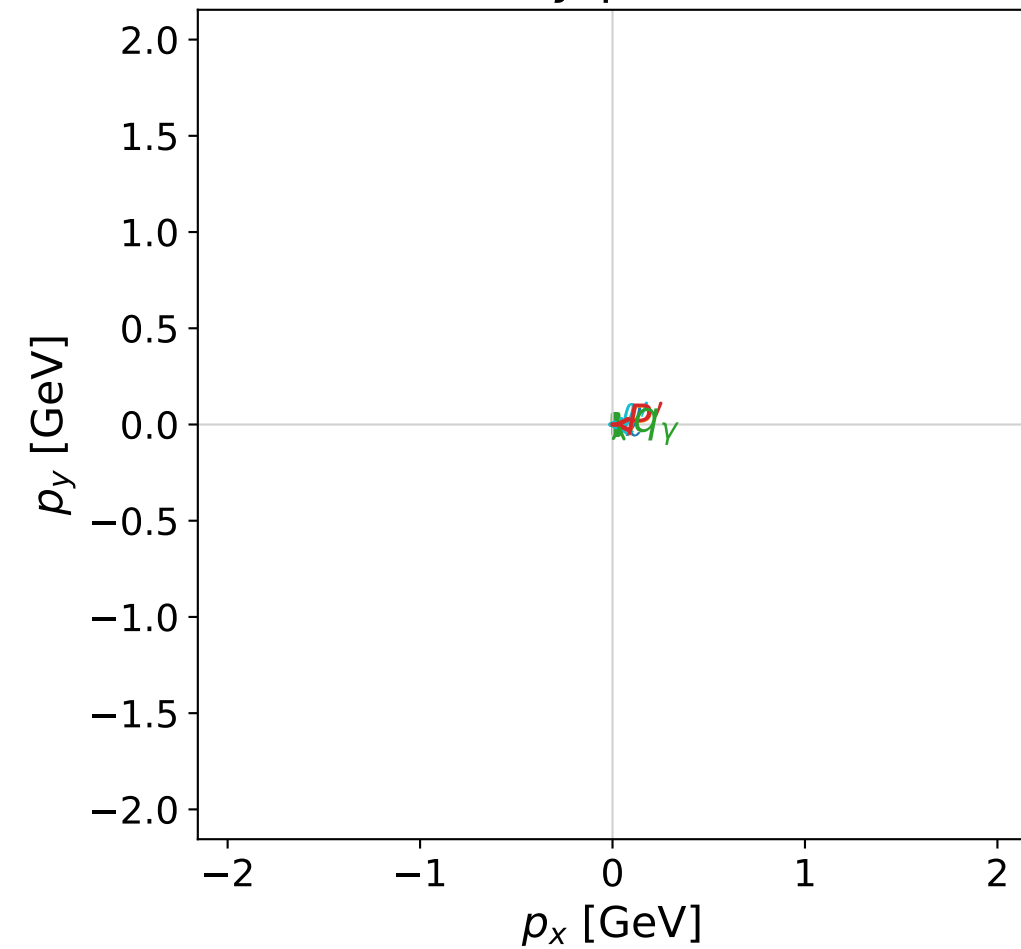


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_01_local_minimum_207

$d_{\text{GHZ}}=2.62949\text{e-}05$

region: polarization_cluster_01_local_minimum_207

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0207

lepton mixing angle: $\alpha_e=1.0858098$ rad

incoming lepton: $0.466196|+\rangle + 0.884681|-\rangle$

proton mixing angle: $\alpha_p=0.47680636$ rad

incoming proton: $0.888465|+\rangle + 0.458944|-\rangle$

$s=14.6057$, $\sqrt{s}=3.82175$, $|\vec{P}|=1.79576$, $|\vec{P}'|=1.39318$

$\theta_{p'}=0.000700268$, $\theta_\gamma=3.12182$, $\phi_{p'}=3.14146$, $\phi_\gamma=1.35121\text{e-}05$

$E_\gamma=1.7671$, $\theta_{\ell'}=0.0906712$, $\phi_{\ell'}=3.14161$

$Q^2=2.68895$, $x_B=0.198481$, $t=-0.0420382$, $W^2=11.7385$, $y=0.987013$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 1.7958$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7958$

P $E= 2.0260$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7958$

ℓ' $E= 0.3751$ $p_x= -0.0340$ $p_y= -0.0000$ $p_z= 0.3736$

P' $E= 1.6795$ $p_x= -0.0010$ $p_y= 0.0000$ $p_z= 1.3932$

q_γ $E= 1.7671$ $p_x= 0.0349$ $p_y= 0.0000$ $p_z= -1.7668$

GHZ mechanism: helicity conserving (full H G C)

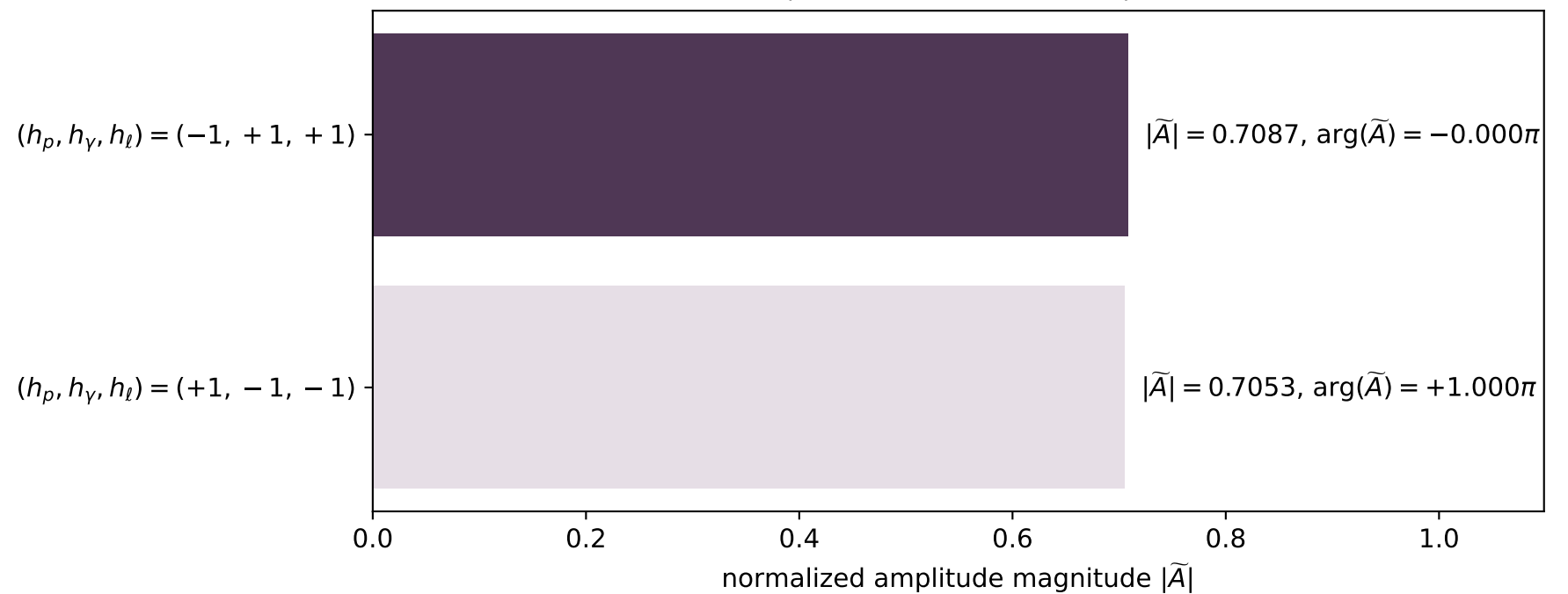
helicity powers: flip 0.039%, conserve 99.961%

$R_{\text{flip/conserve}}=3.858\text{e-}04$; $I_h=-2.807\text{e-}04$

virtual-photon powers: T 99.977%, L 0.023%

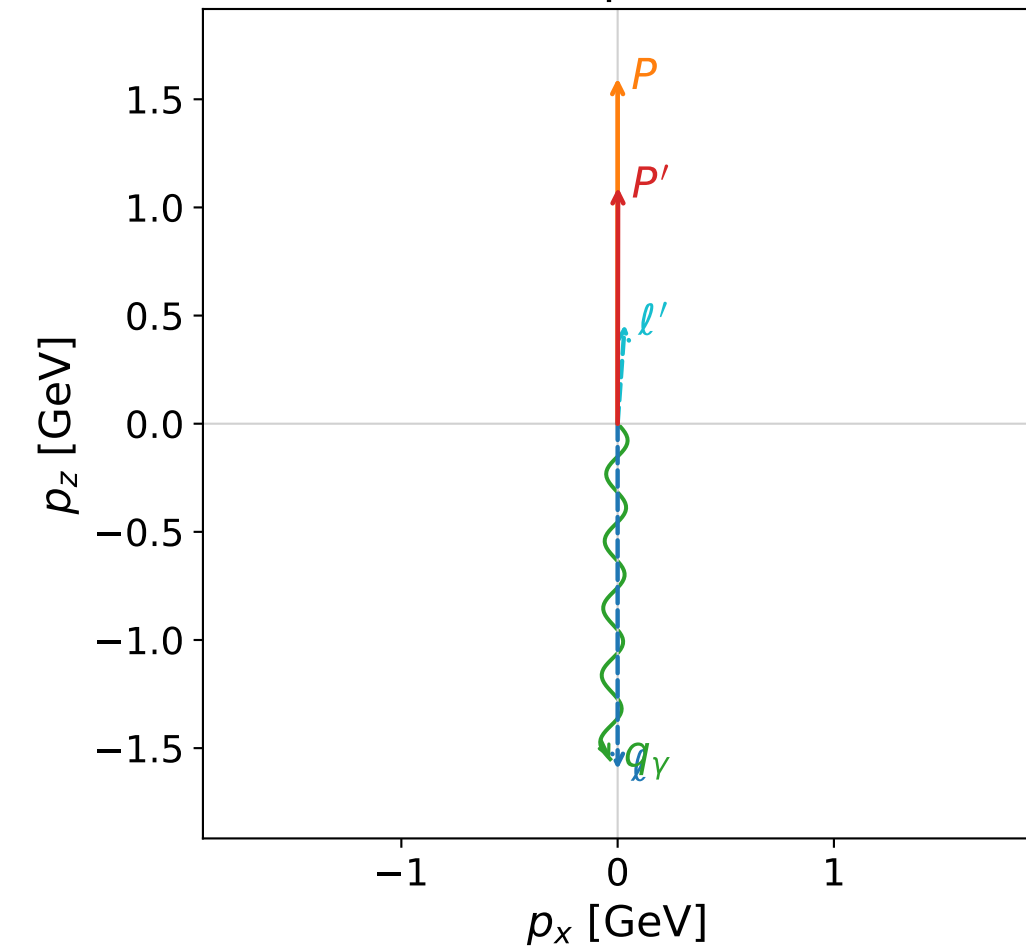
$R_L/T=2.255\text{e-}04$; $I_{TL}=+1.344\text{e-}03$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)

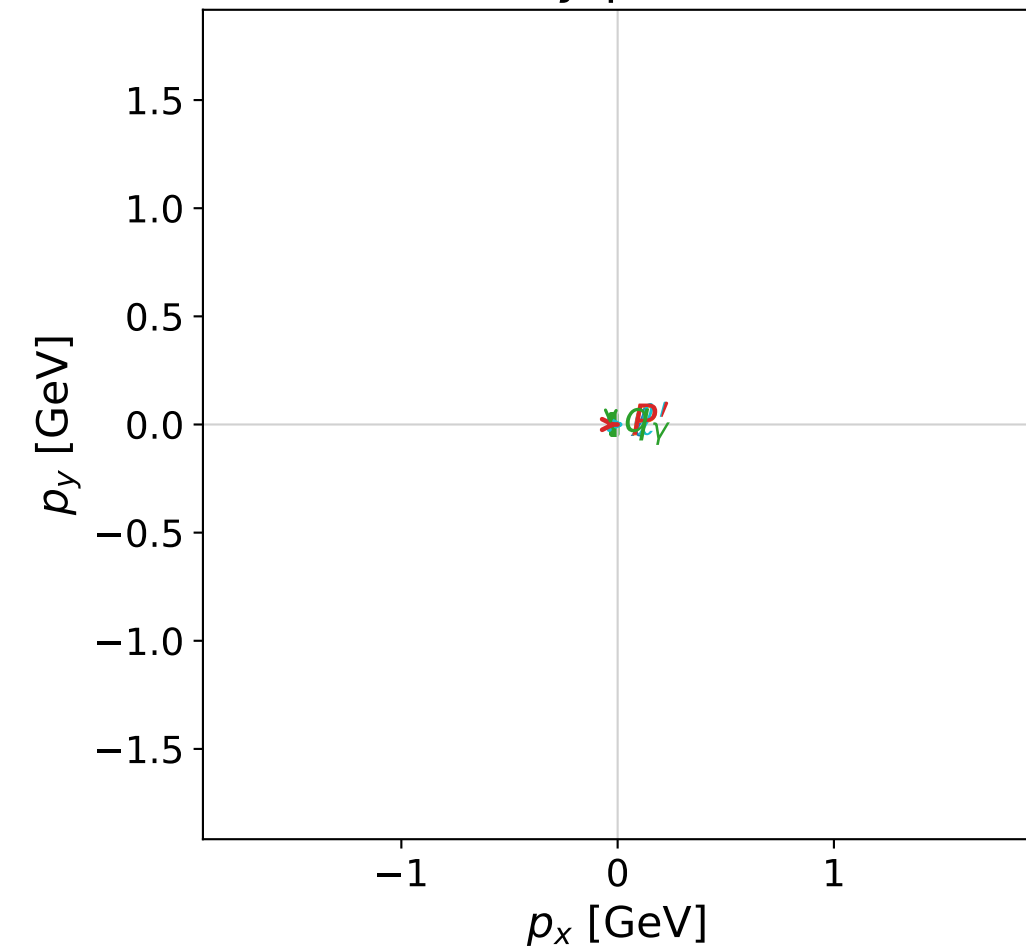


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

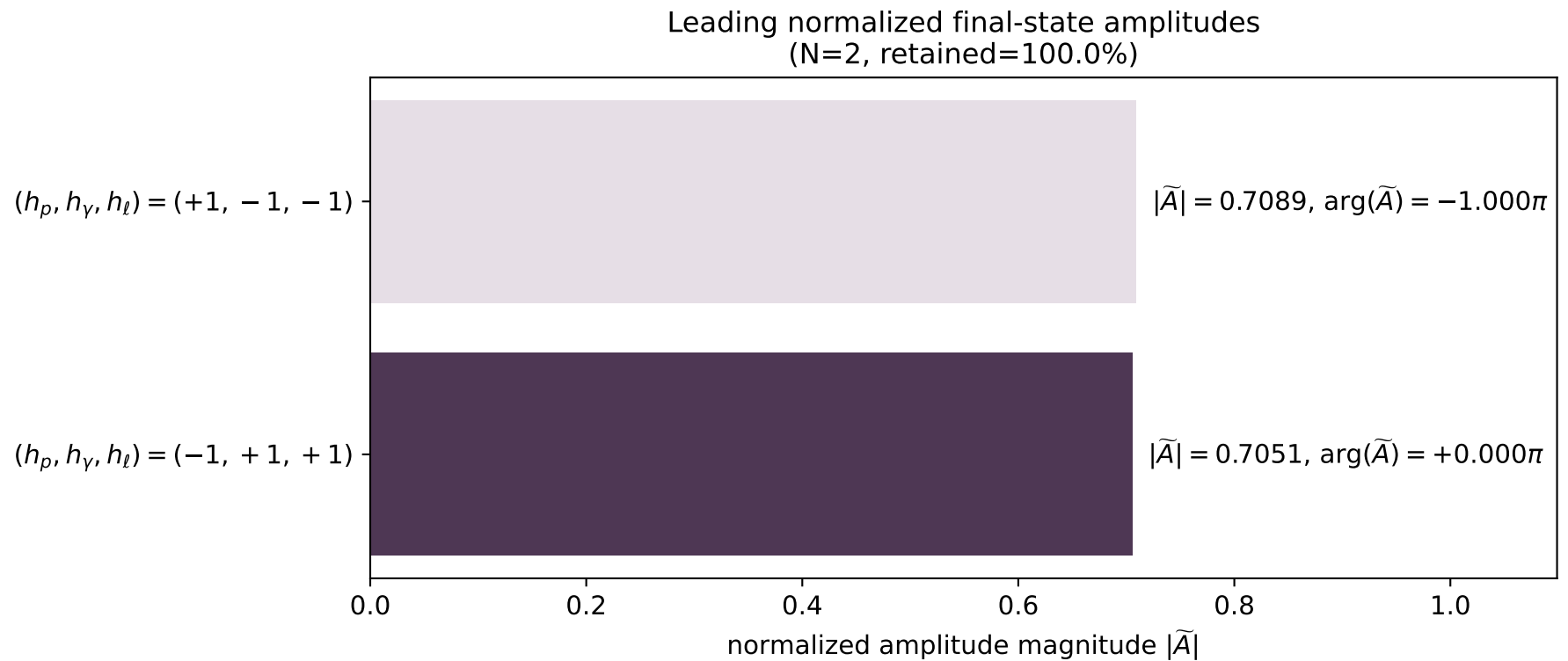


dghz_mixing_angles_polarization_01_local_minimum_216
 $d_{\{\rm GHZ\}}=3.98745\text{e-}05$
 region: polarization_cluster_01_local_minimum_216
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0216
 lepton mixing angle: $\alpha_e=0.32329631$ rad
 incoming lepton: $0.948193|+\rangle +0.317694|-\rangle$
 proton mixing angle: $\alpha_p=1.2539312$ rad
 incoming proton: $0.311589|+\rangle +0.950217|-\rangle$

$s=11.9072$, $\sqrt{s}=3.45069$, $|\vec{P}|=1.59785$, $|\vec{P}'|=1.09272$
 $\theta_{p'}=0.00155077$, $\theta_\gamma=3.1194$, $\phi_{p'}=6.28293$, $\phi_\gamma=3.14161$
 $E_\gamma=1.55126$, $\theta_{l'}=0.0712991$, $\phi_{l'}=2.66731\text{e-}05$
 $Q^2=2.93205$, $x_B=0.271753$, $t=-0.0848181$, $W^2=8.73722$, $y=0.97842$

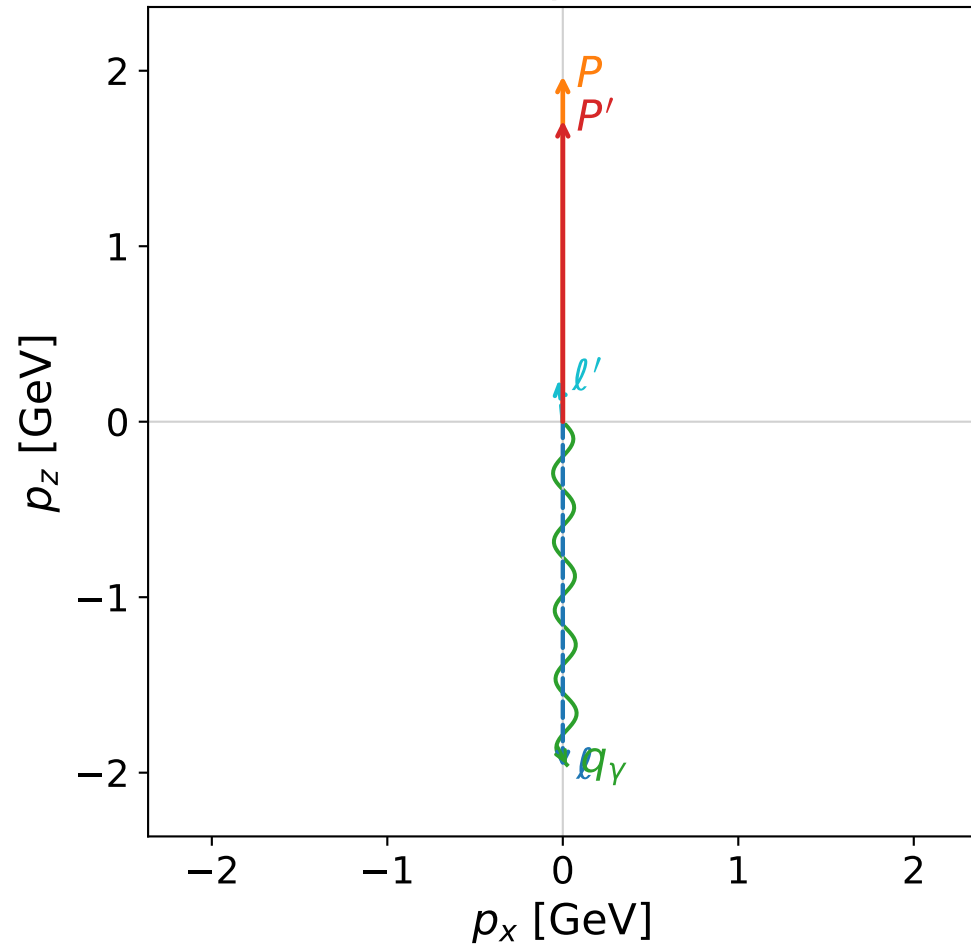
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.5979$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5979$
 P $E= 1.8528$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5979$
 l' $E= 0.4593$ $p_x= 0.0327$ $p_y= 0.0000$ $p_z= 0.4582$
 P' $E= 1.4401$ $p_x= 0.0017$ $p_y= -0.0000$ $p_z= 1.0927$
 q_γ $E= 1.5513$ $p_x= -0.0344$ $p_y= -0.0000$ $p_z= -1.5509$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.088%, conserve 99.912%
 $R_{\text{flip/conserve}}=8.792\text{e-}04$; $I_h=-1.035\text{e-}03$
 virtual-photon powers: T 99.965%, L 0.035%
 $R_L/T=3.465\text{e-}04$; $I_{TL}=+2.692\text{e-}03$

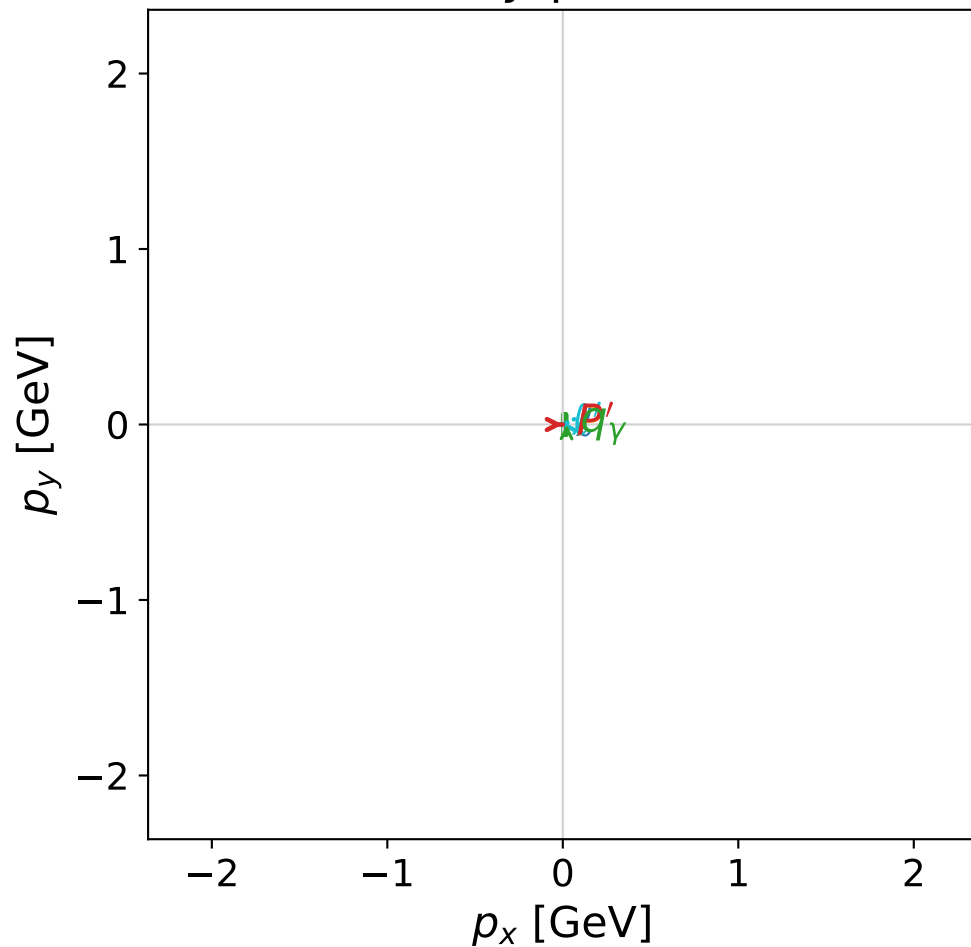


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

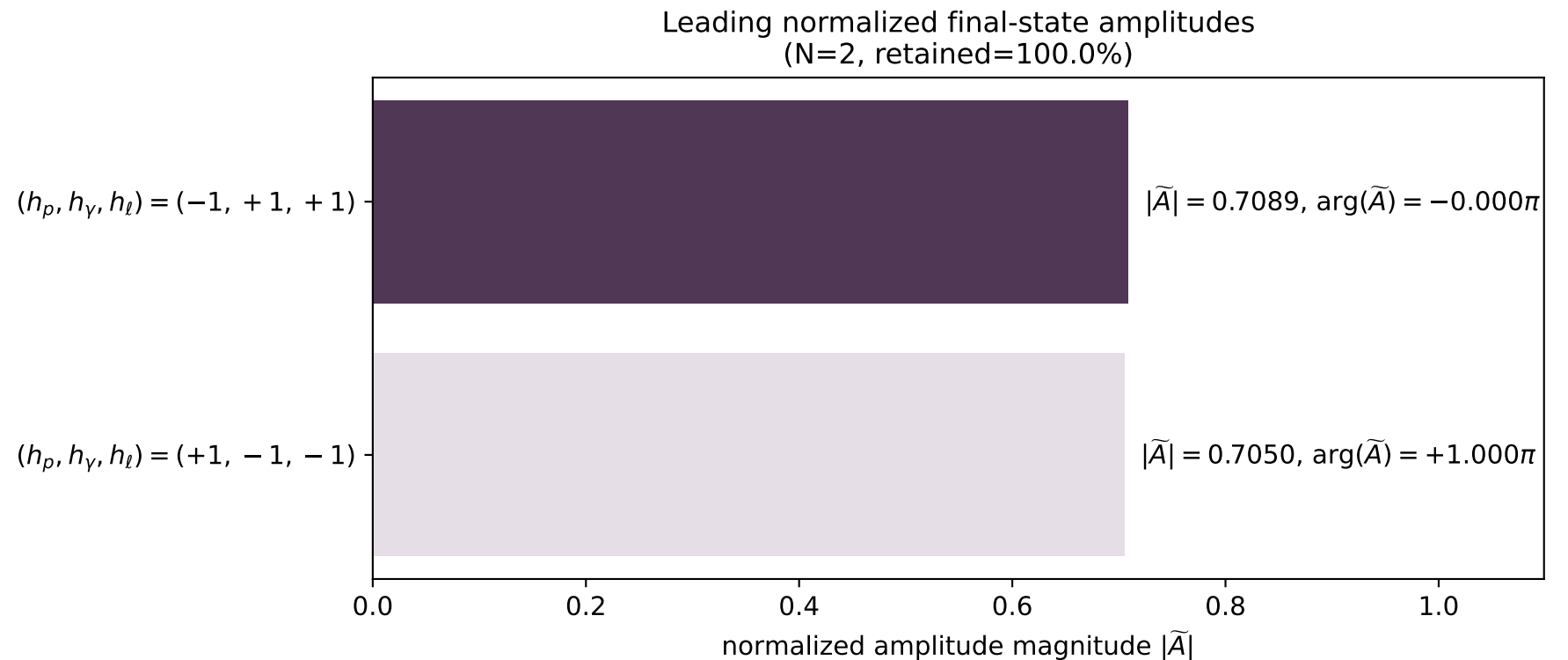


dghz_mixing_angles_polarization_01_local_minimum_219
 $d_{\text{GHZ}}=4.26109\text{e-}05$
 region: polarization_cluster_01_local_minimum_219
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0219
 lepton mixing angle: $\alpha_e=0.90729952$ rad
 incoming lepton: $0.615876|+\rangle + 0.787843|-\rangle$
 proton mixing angle: $\alpha_p=0.66718354$ rad
 incoming proton: $0.785568|+\rangle + 0.618776|-\rangle$

$s=17.2279$, $\sqrt{s}=4.15065$, $|\vec{P}|=1.96934$, $|\vec{P}'|=1.71682$
 $\theta_{p'}=0.00016275$, $\theta_\gamma=3.12967$, $\phi_{p'}=0.000585085$, $\phi_\gamma=1.82782\text{e-}05$
 $E_\gamma=1.95504$, $\theta_{l'}=0.0987232$, $\phi_{l'}=3.14162$
 $Q^2=1.88008$, $x_B=0.115754$, $t=-0.0131571$, $W^2=15.2418$, $y=0.993515$

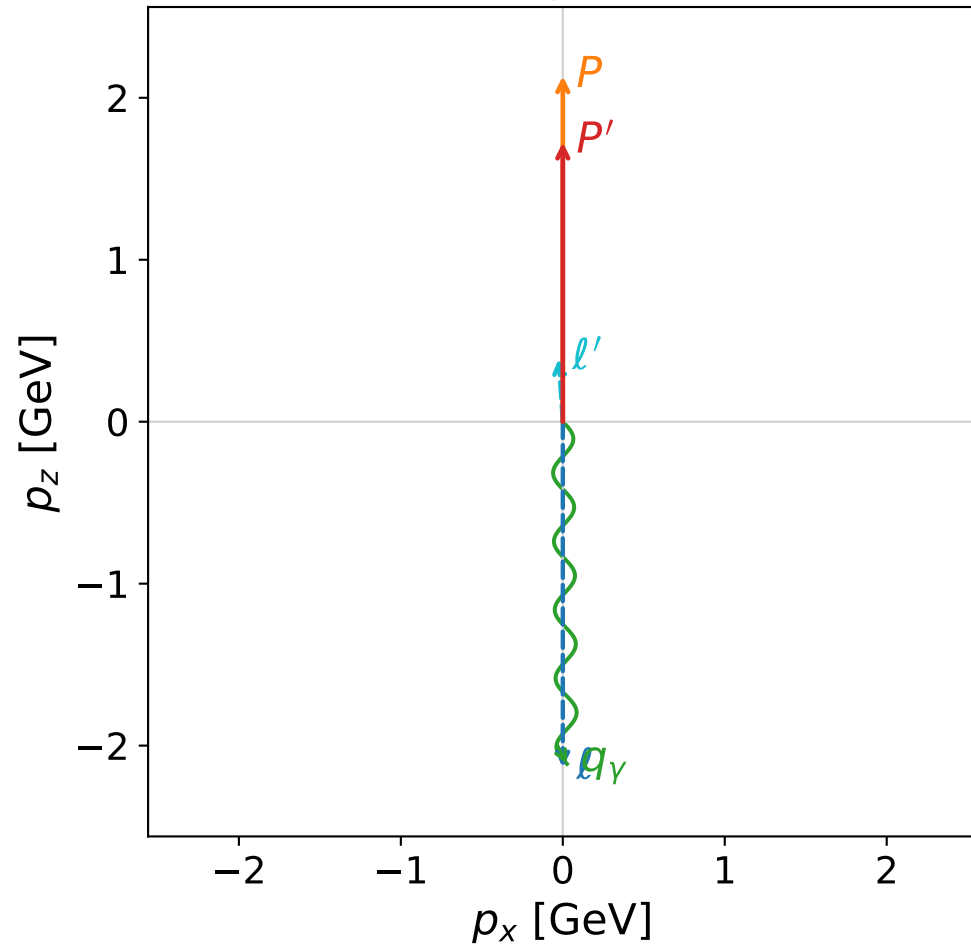
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.9693$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9693$
 P $E= 2.1813$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9693$
 l' $E= 0.2393$ $p_x= -0.0236$ $p_y= -0.0000$ $p_z= 0.2381$
 P' $E= 1.9564$ $p_x= 0.0003$ $p_y= 0.0000$ $p_z= 1.7168$
 q_γ $E= 1.9550$ $p_x= 0.0233$ $p_y= 0.0000$ $p_z= -1.9549$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.052%, conserve 99.948%
 $R_{\text{flip/conserve}}=5.222\text{e-}04$; $I_h=-6.369\text{e-}05$
 virtual-photon powers: T 99.985%, L 0.015%
 $R_{L/T}=1.483\text{e-}04$; $I_{TL}=-3.848\text{e-}04$

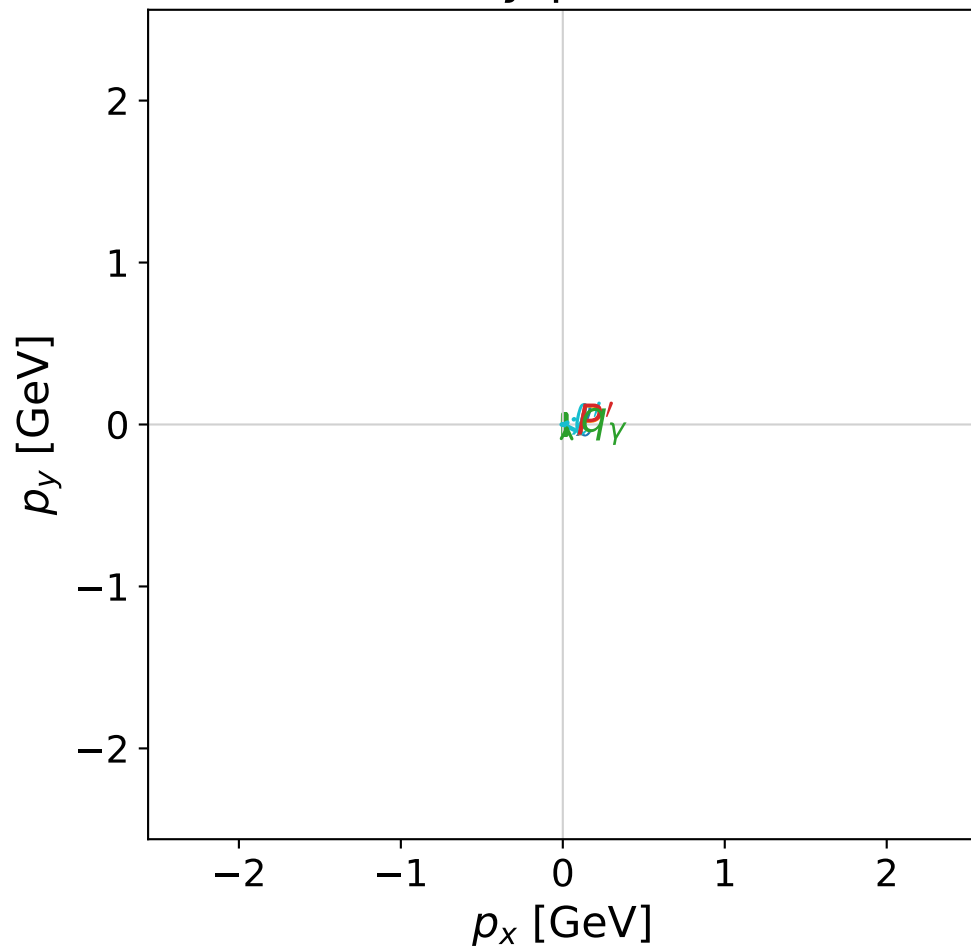


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



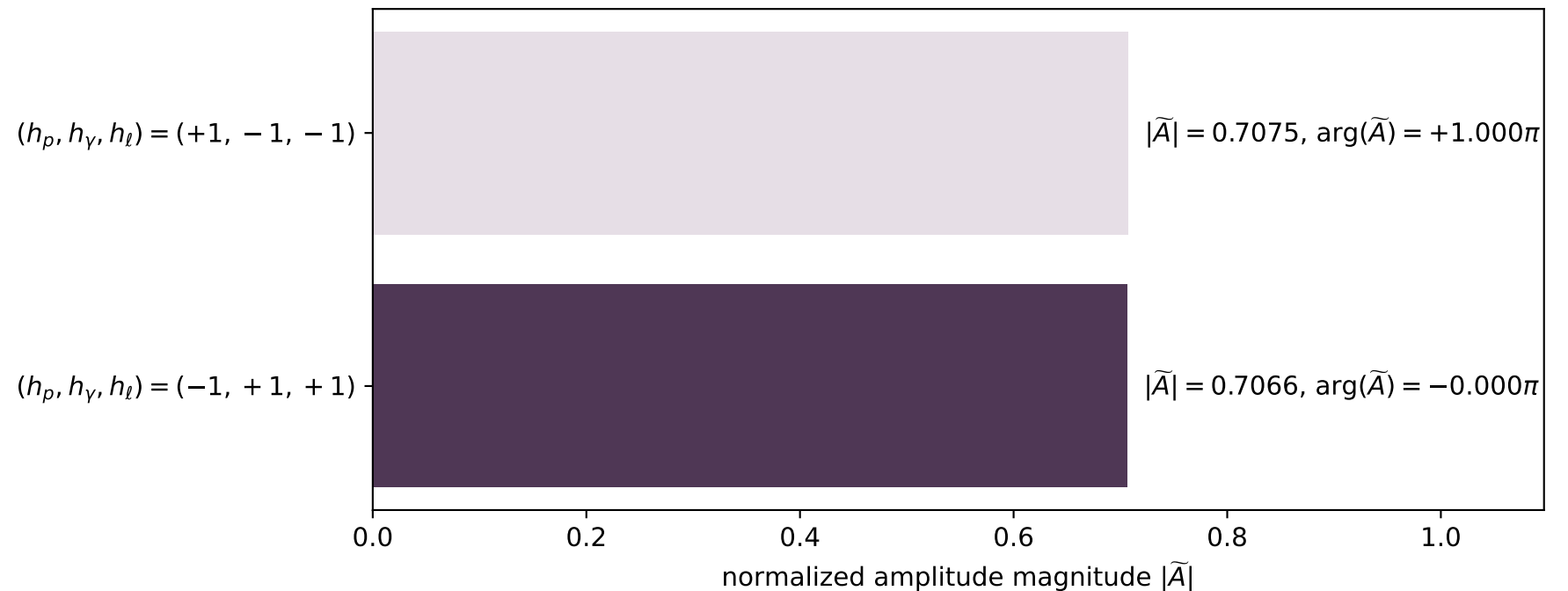
dghz_mixing_angles_polarization_01_local_minimum_231
 $d_{\text{GHZ}}=5.7484\text{e-}05$
 region: polarization_cluster_01_local_minimum_231
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0231
 lepton mixing angle: $\alpha_e=0.97671817$ rad
 incoming lepton: $0.559745|+\rangle + 0.828665|-\rangle$
 proton mixing angle: $\alpha_p=0.59468799$ rad
 incoming proton: $0.828323|+\rangle + 0.56025|-\rangle$

$s=19.9347$, $\sqrt{s}=4.46483$, $|\vec{P}|=2.13388$, $|\vec{P}'|=1.72462$
 $\theta_{P'}=0$, $\theta_\gamma=3.12944$, $\phi_{P'}=1.59257$, $\phi_\gamma=6.28318$
 $E_\gamma=2.11278$, $\theta_{l'}=0.066056$, $\phi_{l'}=3.14158$
 $Q^2=3.31544$, $x_B=0.175439$, $t=-0.0322616$, $W^2=16.4624$, $y=0.991767$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.1339$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1339$
 P $E= 2.3309$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1339$
 l' $E= 0.3889$ $p_x= -0.0257$ $p_y= 0.0000$ $p_z= 0.3880$
 P' $E= 1.9632$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.7246$
 q_γ $E= 2.1128$ $p_x= 0.0257$ $p_y= -0.0000$ $p_z= -2.1126$

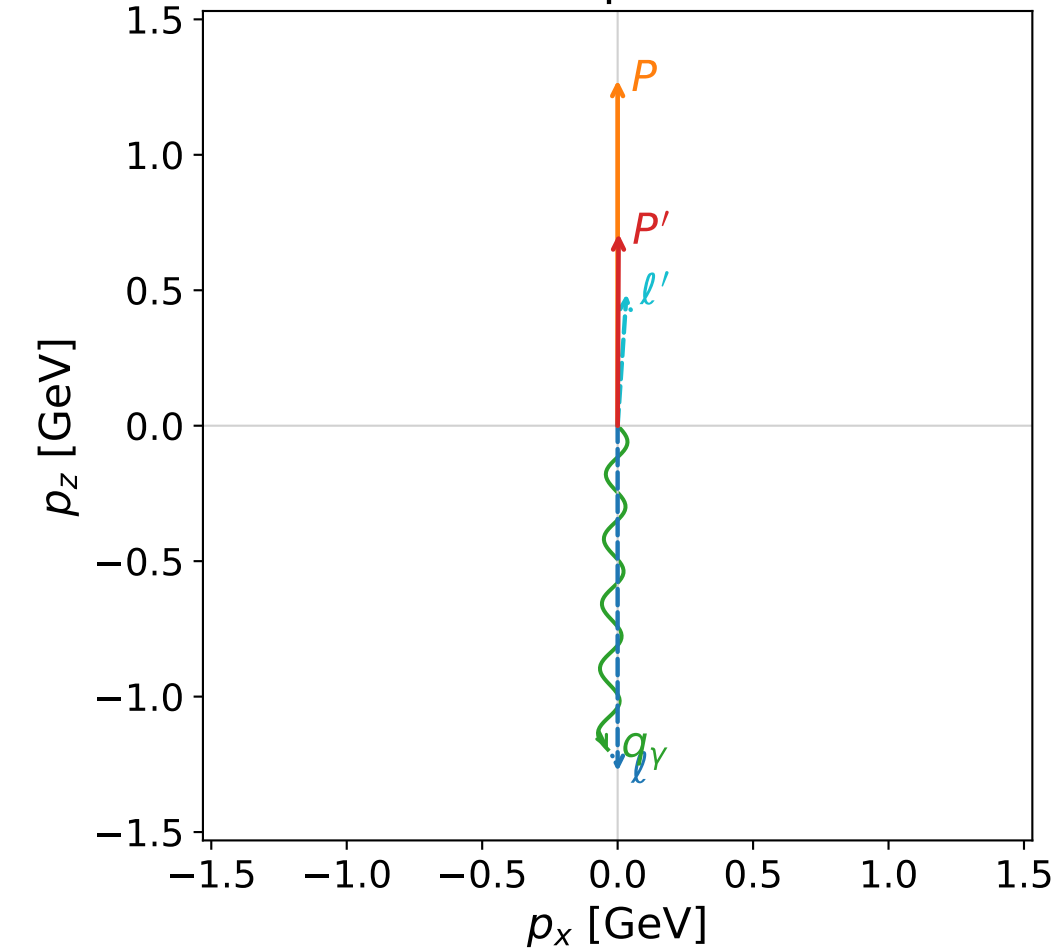
GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.050%, conserve 99.950%
 $R_{\text{flip/conserve}}=5.006\text{e-}04$; $I_h=-1.411\text{e-}04$
 virtual-photon powers: T 99.994%, L 0.006%
 $R_{\text{L/T}}=6.242\text{e-}05$; $I_{\text{TL}}=-3.503\text{e-}05$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)

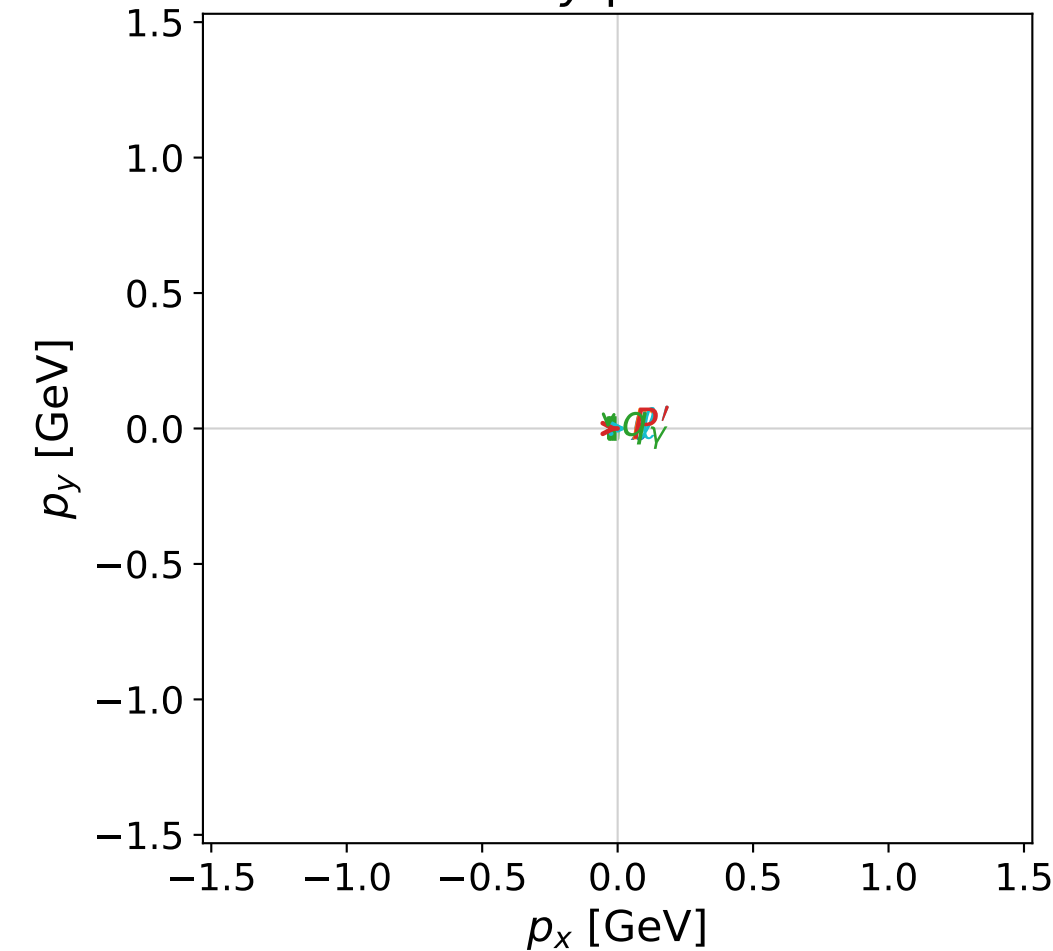


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

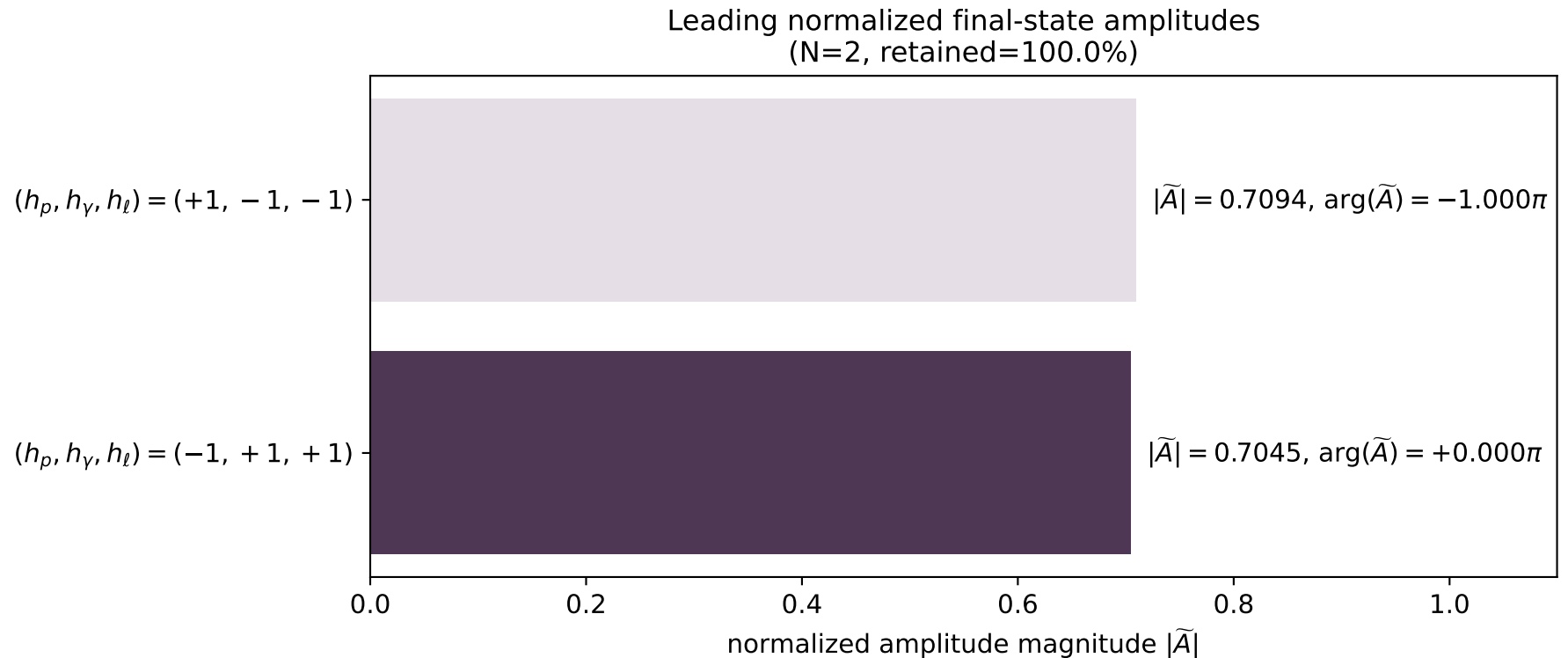


dghz_mixing_angles_polarization_01_local_minimum_236
 $d_{\text{GHZ}}=6.73836\text{e-}05$
 region: polarization_cluster_01_local_minimum_236
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0236
 lepton mixing angle: $\alpha_e=0.17076605$ rad
 incoming lepton: $0.985455|+\rangle +0.169937|-\rangle$
 proton mixing angle: $\alpha_p=1.4059166$ rad
 incoming proton: $0.164134|+\rangle +0.986438|-\rangle$

$s=8.17465$, $\sqrt{s}=2.85913$, $|\vec{P}|=1.2757$, $|\vec{P}'|=0.709145$
 $\theta_{P'}=0.00496652$, $\theta_\gamma=3.11136$, $\phi_{P'}=6.28315$, $\phi_\gamma=3.14159$
 $E_\gamma=1.19591$, $\theta_{l'}=0.0670059$, $\phi_{l'}=4.64873\text{e-}06$
 $Q^2=2.48393$, $x_B=0.355249$, $t=-0.154922$, $W^2=5.38799$, $y=0.958502$

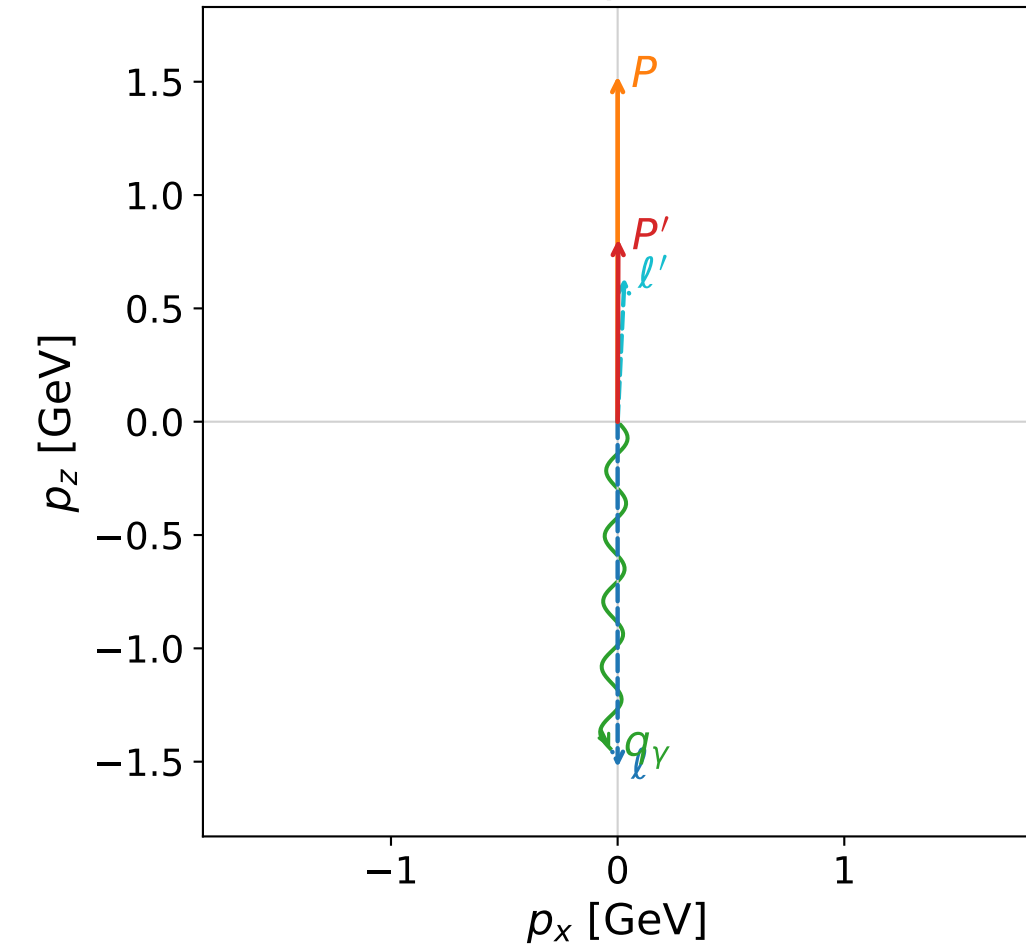
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.2757$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.2757$
 P $E= 1.5834$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2757$
 l' $E= 0.4873$ $p_x= 0.0326$ $p_y= 0.0000$ $p_z= 0.4862$
 P' $E= 1.1759$ $p_x= 0.0035$ $p_y= -0.0000$ $p_z= 0.7091$
 q_γ $E= 1.1959$ $p_x= -0.0362$ $p_y= -0.0000$ $p_z= -1.1954$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.259%, conserve 99.741%
 $R_{\text{flip/conserve}}=2.592\text{e-}03$; $I_h=-4.256\text{e-}03$
 virtual-photon powers: T 99.827%, L 0.173%
 $R_{L/T}=1.731\text{e-}03$; $I_{TL}=+8.687\text{e-}03$

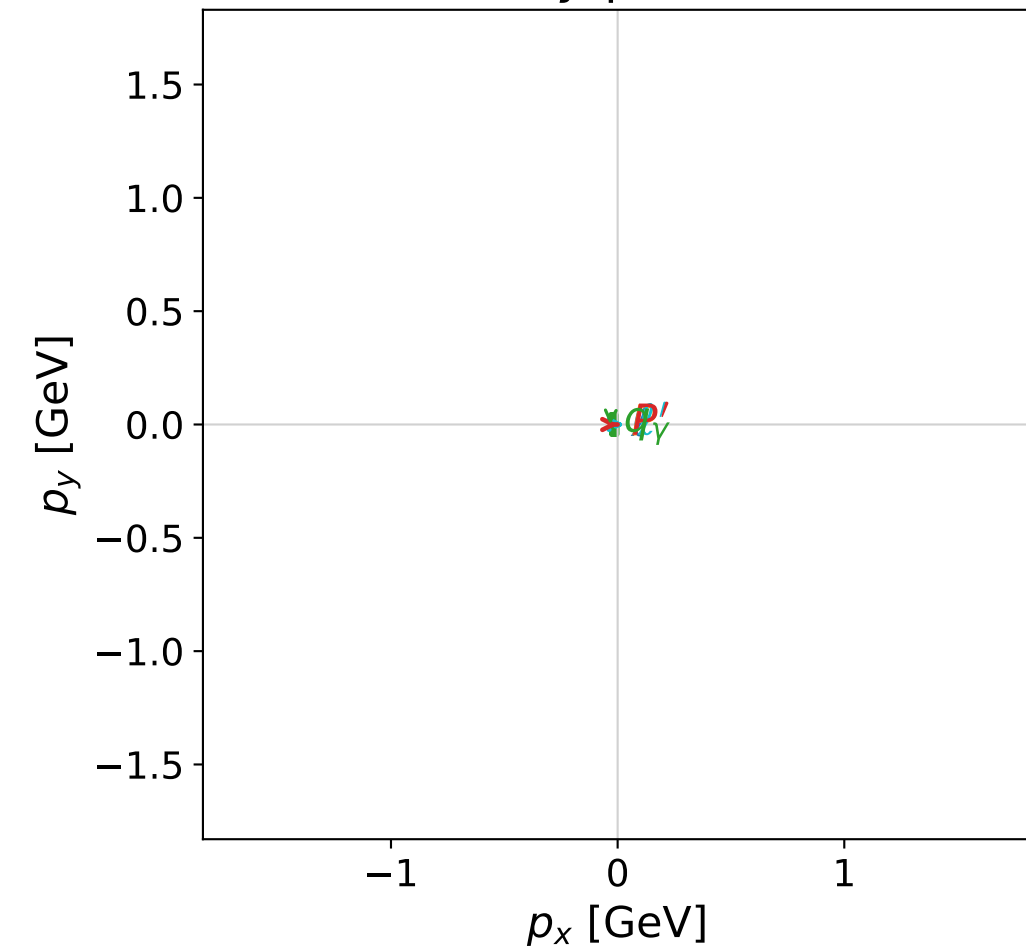


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

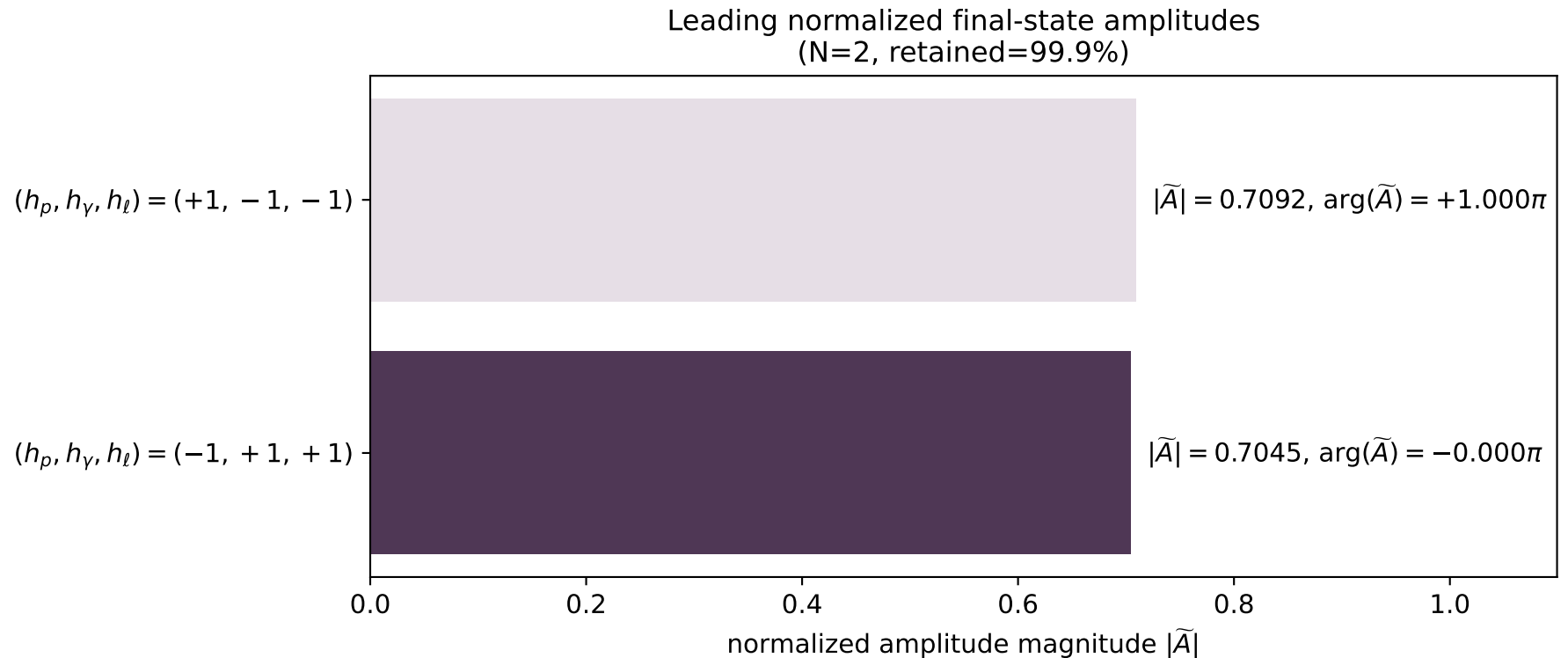


dghz_mixing_angles_polarization_01_local_minimum_241
 $d_{\text{GHZ}}=8.98184\text{e-}05$
 region: polarization_cluster_01_local_minimum_241
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0241
 lepton mixing angle: $\alpha_e=0.17233494$ rad
 incoming lepton: $0.985187|+\rangle +0.171483|-\rangle$
 proton mixing angle: $\alpha_p=1.4019813$ rad
 incoming proton: $0.168014|+\rangle +0.985785|-\rangle$

$s=10.9892$, $\sqrt{s}=3.31499$, $|\vec{P}|=1.52479$, $|\vec{P}'|=0.805866$
 $\theta_{P'}=0.0033848$, $\theta_Y=3.11869$, $\phi_{P'}=0.000438679$, $\phi_Y=3.14159$
 $E_Y=1.44194$, $\theta_{l'}=0.0476231$, $\phi_{l'}=6.28315$
 $Q^2=3.87942$, $x_B=0.397103$, $t=-0.210426$, $W^2=6.76972$, $y=0.966365$

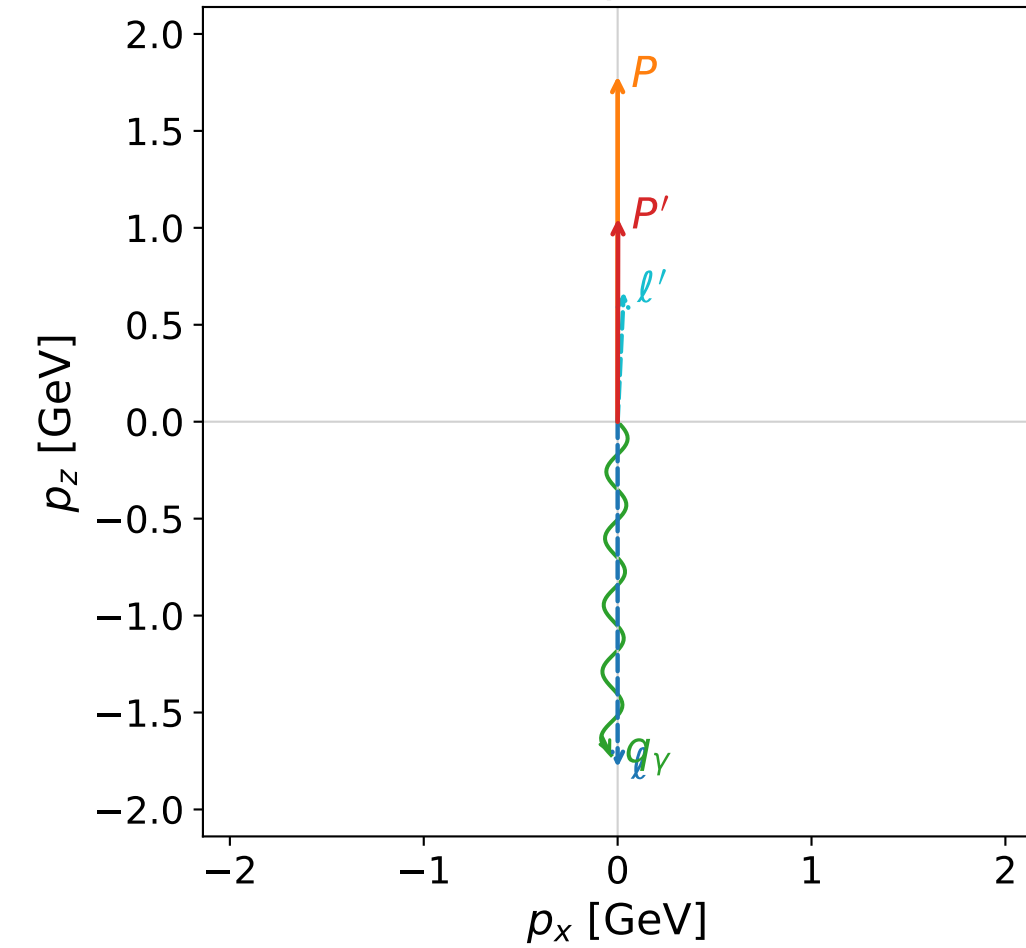
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.5248$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5248$
 P $E= 1.7902$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5248$
 l' $E= 0.6364$ $p_x= 0.0303$ $p_y= -0.0000$ $p_z= 0.6357$
 P' $E= 1.2366$ $p_x= 0.0027$ $p_y= 0.0000$ $p_z= 0.8059$
 q_Y $E= 1.4419$ $p_x= -0.0330$ $p_y= -0.0000$ $p_z= -1.4416$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.315%, conserve 99.685%
 $R_{\text{flip/conserve}}=3.165\text{e-}03$; $I_h=-4.125\text{e-}03$
 virtual-photon powers: T 99.943%, L 0.057%
 $R_L/T=5.654\text{e-}04$; $I_{TL}=+3.842\text{e-}03$

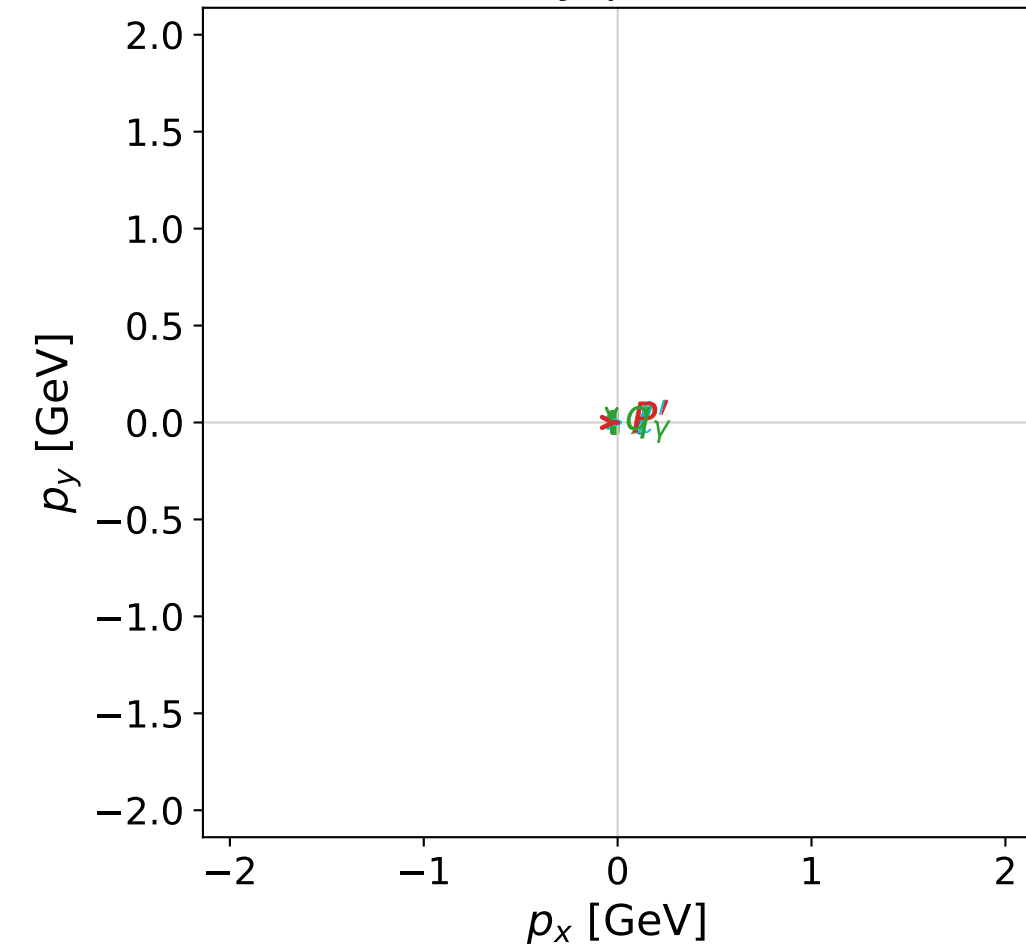


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

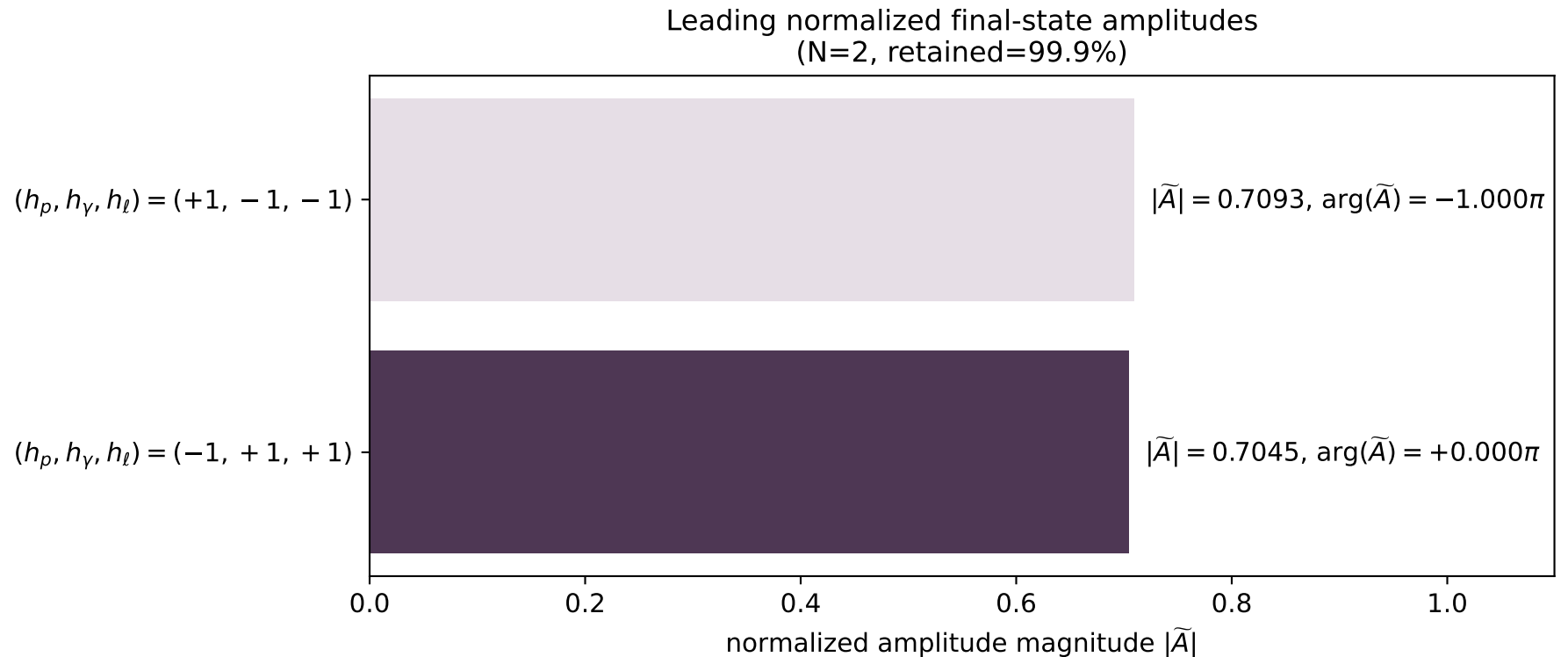


dghz_mixing_angles_polarization_01_local_minimum_242
 $d_{\text{GHZ}}=0.000102039$
 region: polarization_cluster_01_local_minimum_242
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0242
 lepton mixing angle: $\alpha_e=0.24367421$ rad
 incoming lepton: $0.970458|+\rangle +0.24127|-\rangle$
 proton mixing angle: $\alpha_p=1.3312341$ rad
 incoming proton: $0.237277|+\rangle +0.971442|-\rangle$

$s=14.4155$, $\sqrt{s}=3.79678$, $|\vec{P}|=1.78252$, $|\vec{P}'|=1.0492$
 $\theta_{P'}=0.00202721$, $\theta_\gamma=3.12202$, $\phi_{P'}=6.28087$, $\phi_\gamma=3.14159$
 $E_\gamma=1.7191$, $\theta_{l'}=0.0470478$, $\phi_{l'}=0.000148497$
 $Q^2=4.77679$, $x_B=0.361266$, $t=-0.169449$, $W^2=9.32542$, $y=0.976852$

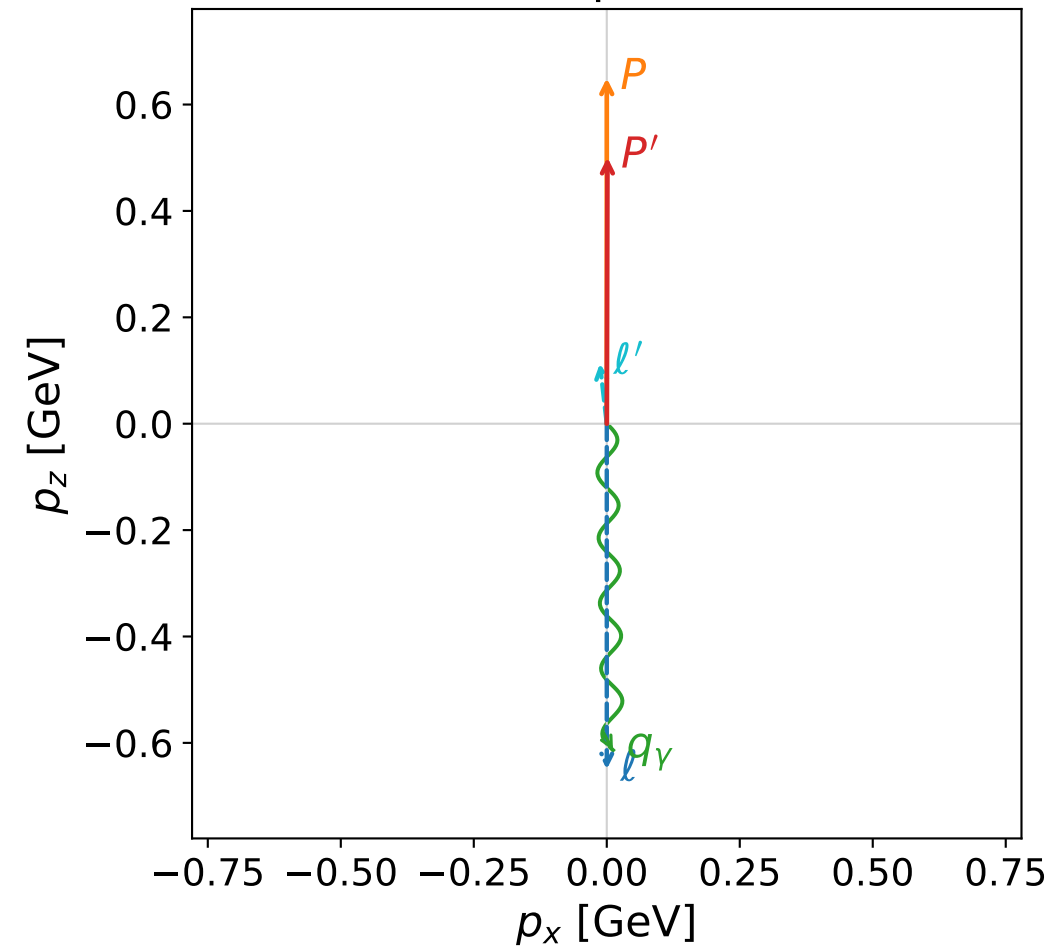
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.7825$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7825$
 P $E= 2.0143$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7825$
 l' $E= 0.6703$ $p_x= 0.0315$ $p_y= 0.0000$ $p_z= 0.6696$
 P' $E= 1.4074$ $p_x= 0.0021$ $p_y= -0.0000$ $p_z= 1.0492$
 q_γ $E= 1.7191$ $p_x= -0.0337$ $p_y= 0.0000$ $p_z= -1.7188$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.162%, conserve 99.838%
 $R_{\text{flip}}/\text{conserve}=1.622\text{e-}03$; $I_h=-1.865\text{e-}03$
 virtual-photon powers: T 99.980%, L 0.020%
 $R_L/T=2.036\text{e-}04$; $I_{TL}=+2.115\text{e-}03$

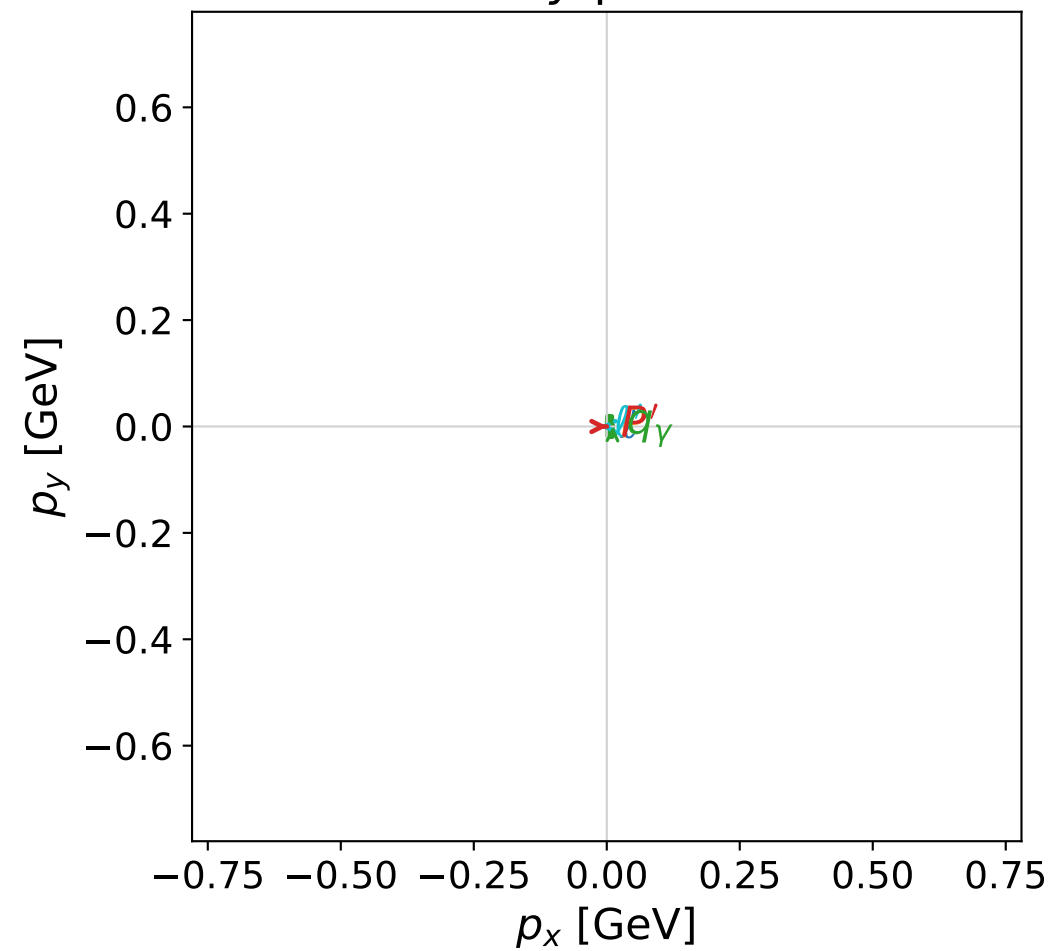


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



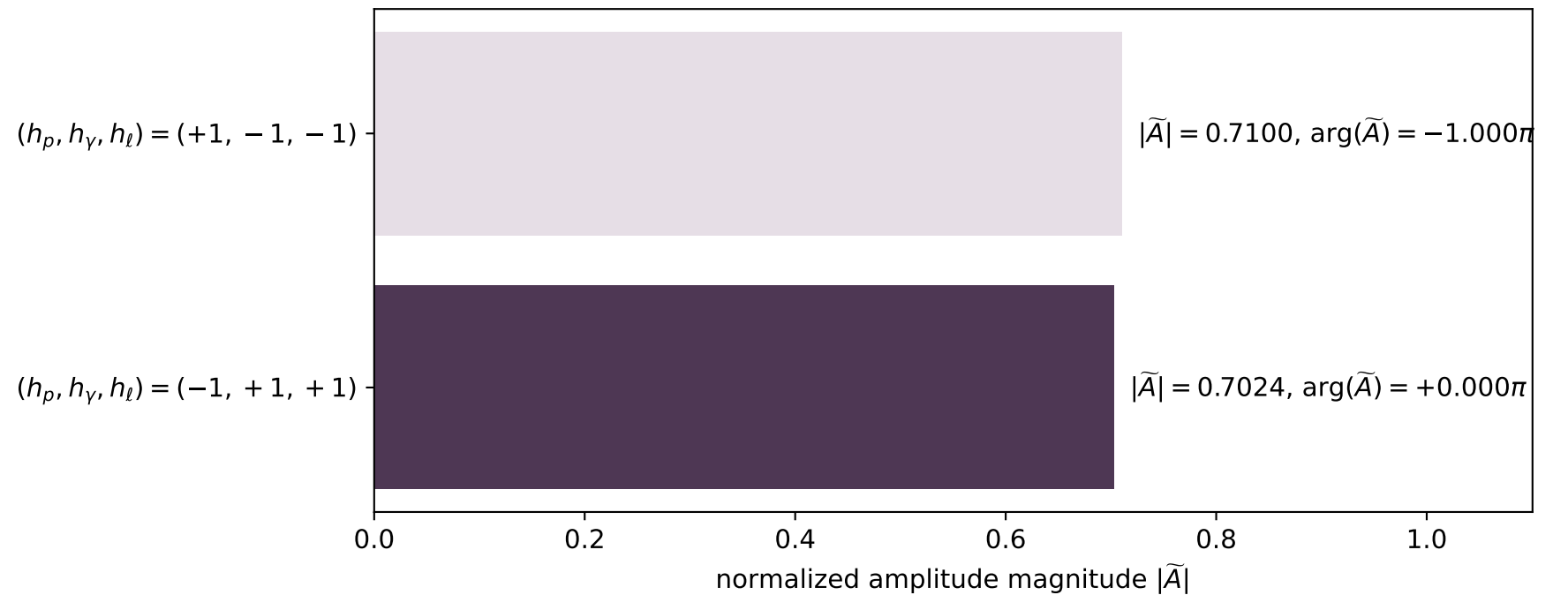
dghz_mixing_angles_polarization_01_local_minimum_249
 $d_{\text{GHZ}}=0.000132871$
 region: polarization_cluster_01_local_minimum_249
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0249
 lepton mixing angle: $\alpha_e=1.1524667$ rad
 incoming lepton: $0.406235|+\rangle +0.913769|-\rangle$
 proton mixing angle: $\alpha_p=0.44527808$ rad
 incoming proton: $0.902491|+\rangle +0.430709|-\rangle$

$s=3.20651$, $\sqrt{s}=1.79067$, $|\vec{P}|=0.649662$, $|\vec{P}'|=0.500629$
 $\theta_{p'}=0.0025126$, $\theta_\gamma=3.12259$, $\phi_{p'}=6.28317$, $\phi_\gamma=6.28316$
 $E_\gamma=0.613719$, $\theta_{l'}=0.113834$, $\phi_{l'}=3.14157$
 $Q^2=0.294553$, $x_B=0.133043$, $t=-0.0161642$, $W^2=2.79925$, $y=0.951558$

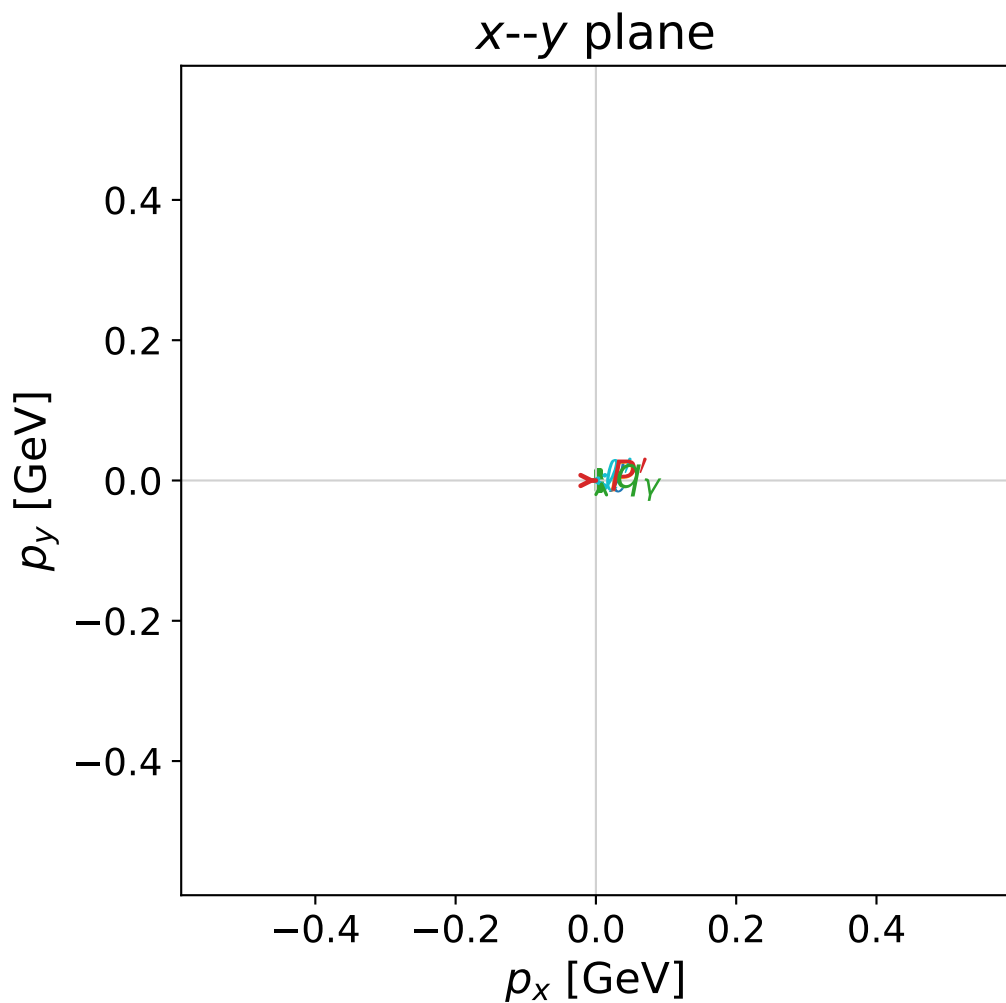
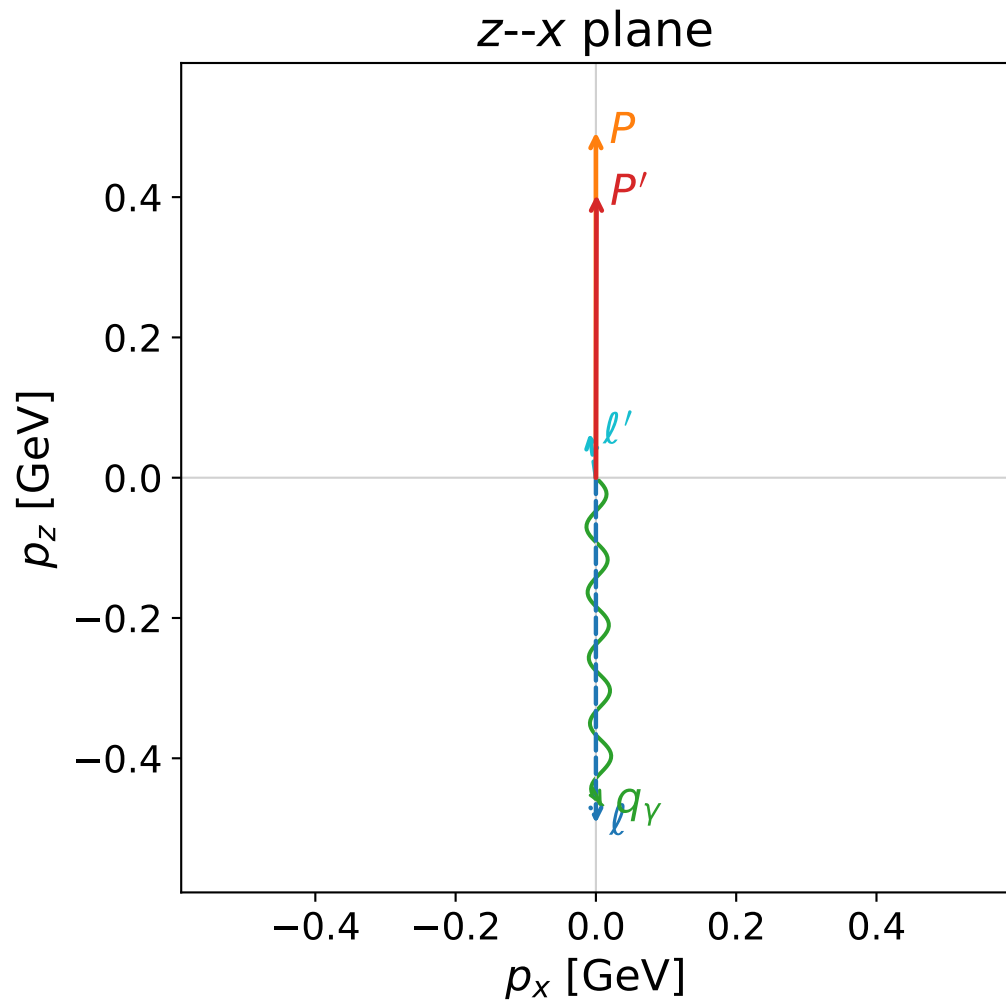
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.6497$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6497$
 P $E= 1.1410$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.6497$
 l' $E= 0.1137$ $p_x= -0.0129$ $p_y= 0.0000$ $p_z= 0.1130$
 P' $E= 1.0632$ $p_x= 0.0013$ $p_y= -0.0000$ $p_z= 0.5006$
 q_γ $E= 0.6137$ $p_x= 0.0117$ $p_y= -0.0000$ $p_z= -0.6136$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.379%, conserve 99.621%
 $R_{\text{flip/conserve}}=3.807\text{e-}03$; $I_h=-3.235\text{e-}03$
 virtual-photon powers: T 99.685%, L 0.315%
 $R_L/T=3.157\text{e-}03$; $I_{TL}=-2.381\text{e-}02$

Leading normalized final-state amplitudes
 (N=2, retained=99.8%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)



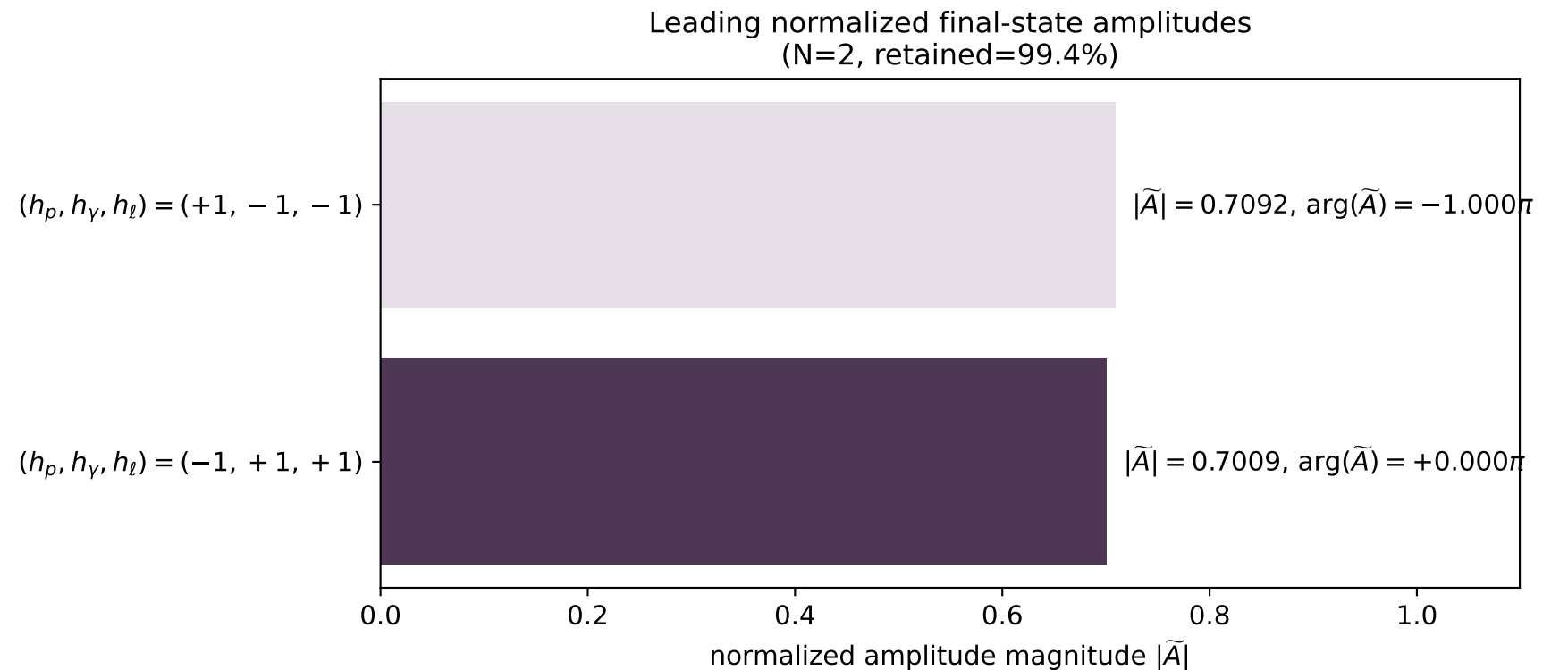
dghz_mixing_angles_polarization_01_local_minimum_253
 $d_{\text{GHZ}}=0.000141867$
 region: polarization_cluster_01_local_minimum_253
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0253
 lepton mixing angle: $\alpha_e=1.0645582$ rad
 incoming lepton: $0.484891|+\rangle +0.874575|-\rangle$
 proton mixing angle: $\alpha_p=0.55250545$ rad
 incoming proton: $0.851212|+\rangle +0.524822|-\rangle$

$s=2.40938$, $\sqrt{s}=1.55222$, $|\vec{P}|=0.492693$, $|\vec{P}'|=0.403492$
 $\theta_{P'}=0.00213162$, $\theta_Y=3.12381$, $\phi_{P'}=6.28318$, $\phi_Y=2.67404e-05$
 $E_Y=0.46701$, $\theta_{l'}=0.143434$, $\phi_{l'}=3.14162$
 $Q^2=0.125686$, $x_B=0.0863101$, $t=-0.00648154$, $W^2=2.21037$, $y=0.952061$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

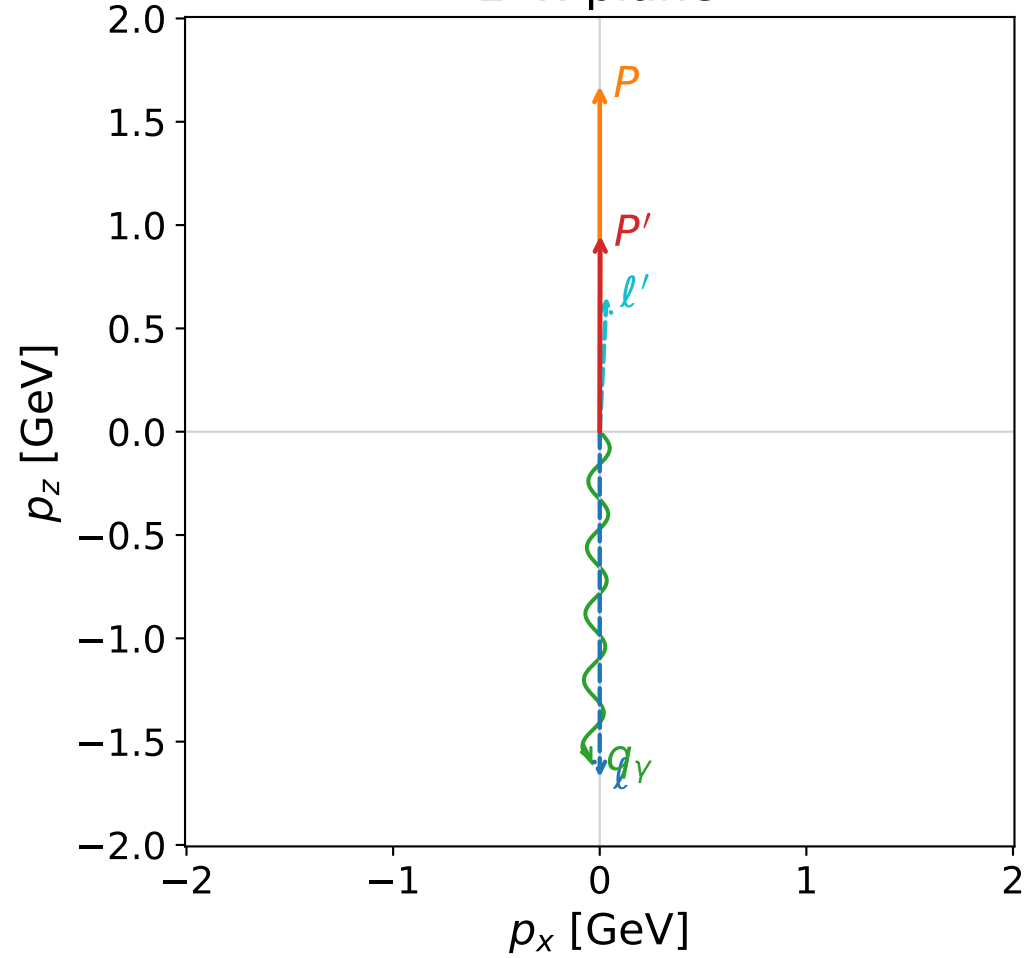
Particle	E	p_x	p_y	p_z
l	0.4927	0.0000	0.0000	-0.4927
P	1.0595	0.0000	0.0000	0.4927
l'	0.0641	-0.0092	-0.0000	0.0634
P'	1.0211	0.0009	-0.0000	0.4035
q_Y	0.4670	0.0083	0.0000	-0.4669

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.497%, conserve 99.503%
 $R_{\text{flip/conserve}}=4.998e-03$; $I_h=-2.272e-03$
 virtual-photon powers: T 99.274%, L 0.726%
 $R_{L/T}=7.308e-03$; $I_{TL}=-3.115e-02$

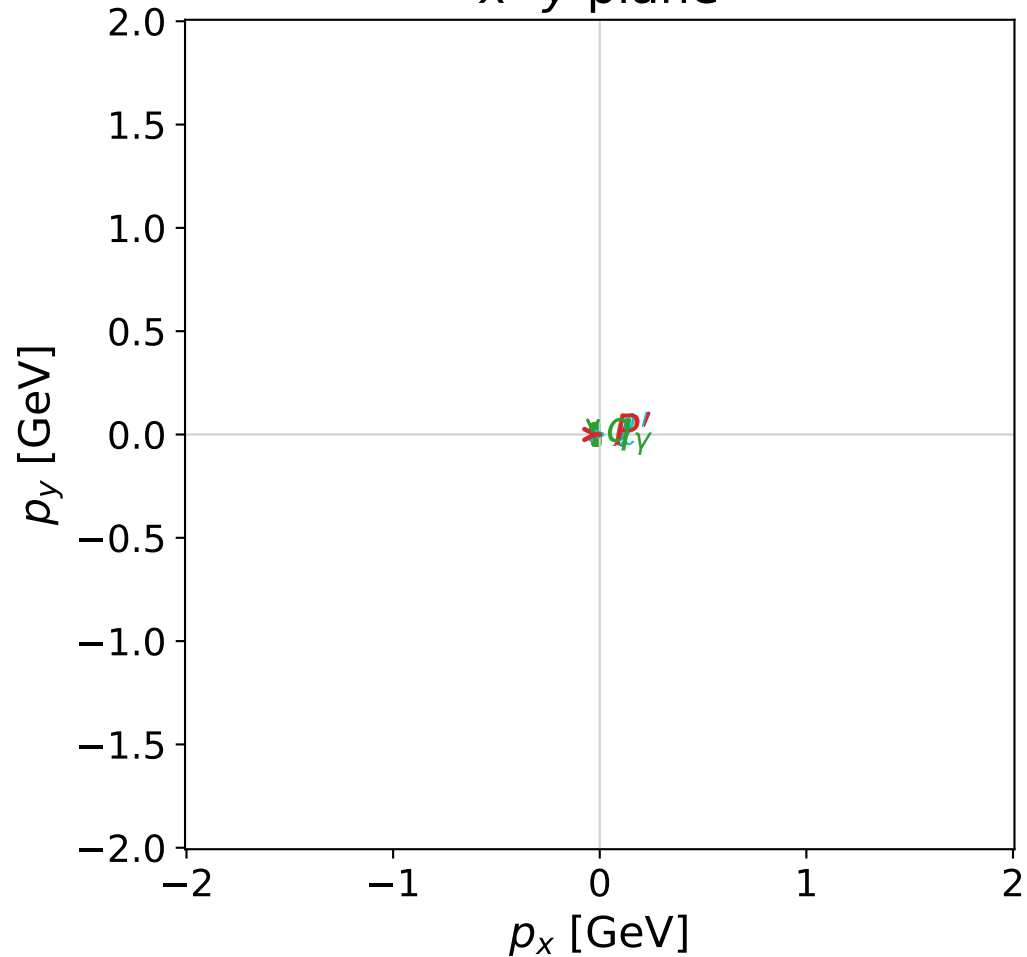


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

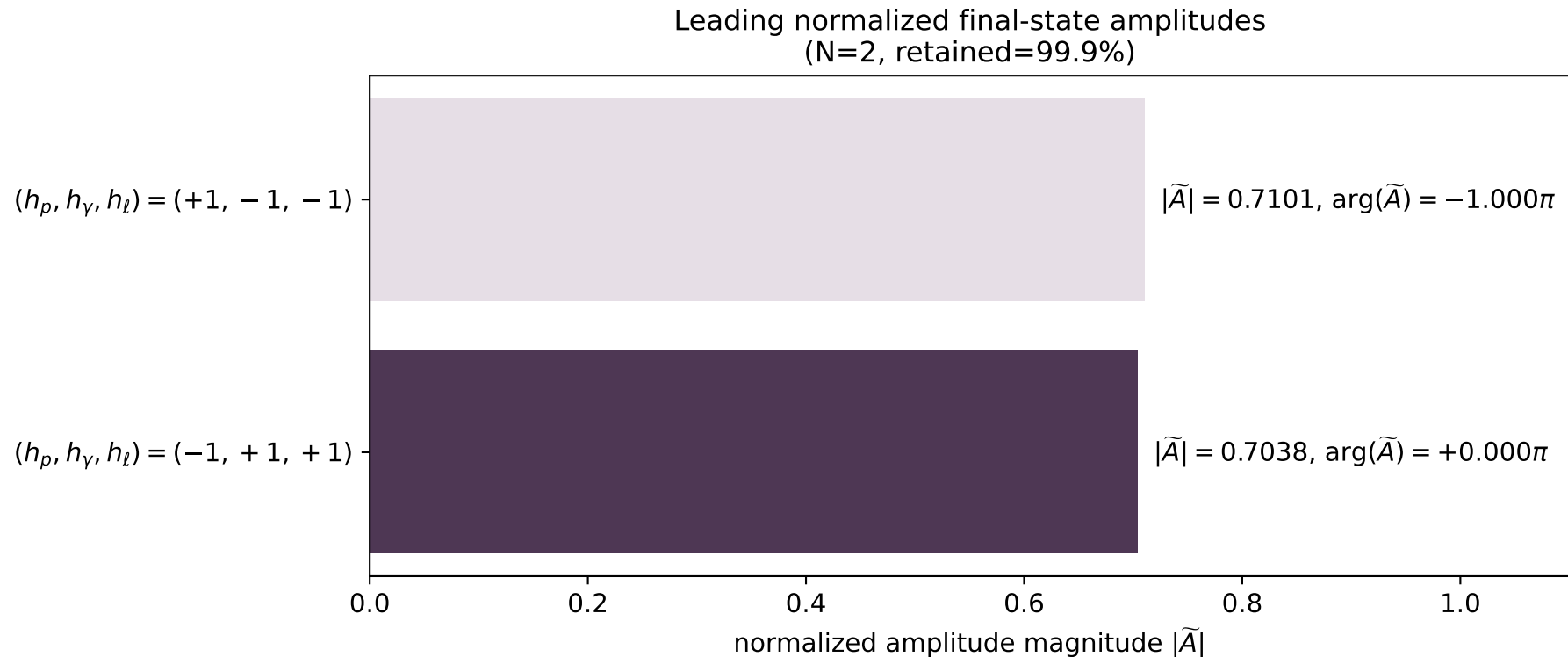


dghz_mixing_angles_polarization_01_local_minimum_266
 $d_{\text{GHZ}}=0.000189044$
 region: polarization_cluster_01_local_minimum_266
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0266
 lepton mixing angle: $\alpha_e=0.20844456$ rad
 incoming lepton: $0.978354|+\rangle + 0.206938|-\rangle$
 proton mixing angle: $\alpha_p=1.3671115$ rad
 incoming proton: $0.202279|+\rangle + 0.979328|-\rangle$

$s=12.8901$, $\sqrt{s}=3.59027$, $|\vec{P}|=1.67261$, $|\vec{P}'|=0.948936$
 $\theta_{P'}=0.00291456$, $\theta_\gamma=3.11931$, $\phi_{P'}=6.2819$, $\phi_\gamma=3.14164$
 $E_\gamma=1.60224$, $\theta_{l'}=0.0504001$, $\phi_{l'}=0.000160702$
 $Q^2=4.37105$, $x_B=0.374008$, $t=-0.183378$, $W^2=8.19583$, $y=0.97309$

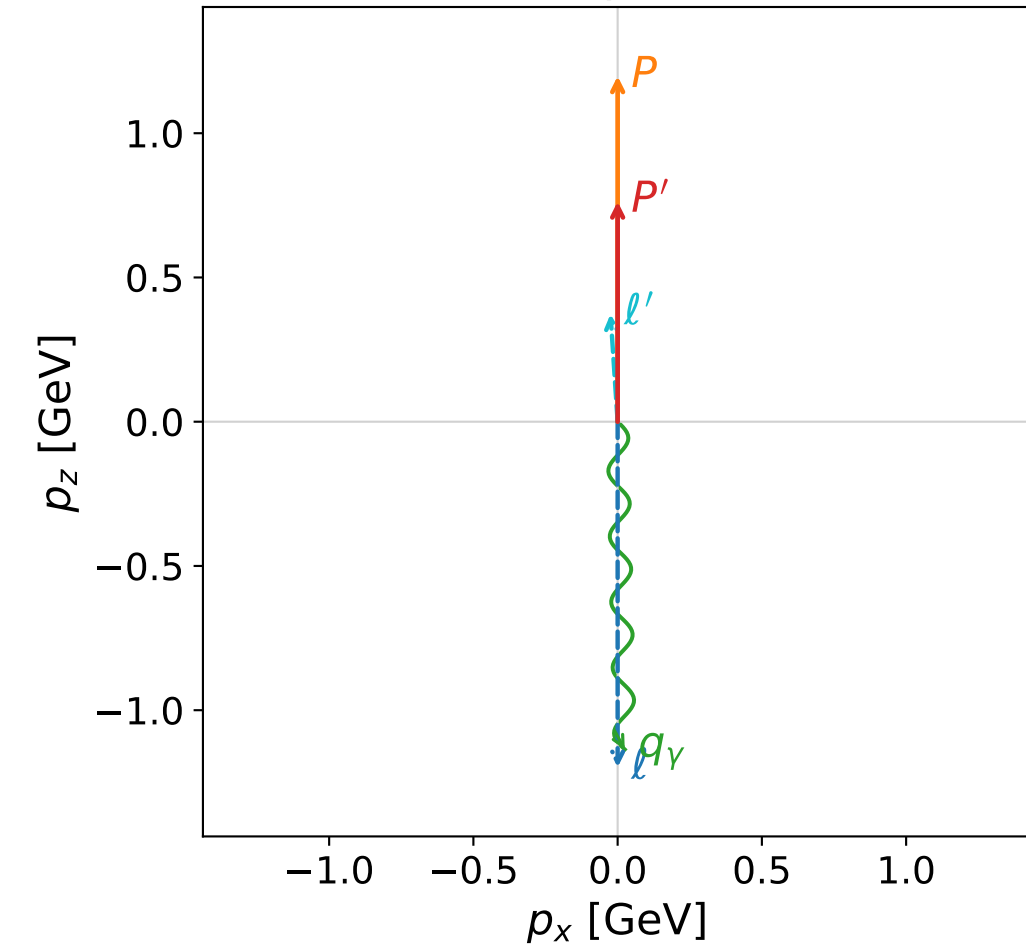
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.6726$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.6726$
 P $E= 1.9177$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.6726$
 l' $E= 0.6537$ $p_x= 0.0329$ $p_y= 0.0000$ $p_z= 0.6529$
 P' $E= 1.3343$ $p_x= 0.0028$ $p_y= -0.0000$ $p_z= 0.9489$
 q_γ $E= 1.6022$ $p_x= -0.0357$ $p_y= -0.0000$ $p_z= -1.6018$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.193%, conserve 99.807%
 $R_{\text{flip/conserve}}=1.938\text{e-}03$; $I_h=-2.458\text{e-}03$
 virtual-photon powers: T 99.963%, L 0.037%
 $R_L/T=3.675\text{e-}04$; $I_{TL}=+3.397\text{e-}03$

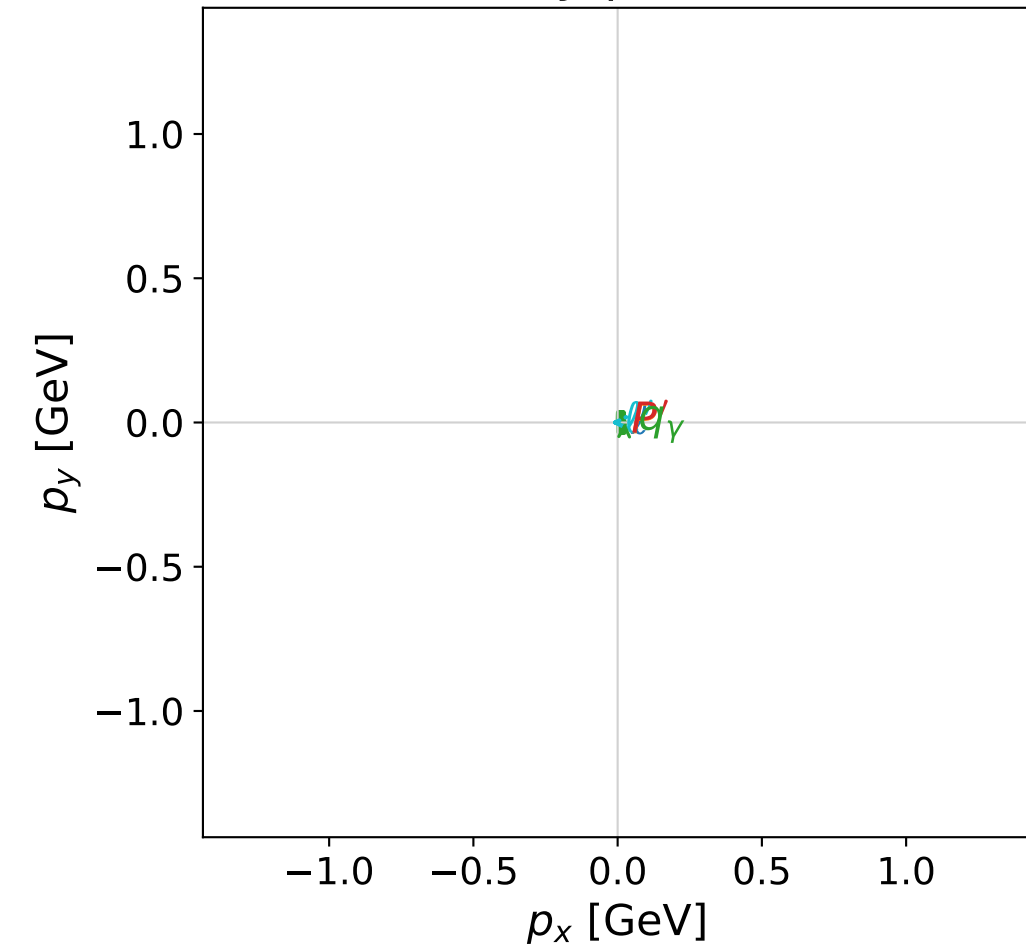


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

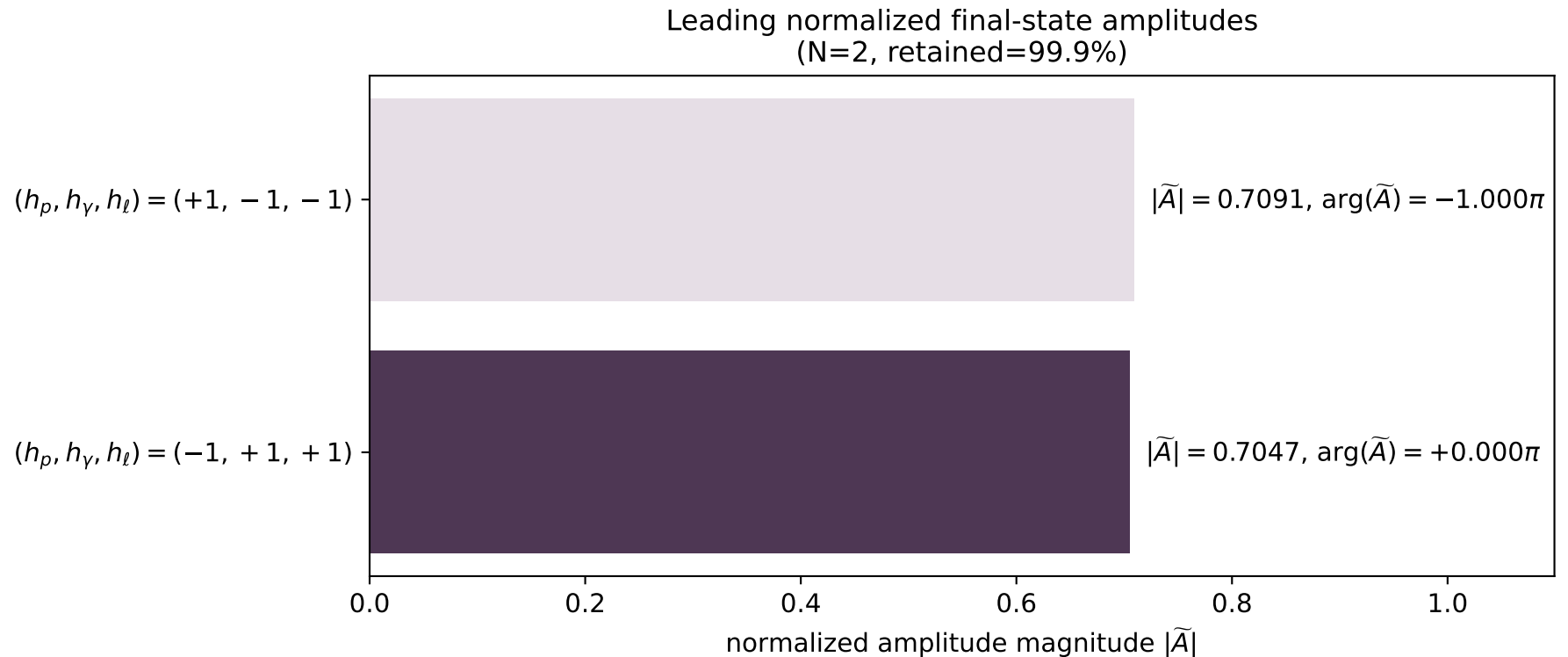


dghz_mixing_angles_polarization_01_local_minimum_314
 $d_{\text{GHZ}}=0.000387584$
 region: polarization_cluster_01_local_minimum_314
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0314
 lepton mixing angle: $\alpha_e=1.3236924$ rad
 incoming lepton: $0.244597|+\rangle +0.969625|-\rangle$
 proton mixing angle: $\alpha_p=0.24858017$ rad
 incoming proton: $0.969263|+\rangle +0.246028|-\rangle$

$s=7.3952$, $\sqrt{s}=2.71941$, $|\vec{P}|=1.19794$, $|\vec{P}'|=0.763049$
 $\theta_{P'}=0$, $\theta_\gamma=3.11984$, $\phi_{P'}=3.82165$, $\phi_\gamma=2.0169\text{e-}05$
 $E_\gamma=1.13637$, $\theta_{l'}=0.0661538$, $\phi_{l'}=3.14161$
 $Q^2=1.78954$, $x_B=0.285346$, $t=-0.091589$, $W^2=5.36178$, $y=0.962567$

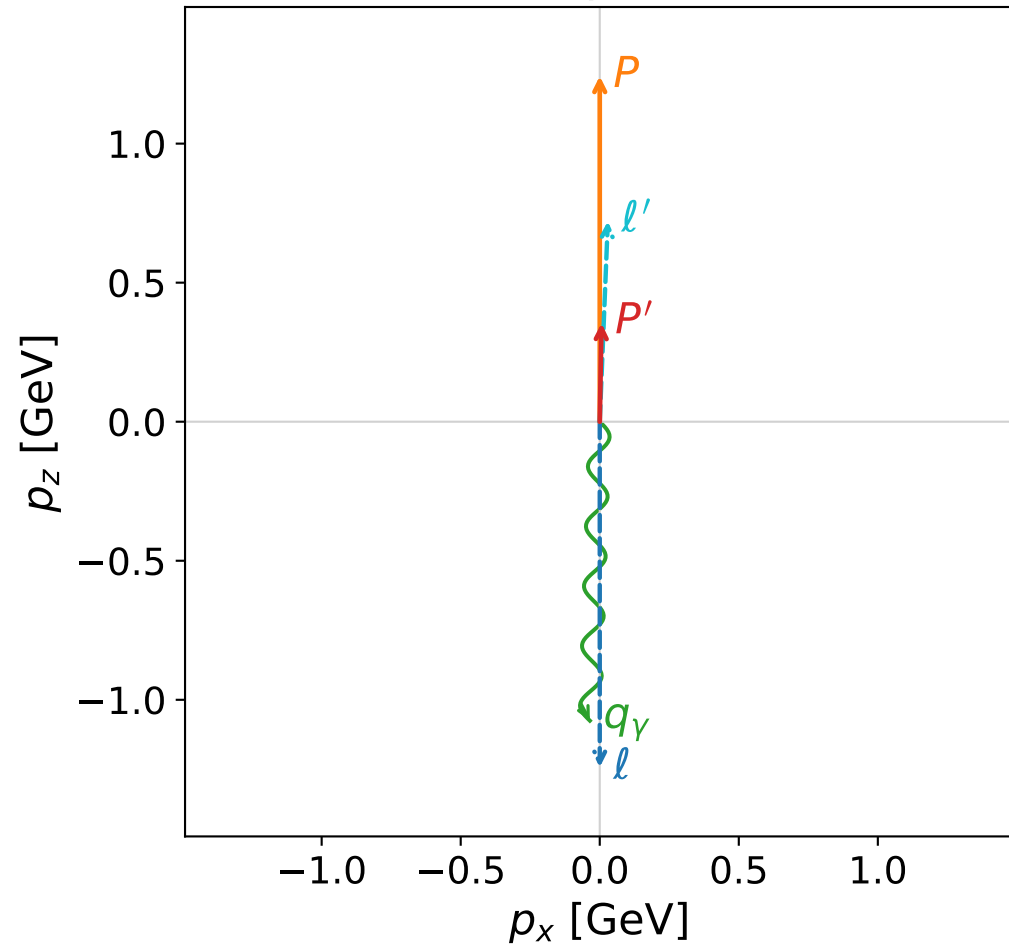
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.1979$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.1979$
 P $E= 1.5215$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.1979$
 l' $E= 0.3739$ $p_x= -0.0247$ $p_y= -0.0000$ $p_z= 0.3731$
 P' $E= 1.2092$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.7630$
 q_γ $E= 1.1364$ $p_x= 0.0247$ $p_y= 0.0000$ $p_z= -1.1361$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.268%, conserve 99.732%
 $R_{\text{flip/conserve}}=2.684\text{e-}03$; $I_h=-3.390\text{e-}03$
 virtual-photon powers: T 99.925%, L 0.075%
 $R_{L/T}=7.519\text{e-}04$; $I_{TL}=-1.684\text{e-}03$

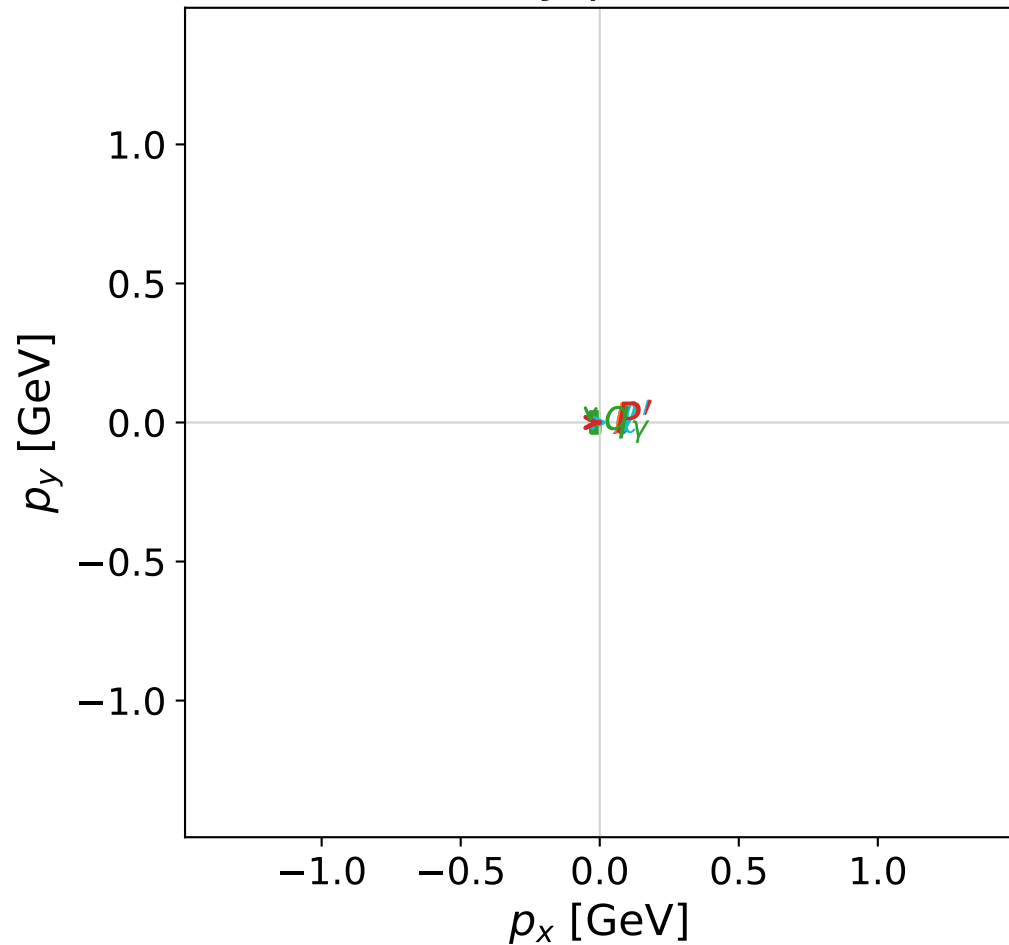


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_01_local_minimum_331
 $d_{\text{GHZ}}=0.000498586$
 region: polarization_cluster_01_local_minimum_331
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0331
 lepton mixing angle: $\alpha_e=0.063309233$ rad
 incoming lepton: $0.997997|+\rangle + 0.063267|-\rangle$
 proton mixing angle: $\alpha_p=1.5095946$ rad
 incoming proton: $0.0611635|+\rangle + 0.998128|-\rangle$

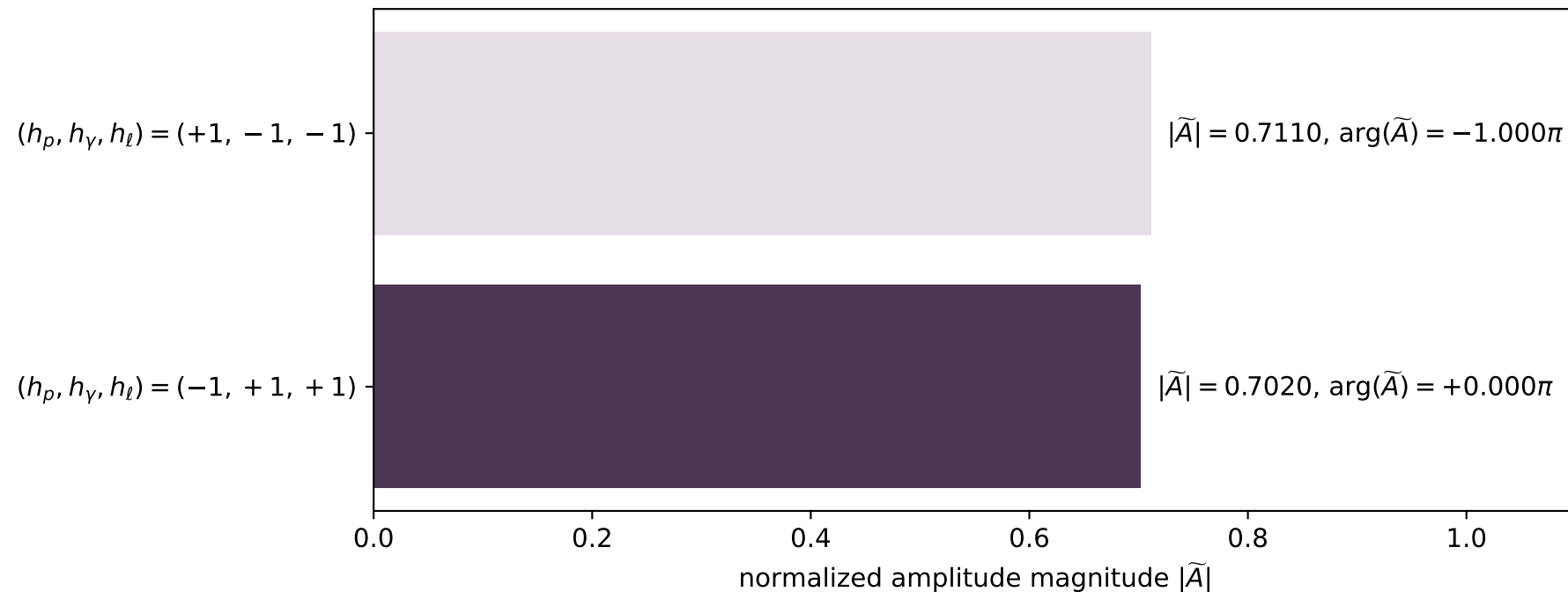
$s=7.83845$, $\sqrt{s}=2.79972$, $|\vec{P}|=1.24273$, $|\vec{P}'|=0.354558$
 $\theta_{P'}=0.0179191$, $\theta_\gamma=3.1086$, $\phi_{P'}=6.28227$, $\phi_\gamma=3.14163$
 $E_\gamma=1.07572$, $\theta_{\ell'}=0.0404006$, $\phi_{\ell'}=0.000247116$
 $Q^2=3.5837$, $x_B=0.551013$, $t=-0.481835$, $W^2=3.79999$, $y=0.934646$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

Particle	E	p_x	p_y	p_z
ℓ	1.2427	0.0000	0.0000	-1.2427
P	1.5570	0.0000	0.0000	1.2427
ℓ'	0.7212	0.0291	0.0000	0.7206
P'	1.0028	0.0064	-0.0000	0.3545
q_γ	1.0757	-0.0355	-0.0000	-1.0751

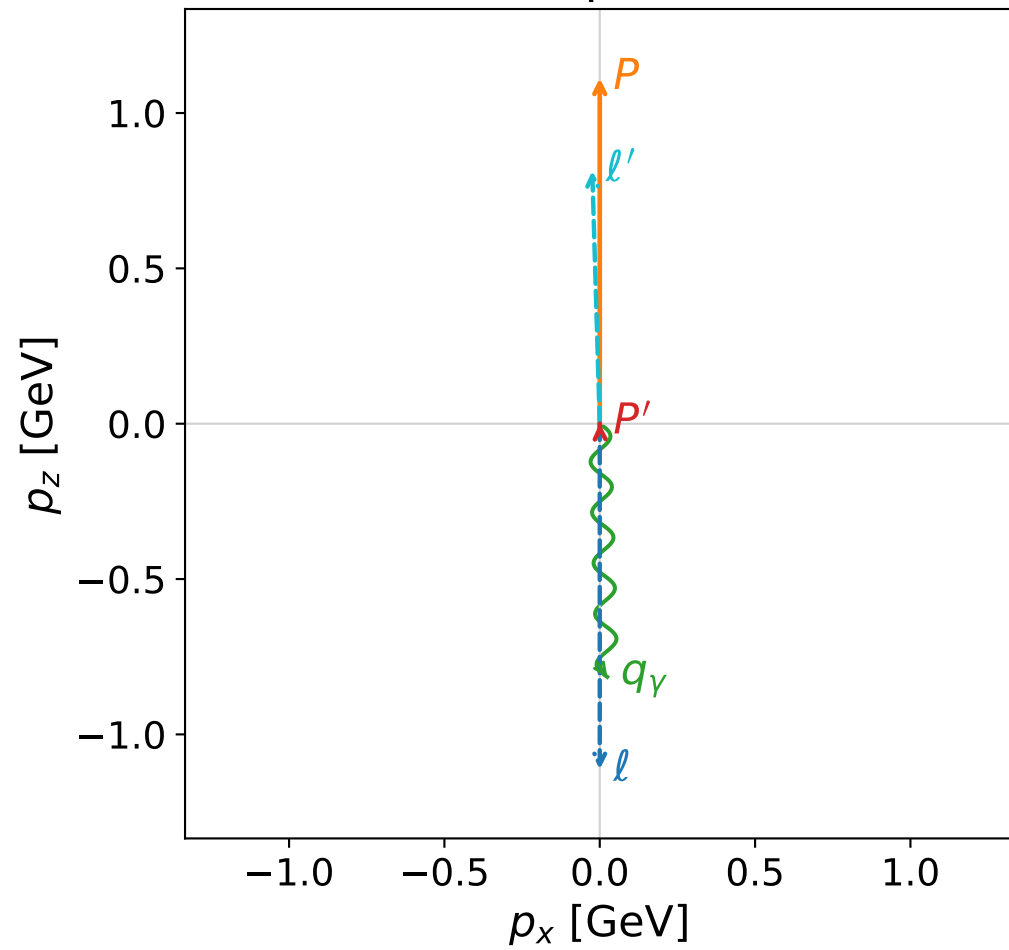
GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 1.429%, conserve 98.571%
 $R_{\text{flip/conserve}}=1.449\text{e-}02$; $I_h=-2.234\text{e-}02$
 virtual-photon powers: T 99.502%, L 0.498%
 $R_L/T=5.008\text{e-}03$; $I_{TL}=+1.176\text{e-}02$

Leading normalized final-state amplitudes (N=2, retained=99.8%)

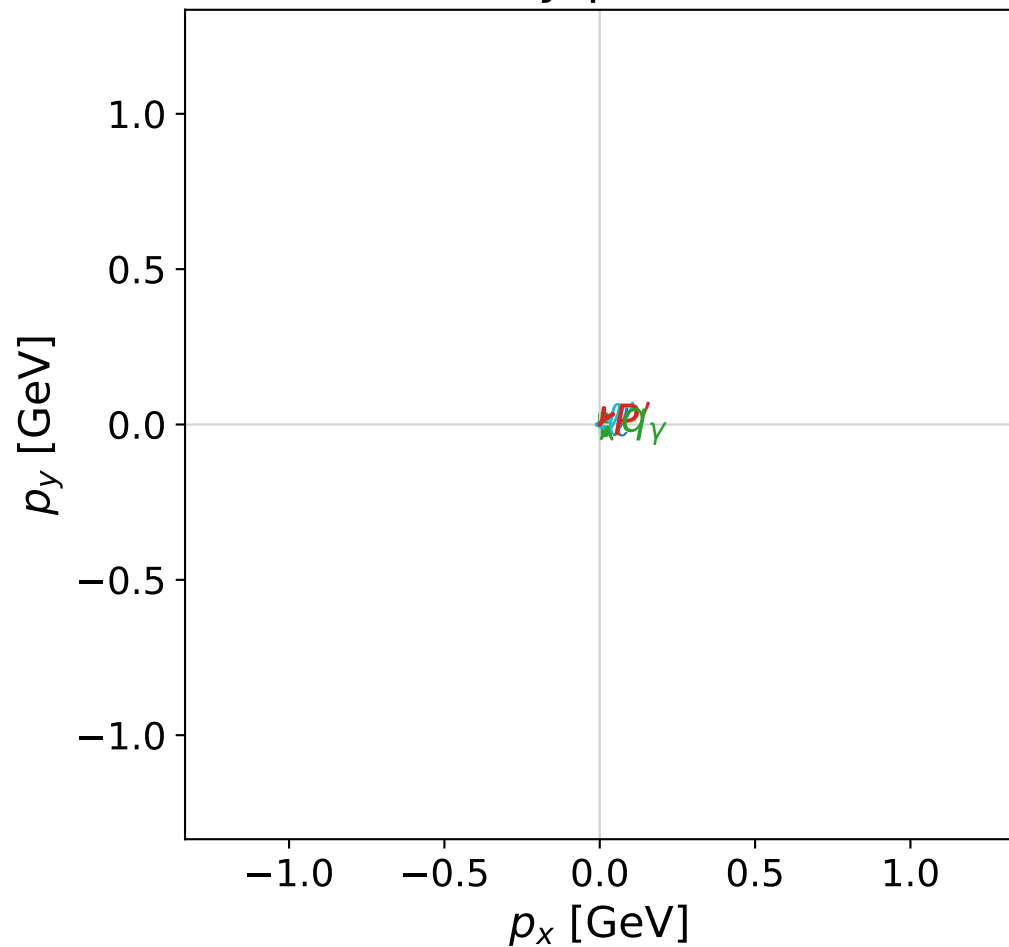


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

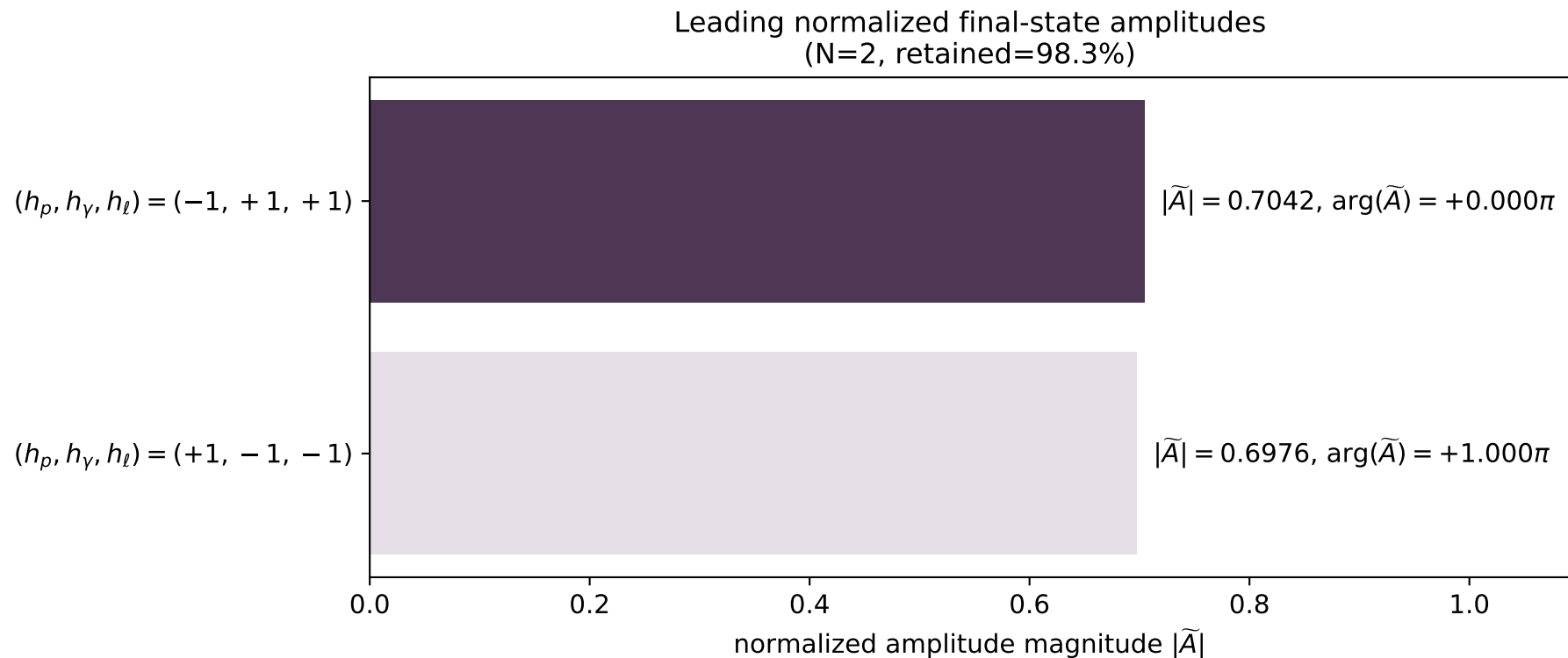


dghz_mixing_angles_polarization_01_local_minimum_333
 $d_{\text{GHZ}}=0.000507492$
 region: polarization_cluster_01_local_minimum_333
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0333
 lepton mixing angle: $\alpha_e=1.5422293$ rad
 incoming lepton: $0.0285631|+\rangle +0.999592|-\rangle$
 proton mixing angle: $\alpha_p=0.028304256$ rad
 incoming proton: $0.999599|+\rangle +0.0283005|-\rangle$

$s=6.59164$, $\sqrt{s}=2.56742$, $|\vec{P}|=1.11236$, $|\vec{P}'|=0.00112869$
 $\theta_{P'}=0.054477$, $\theta_Y=3.11194$, $\phi_{P'}=4.13085$, $\phi_Y=7.31348e-07$
 $E_Y=0.815273$, $\theta_{\ell'}=0.0296545$, $\phi_{\ell'}=3.13947$
 $Q^2=3.6217$, $x_B=0.702834$, $t=-0.967495$, $W^2=2.41114$, $y=0.902166$

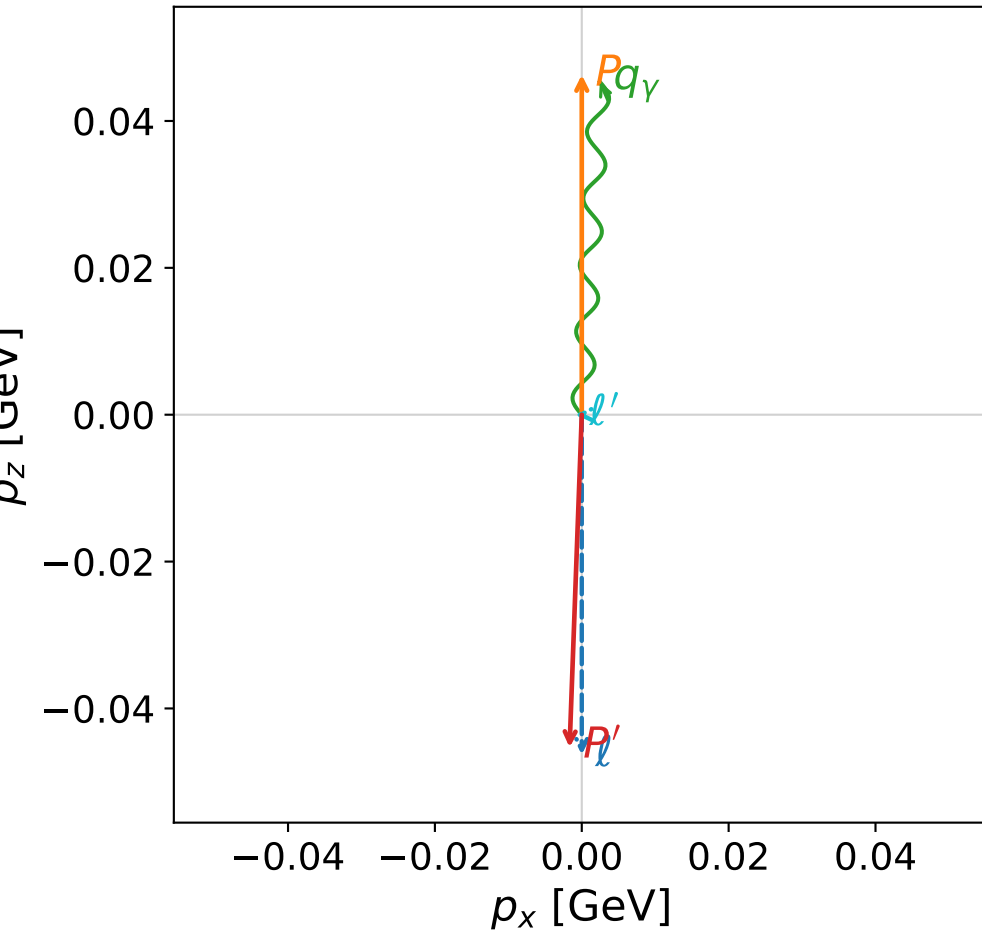
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.1124$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.1124$
 P $E= 1.4551$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.1124$
 ℓ' $E= 0.8141$ $p_x= -0.0241$ $p_y= 0.0001$ $p_z= 0.8138$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0001$ $p_z= 0.0011$
 q_Y $E= 0.8153$ $p_x= 0.0242$ $p_y= 0.0000$ $p_z= -0.8149$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 7.080%, conserve 92.920%
 $R_{\text{flip/conserve}}=7.620e-02$; $I_h=-9.492e-02$
 virtual-photon powers: T 99.107%, L 0.893%
 $R_L/T=9.007e-03$; $I_{TL}=-3.422e-02$

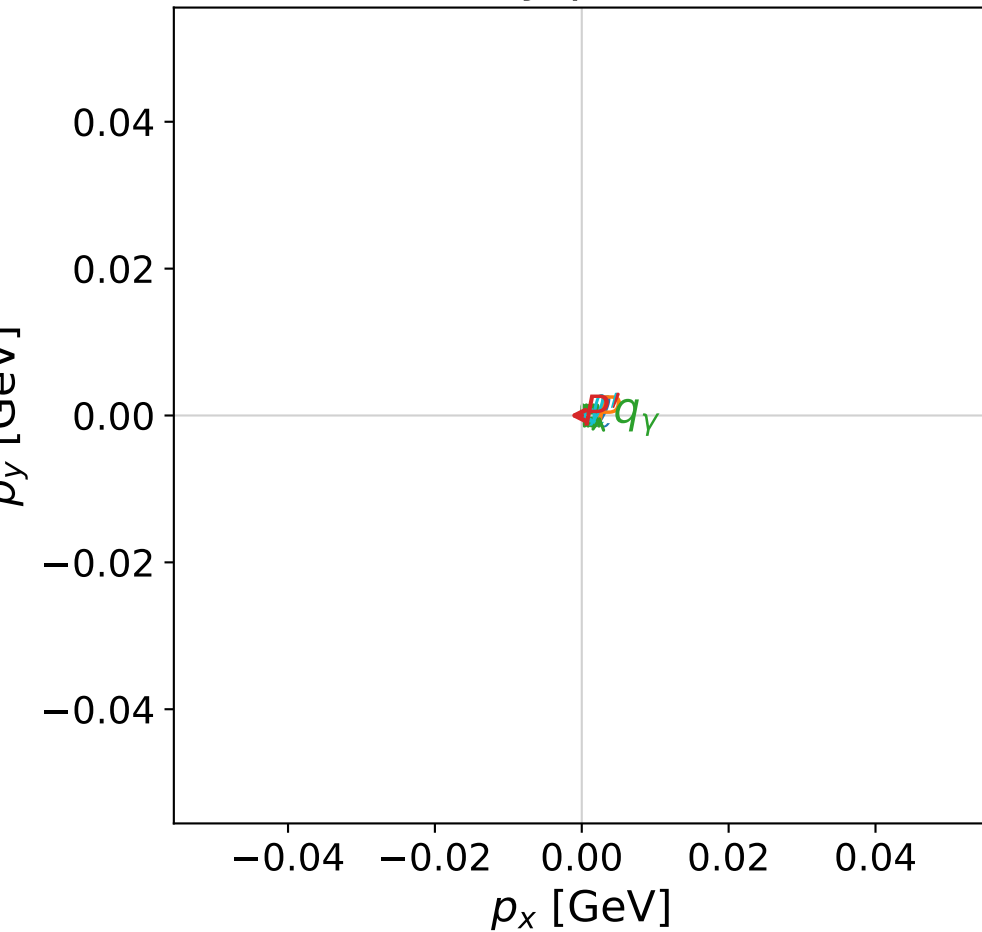


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



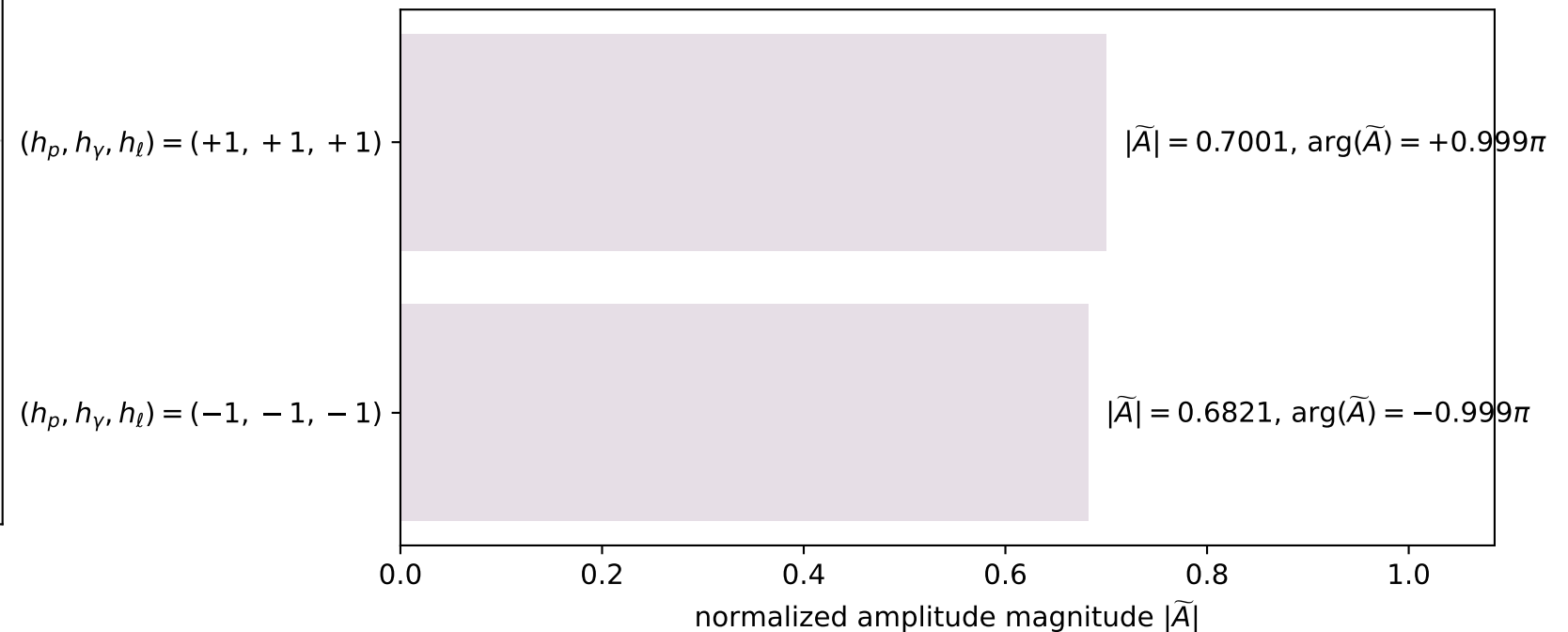
dghz_mixing_angles_polarization_01_local_minimum_351
 $d_{\text{GHZ}}=0.000879976$
 region: polarization_cluster_01_local_minimum_351
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0351
 lepton mixing angle: $\alpha_e=0.76448436$ rad
 incoming lepton: $0.721739|+\rangle +0.692165|-\rangle$
 proton mixing angle: $\alpha_p=0.73414056$ rad
 incoming proton: $0.742407|+\rangle +0.669949|-\rangle$

$s=0.971085$, $\sqrt{s}=0.985437$, $|\vec{P}|=0.0462921$, $|\vec{P}'|=0.0453374$
 $\theta_p=3.10496$, $\theta_\gamma=0.0542554$, $\phi_p=3.141$, $\phi_\gamma=0.000630486$
 $E_\gamma=0.045391$, $\theta_{\ell'}=1.59229$, $\phi_{\ell'}=3.14476$
 $Q^2=8.58888\text{e-}05$, $x_B=0.000960145$, $t=-0.00839316$, $W^2=0.969212$, $y=0.980410$

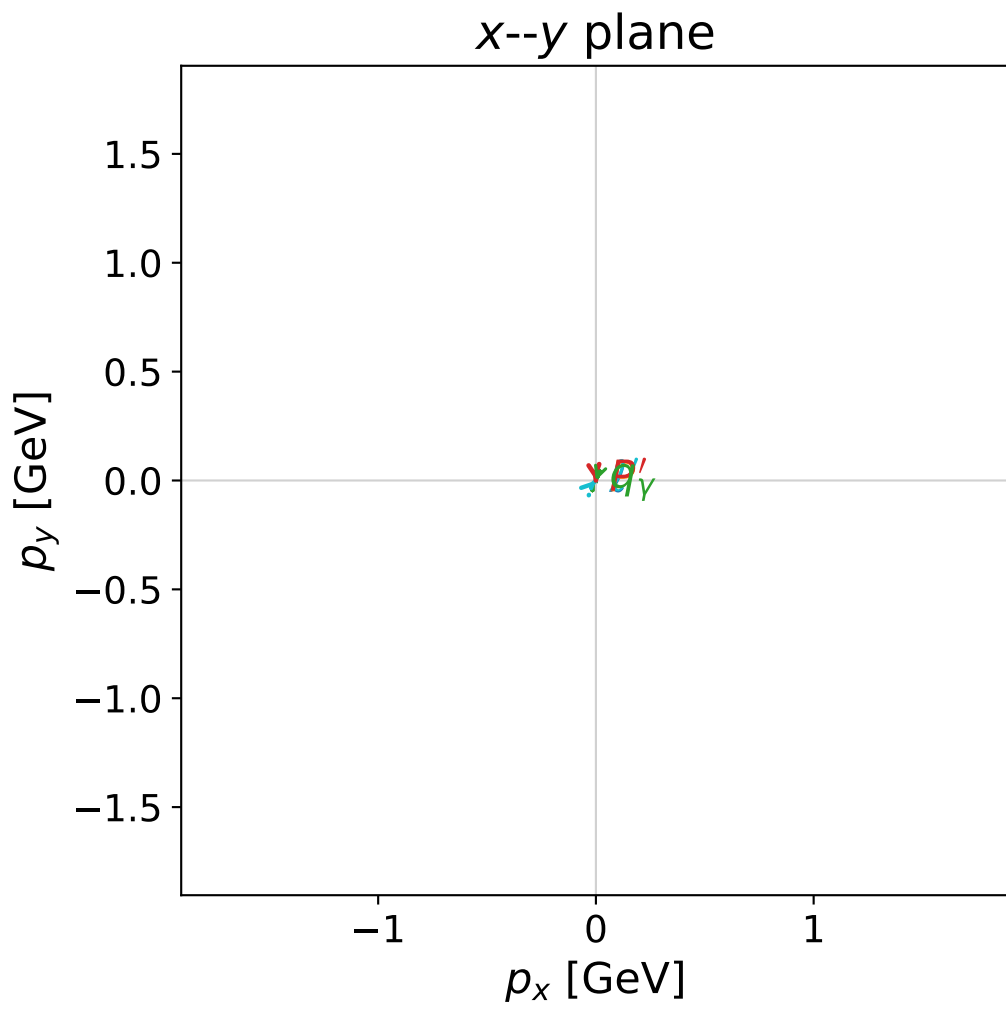
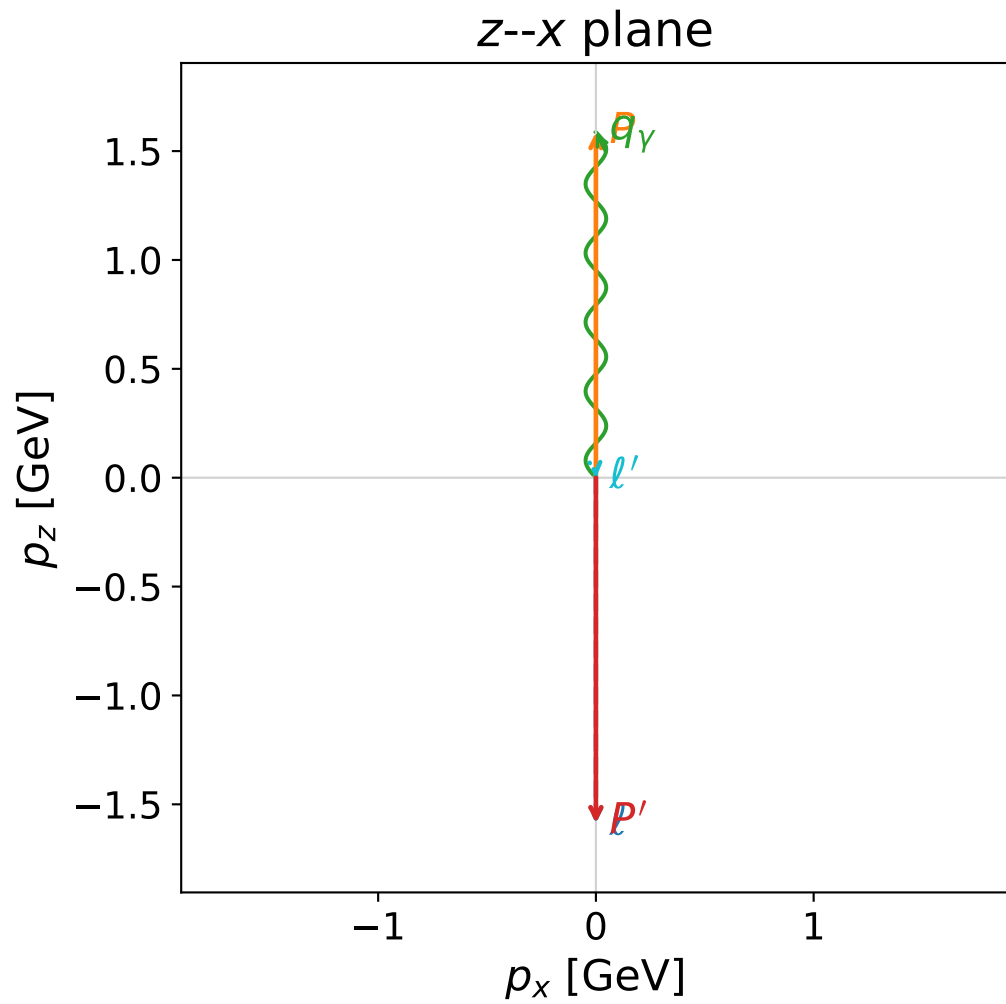
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.0463$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0463$
 P $E= 0.9391$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0463$
 ℓ' $E= 0.0010$ $p_x= -0.0008$ $p_y= -0.0000$ $p_z= -0.0000$
 P' $E= 0.9391$ $p_x= -0.0017$ $p_y= 0.0000$ $p_z= -0.0453$
 q_γ $E= 0.0454$ $p_x= 0.0025$ $p_y= 0.0000$ $p_z= 0.0453$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 6.591%, conserve 93.409%
 $R_{\text{flip/conserve}}=7.056\text{e-}02$; $I_h=+1.147\text{e-}03$
 virtual-photon powers: T 98.347%, L 1.653%
 $R_L/T=1.681\text{e-}02$; $I_{TL}=-3.444\text{e-}03$

Leading normalized final-state amplitudes
 (N=2, retained=95.5%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)



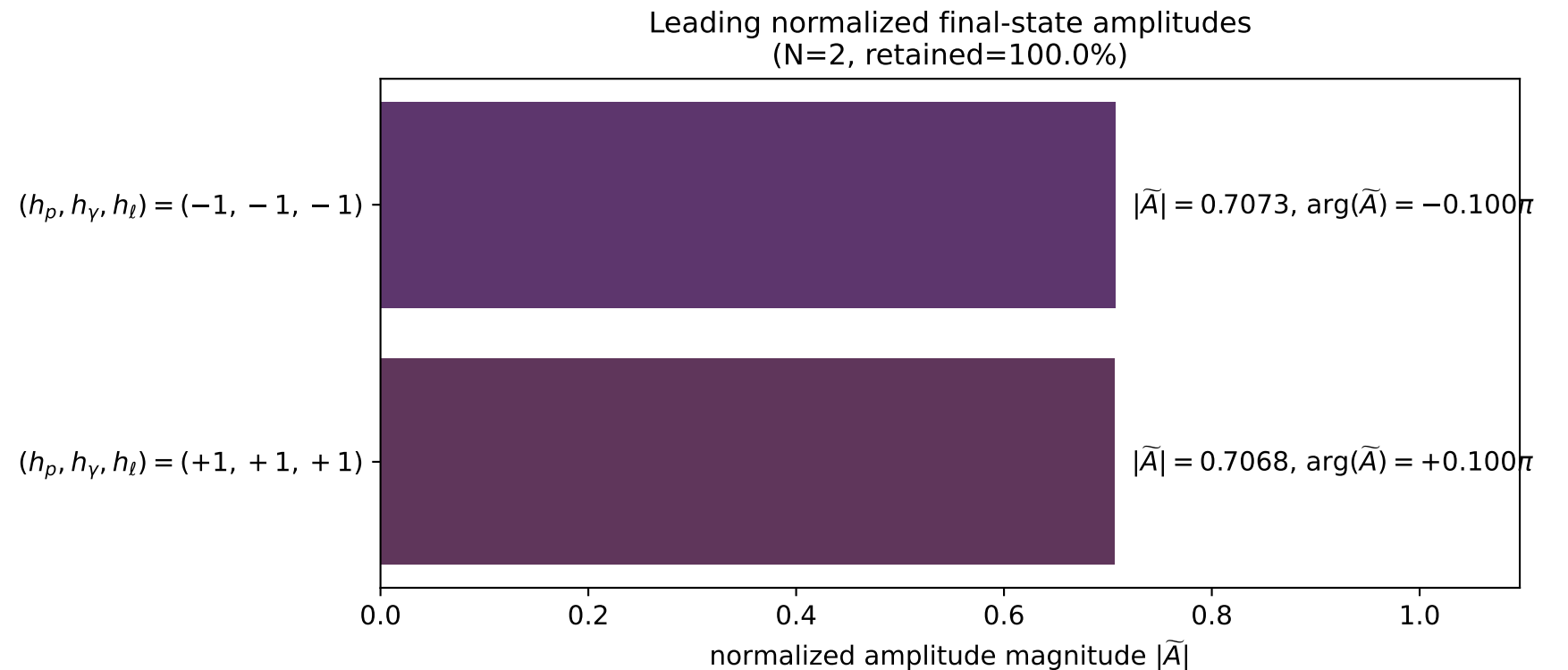
dghz_mixing_angles_polarization_01_local_minimum_356
 $d_{\{\text{rm GHZ}\}}=0.0011903$
 region: polarization_cluster_01_local_minimum_356
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0356
 lepton mixing angle: $\alpha_e=0.78547747$ rad
 incoming lepton: $0.707051|+\rangle +0.707163|-\rangle$
 proton mixing angle: $\alpha_p=0.78569818$ rad
 incoming proton: $0.706895|+\rangle +0.707319|-\rangle$

$s=11.7704$, $\sqrt{s}=3.4308$, $|\vec{P}|=1.58717$, $|\vec{P}'|=1.58616$
 $\theta_p=3.14154$, $\theta_\gamma=4.27948\text{e-}05$, $\phi_p=4.8388$, $\phi_\gamma=2.80317$
 $E_\gamma=1.58703$, $\theta_{\ell'}=3.05347$, $\phi_{\ell'}=0.792465$
 $Q^2=0.000448243$, $x_B=4.11835\text{e-}05$, $t=-10.07$, $W^2=11.7634$, $y=0.999402$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

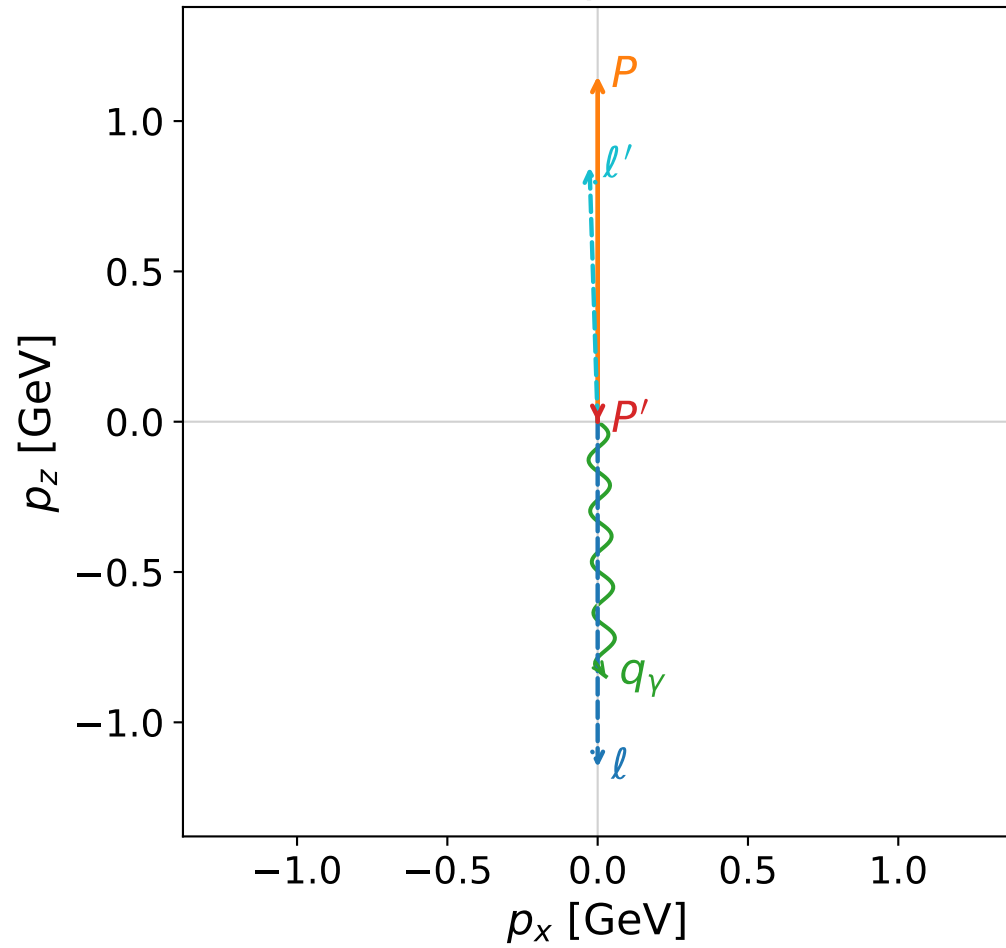
Particle	E	p_x	p_y	p_z
ℓ	1.5872	0.0000	0.0000	-1.5872
P	1.8436	0.0000	0.0000	1.5872
ℓ'	0.0010	0.0001	0.0001	-0.0009
P'	1.8428	0.0000	-0.0001	-1.5862
q_γ	1.5870	-0.0001	0.0000	1.5870

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.014%, conserve 99.986%
 $R_{\text{flip/conserve}}=1.415\text{e-}04$; $I_h=+1.727\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=3.671\text{e-}13$; $I_{TL}=+5.250\text{e-}10$

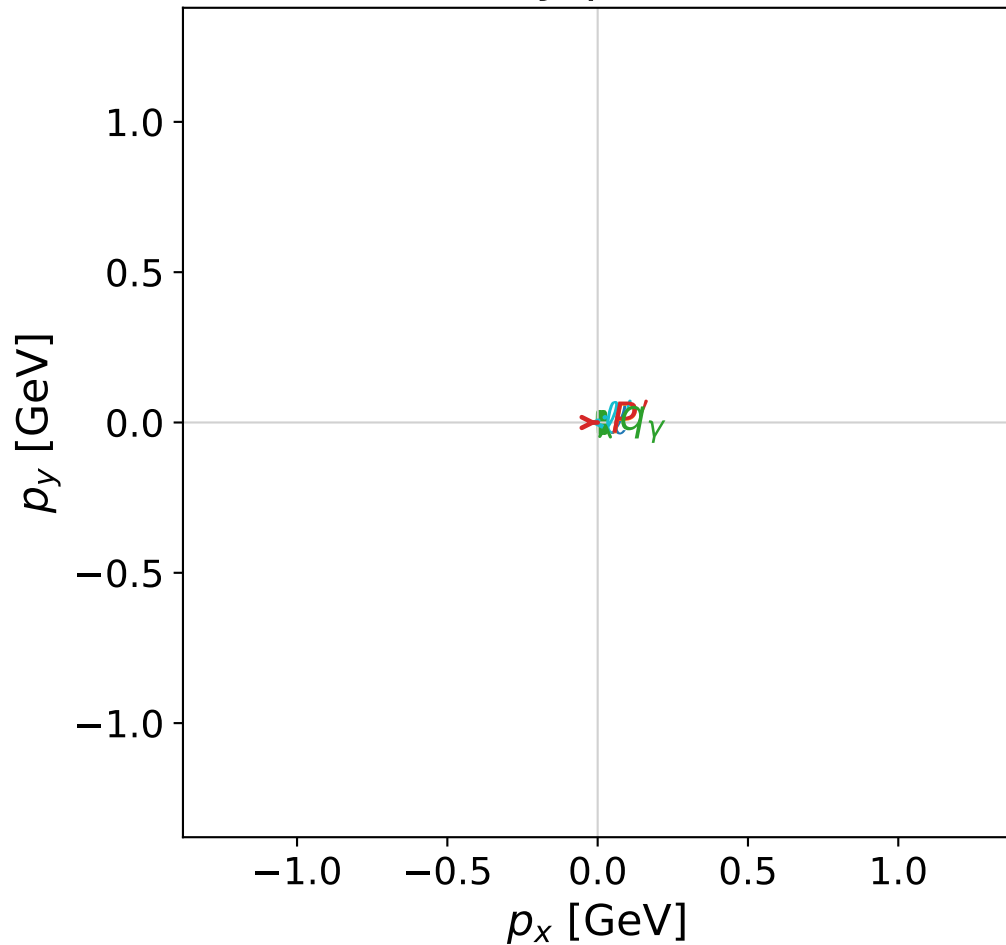


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



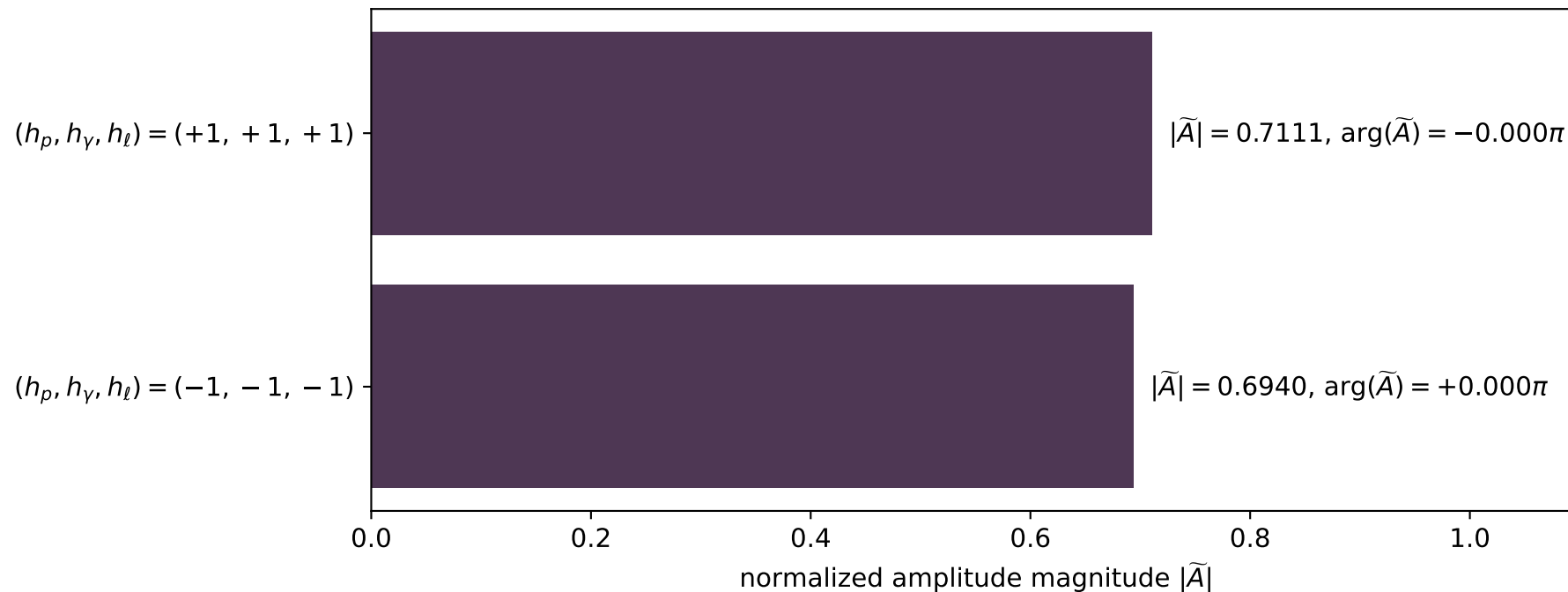
dghz_mixing_angles_polarization_01_local_minimum_359
 $d_{\text{GHZ}}=0.00157373$
 region: polarization_cluster_01_local_minimum_359
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0359
 lepton mixing angle: $\alpha_e=1.5434269$ rad
 incoming lepton: $0.027366|+\rangle + 0.999625|-\rangle$
 proton mixing angle: $\alpha_p=0.026716948$ rad
 incoming proton: $0.999643|+\rangle + 0.0267138|-\rangle$

$s=6.9343$, $\sqrt{s}=2.63331$, $|\vec{P}|=1.14959$, $|\vec{P}'|=3.60035\text{e-}07$
 $\theta_{p'}=3.14082$, $\theta_\gamma=3.10976$, $\phi_{p'}=0.000770639$, $\phi_\gamma=6.28315$
 $E_\gamma=0.847653$, $\theta_{\ell'}=0.0318365$, $\phi_{\ell'}=3.14156$
 $Q^2=3.89684$, $x_B=0.71019$, $t=-1.02376$, $W^2=2.47004$, $y=0.90628$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.1496$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.1496$
 P $E= 1.4837$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.1496$
 ℓ' $E= 0.8477$ $p_x= -0.0270$ $p_y= 0.0000$ $p_z= 0.8472$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0000$
 q_γ $E= 0.8477$ $p_x= 0.0270$ $p_y= -0.0000$ $p_z= -0.8472$

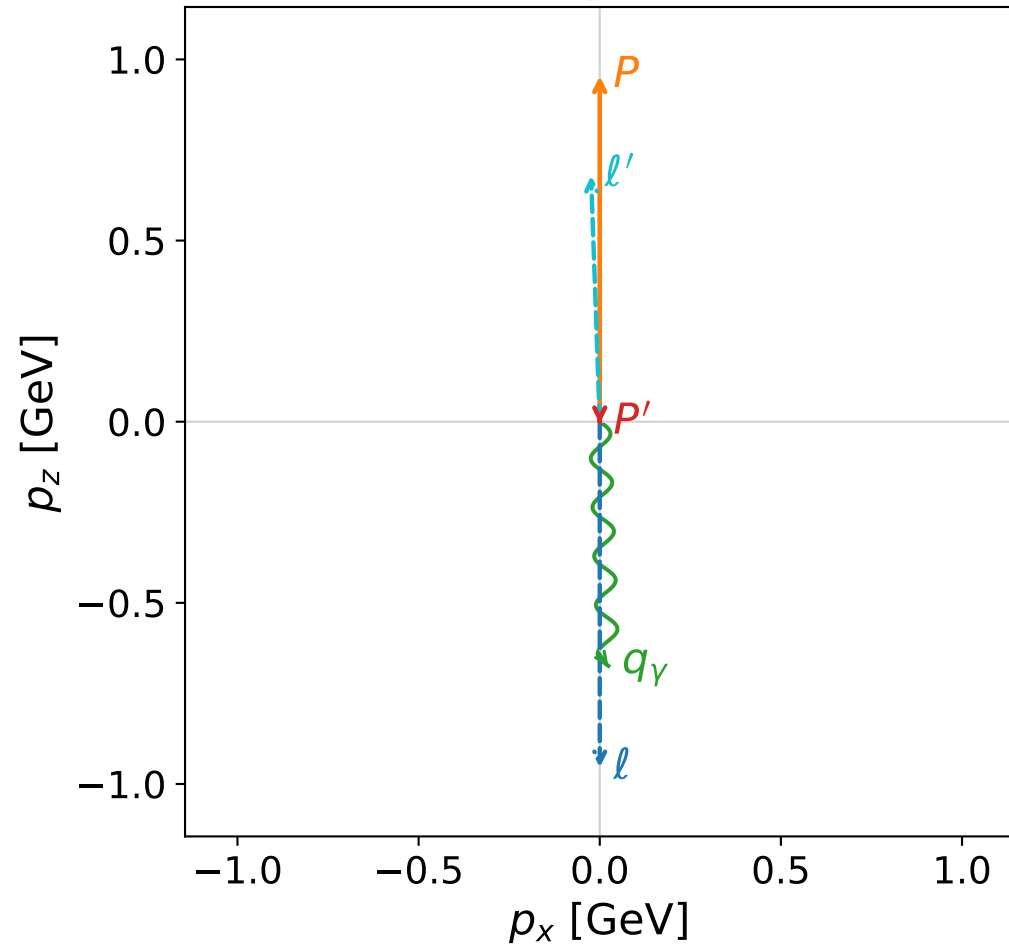
GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 6.493%, conserve 93.507%
 $R_{\text{flip/conserve}}=6.944\text{e-}02$; $I_h=-9.045\text{e-}02$
 virtual-photon powers: T 98.954%, L 1.046%
 $R_L/T=1.057\text{e-}02$; $I_{TL}=-3.640\text{e-}02$

Leading normalized final-state amplitudes
(N=2, retained=98.7%)

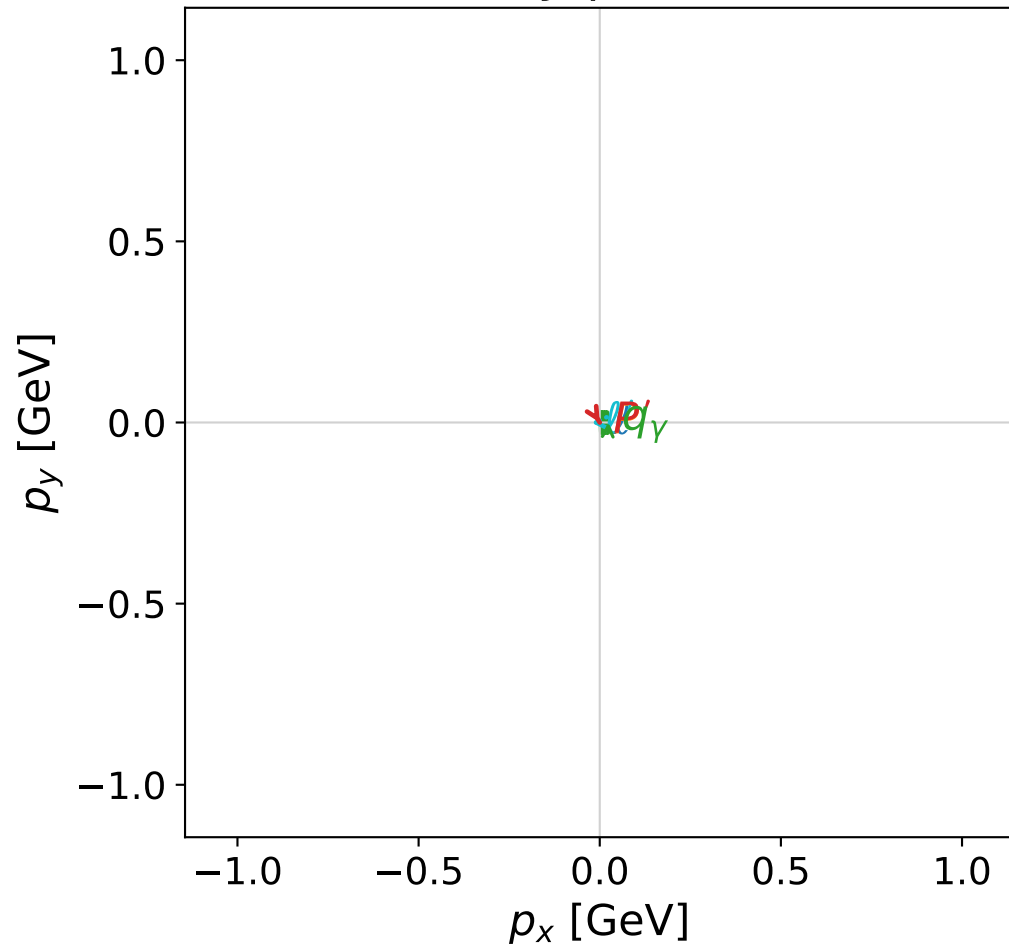


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

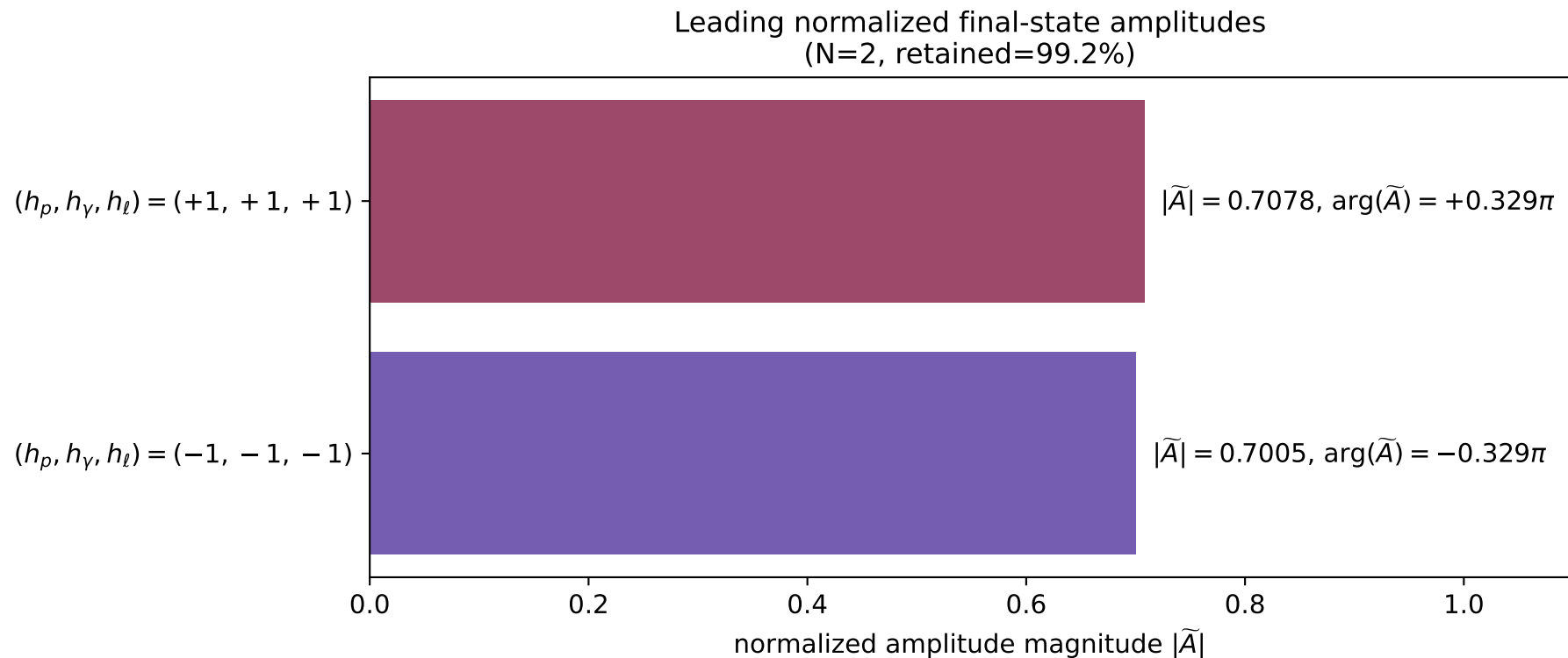


dghz_mixing_angles_polarization_01_local_minimum_360
 $d_{\text{GHZ}}=0.00187001$
 region: polarization_cluster_01_local_minimum_360
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0360
 lepton mixing angle: $\alpha_e=1.5430613$ rad
 incoming lepton: $0.0277314|+\rangle + 0.999615|-\rangle$
 proton mixing angle: $\alpha_p=0.027451861$ rad
 incoming proton: $0.999623|+\rangle + 0.0274484|-\rangle$

$s=5.25221$, $\sqrt{s}=2.29177$, $|\vec{P}|=0.953927$, $|\vec{P}'|=0.00502558$
 $\theta_{p'}=3.14128$, $\theta_\gamma=3.10607$, $\phi_{p'}=5.2495$, $\phi_\gamma=6.28316$
 $E_\gamma=0.674367$, $\theta_{\ell'}=0.0352562$, $\phi_{\ell'}=3.14151$
 $Q^2=2.59155$, $x_B=0.673146$, $t=-0.759728$, $W^2=2.1382$, $y=0.880508$

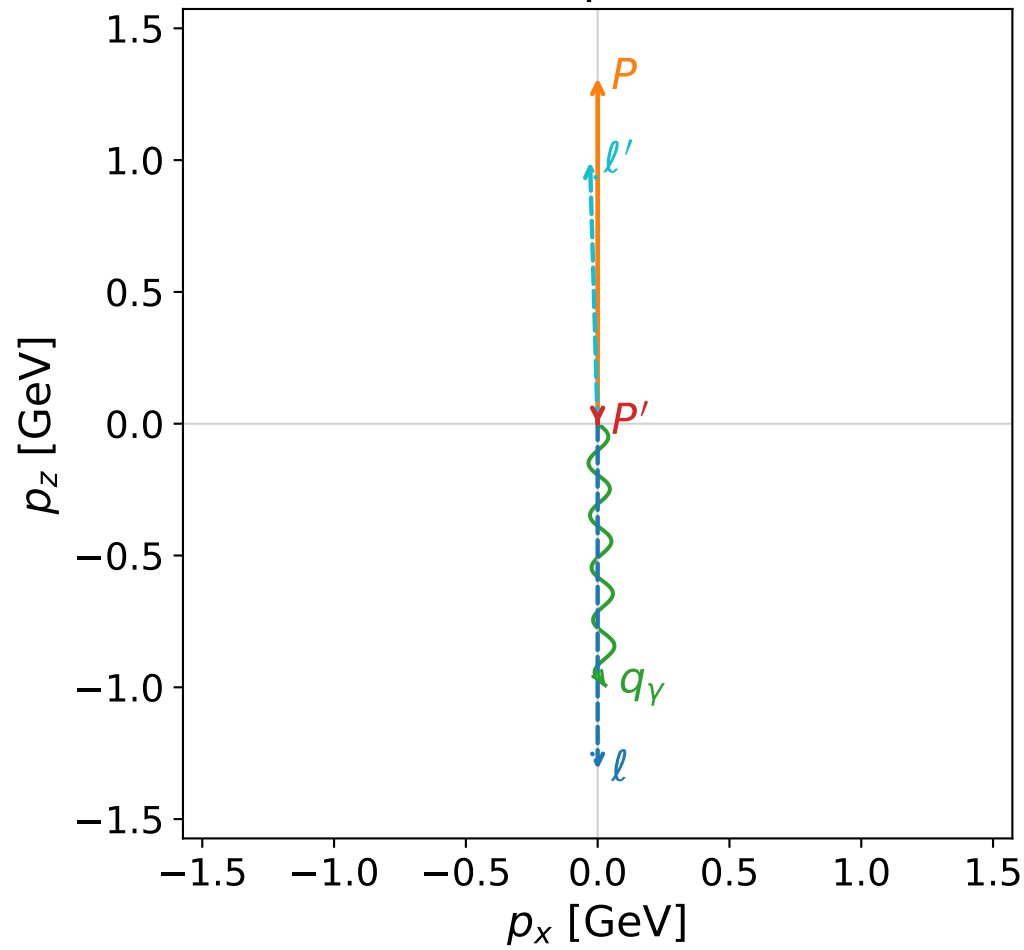
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.9539$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.9539$
 P $E= 1.3378$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.9539$
 ℓ' $E= 0.6794$ $p_x= -0.0239$ $p_y= 0.0000$ $p_z= 0.6790$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0050$
 q_γ $E= 0.6744$ $p_x= 0.0239$ $p_y= -0.0000$ $p_z= -0.6739$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 6.640%, conserve 93.360%
 $R_{\text{flip/conserve}}=7.113\text{e-}02$; $I_h=-1.009\text{e-}01$
 virtual-photon powers: T 98.089%, L 1.911%
 $R_L/T=1.948\text{e-}02$; $I_{TL}=-5.484\text{e-}02$

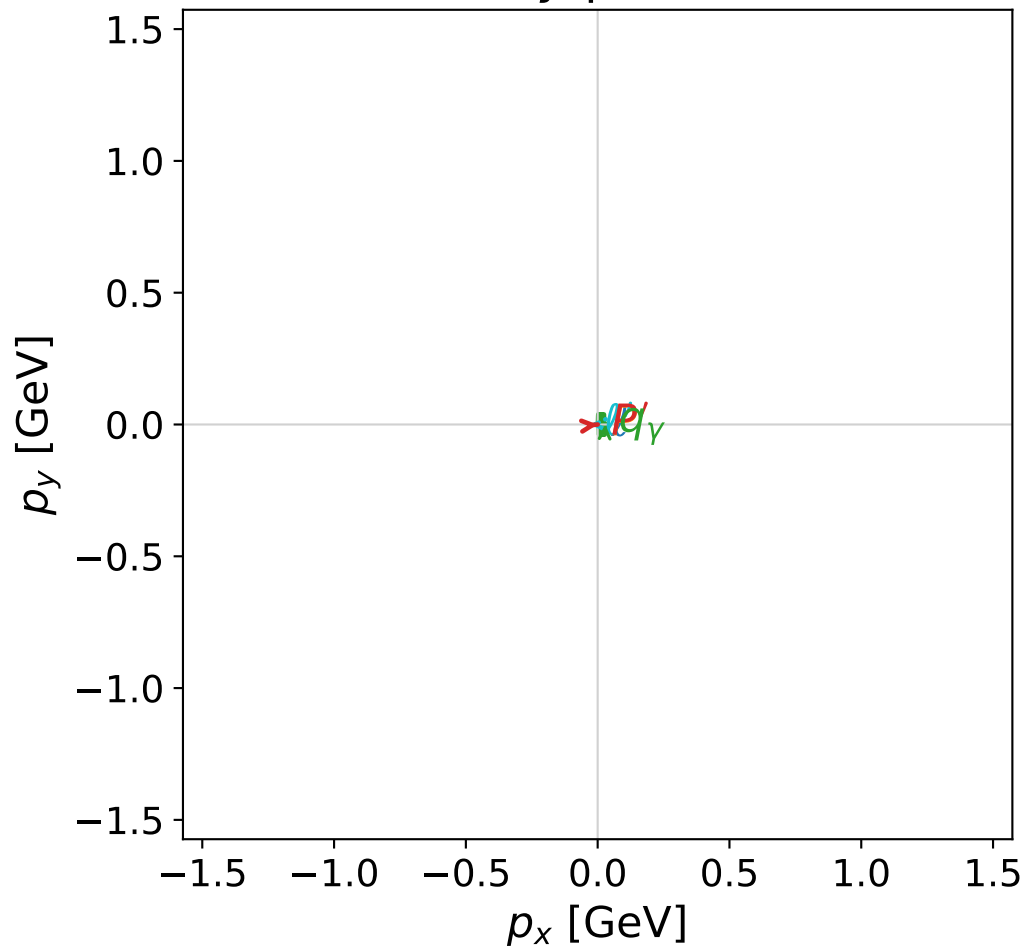


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



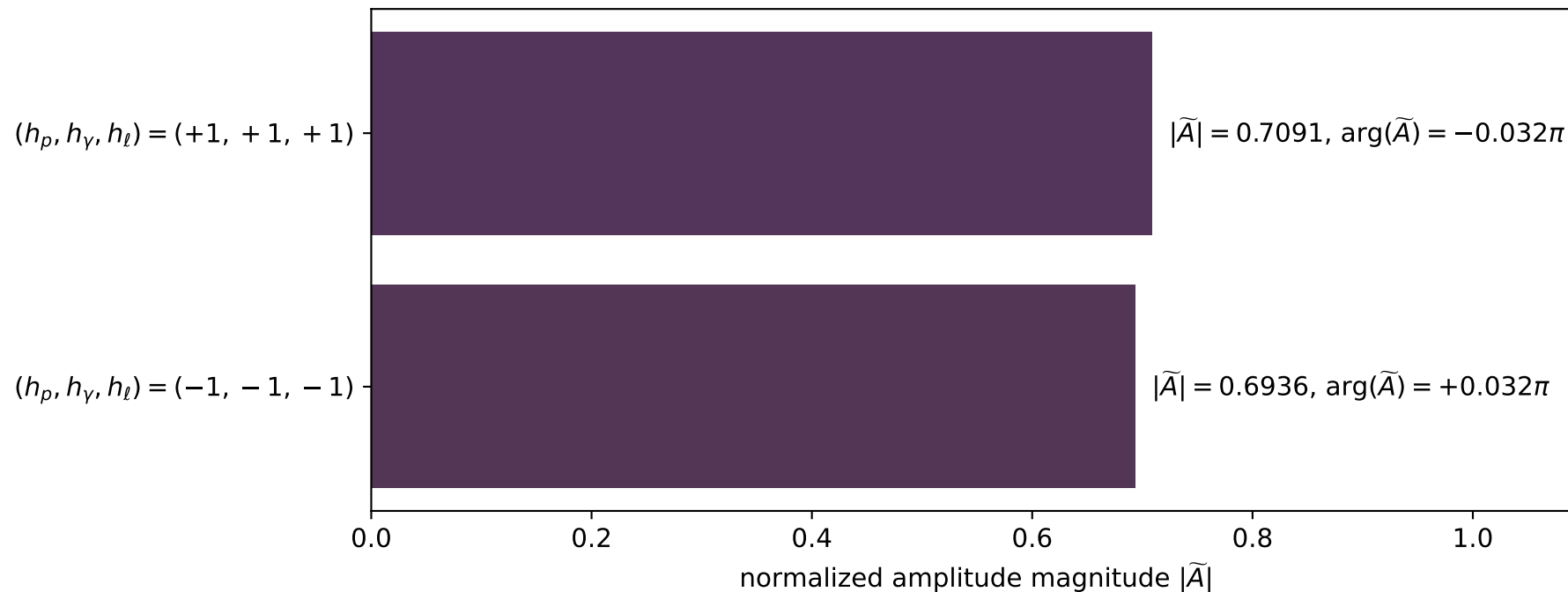
dghz_mixing_angles_polarization_01_local_minimum_361
 $d_{\text{GHZ}}=0.0018947$
 region: polarization_cluster_01_local_minimum_361
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0361
 lepton mixing angle: $\alpha_e=1.5434414$ rad
 incoming lepton: $0.0273516|+\rangle + 0.999626|-\rangle$
 proton mixing angle: $\alpha_p=0.026758462$ rad
 incoming proton: $0.999642|+\rangle + 0.0267553|-\rangle$

$s=8.54507$, $\sqrt{s}=2.9232$, $|\vec{P}|=1.3111$, $|\vec{P}'|=2.76355\text{e-}06$
 $\theta_{p'}=3.13818$, $\theta_\gamma=3.1125$, $\phi_{p'}=0.101143$, $\phi_\gamma=6.28318$
 $E_\gamma=0.992596$, $\theta_{l'}=0.0290944$, $\phi_{l'}=3.14159$
 $Q^2=5.2045$, $x_B=0.736492$, $t=-1.2646$, $W^2=2.74195$, $y=0.921904$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.3111$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3111$
 P $E= 1.6121$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3111$
 l' $E= 0.9926$ $p_x= -0.0289$ $p_y= 0.0000$ $p_z= 0.9922$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0000$
 q_γ $E= 0.9926$ $p_x= 0.0289$ $p_y= -0.0000$ $p_z= -0.9922$

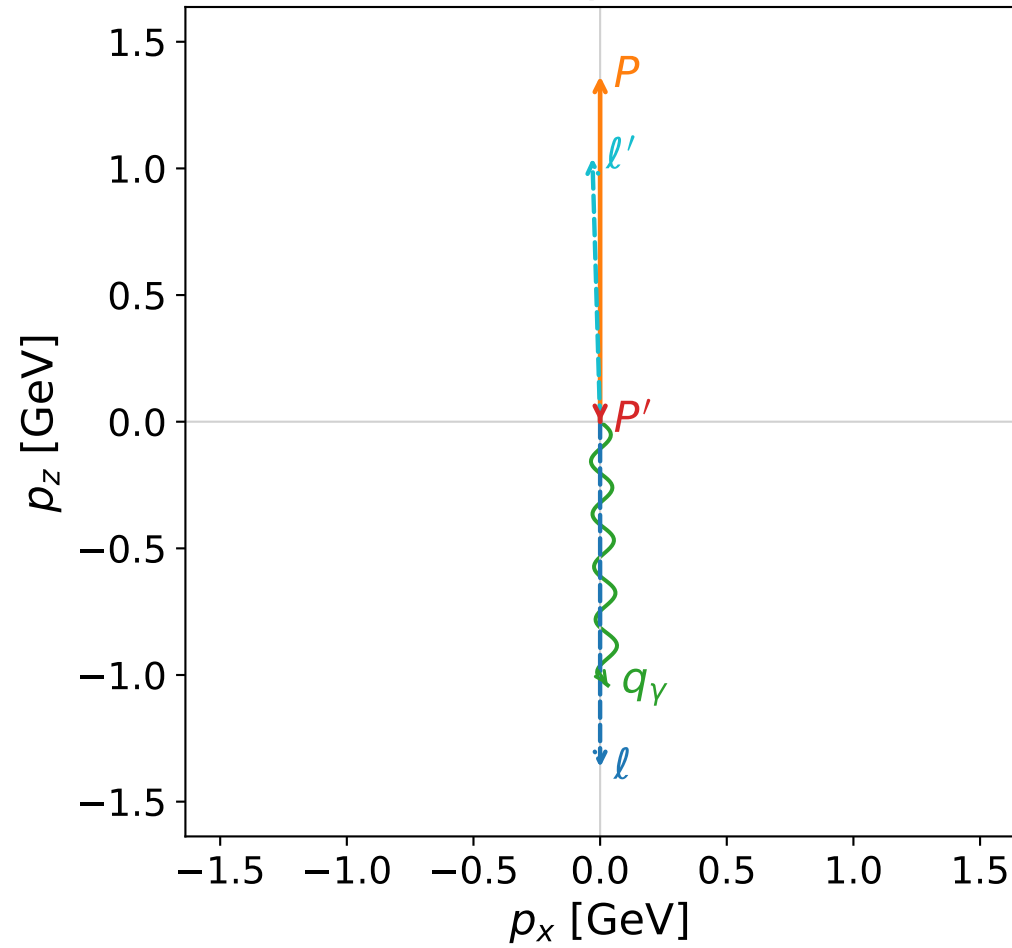
GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 6.329%, conserve 93.671%
 $R_{\text{flip/conserve}}=6.756\text{e-}02$; $I_h=-8.276\text{e-}02$
 virtual-photon powers: T 99.371%, L 0.629%
 $R_L/T=6.334\text{e-}03$; $I_{TL}=-2.625\text{e-}02$

Leading normalized final-state amplitudes
(N=2, retained=98.4%)

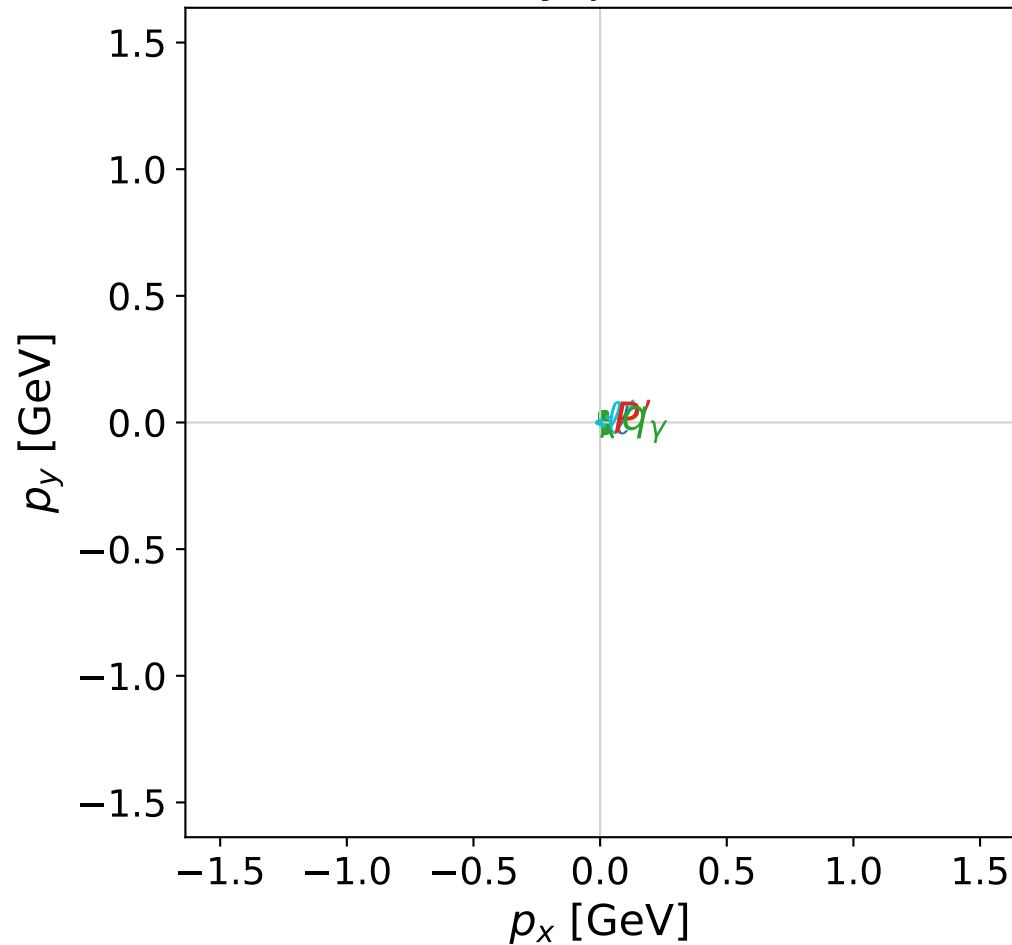


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

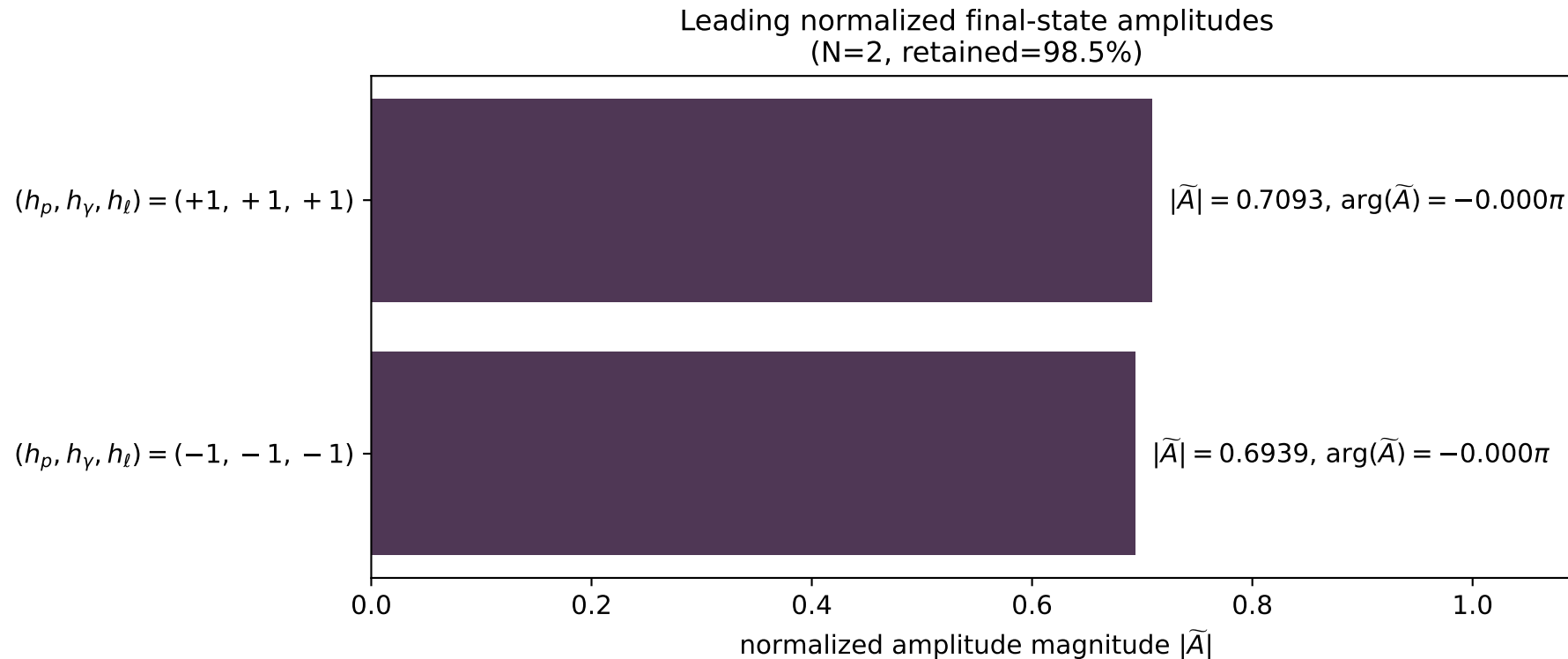


dghz_mixing_angles_polarization_01_local_minimum_364
 $d_{\text{GHZ}}=0.00240137$
 region: polarization_cluster_01_local_minimum_364
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0364
 lepton mixing angle: $\alpha_e=1.5438592$ rad
 incoming lepton: $0.0269338|+\rangle + 0.999637|-\rangle$
 proton mixing angle: $\alpha_p=0.026354725$ rad
 incoming proton: $0.999653|+\rangle + 0.0263517|-\rangle$

$s=9.12098$, $\sqrt{s}=3.0201$, $|\vec{P}|=1.36438$, $|\vec{P}'|=4.15695\text{e-}06$
 $\theta_{P'}=3.14159$, $\theta_\gamma=3.11231$, $\phi_{P'}=0.518887$, $\phi_\gamma=6.28317$
 $E_\gamma=1.04105$, $\theta_{l'}=0.0292827$, $\phi_{l'}=3.14158$
 $Q^2=5.68034$, $x_B=0.74415$, $t=-1.34644$, $W^2=2.83284$, $y=0.926248$

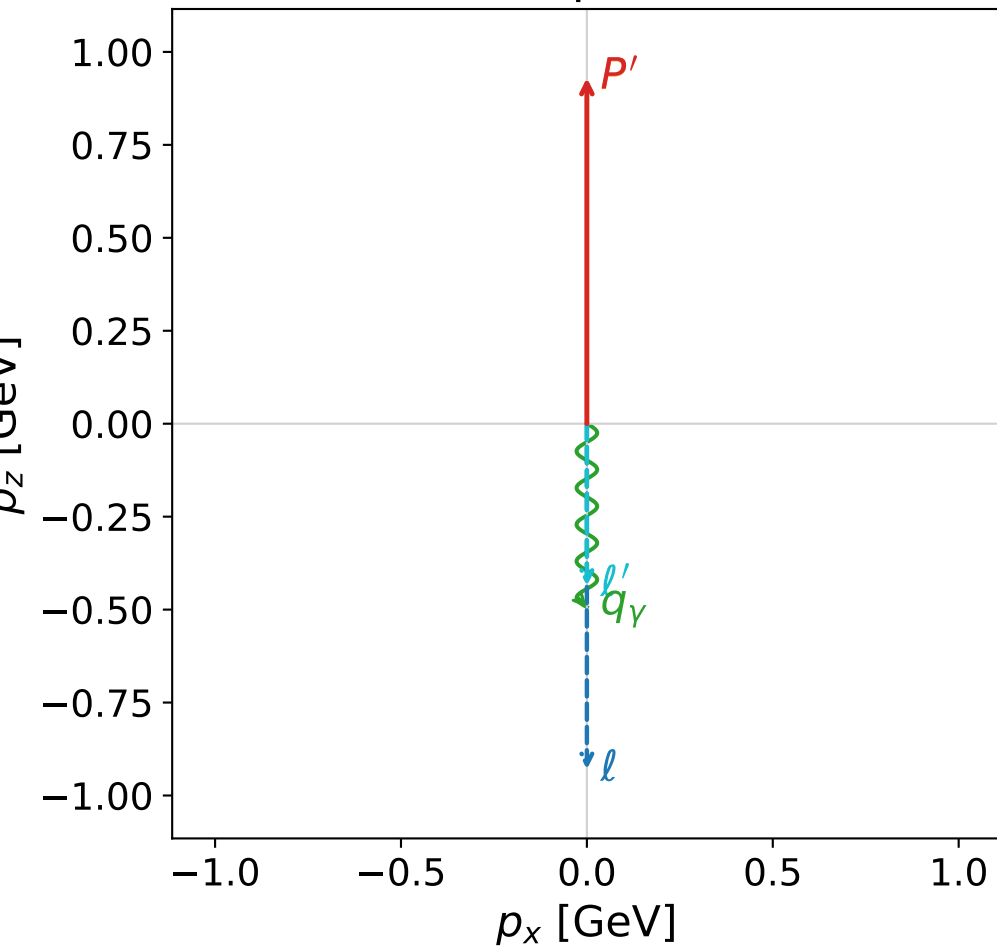
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.3644$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3644$
 P $E= 1.6557$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3644$
 l' $E= 1.0410$ $p_x= -0.0305$ $p_y= 0.0000$ $p_z= 1.0406$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0000$
 q_γ $E= 1.0410$ $p_x= 0.0305$ $p_y= -0.0000$ $p_z= -1.0406$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 6.070%, conserve 93.930%
 $R_{\text{flip/conserve}}=6.462\text{e-}02$; $I_h=-7.924\text{e-}02$
 virtual-photon powers: T 99.406%, L 0.594%
 $R_L/T=5.971\text{e-}03$; $I_{TL}=-2.490\text{e-}02$

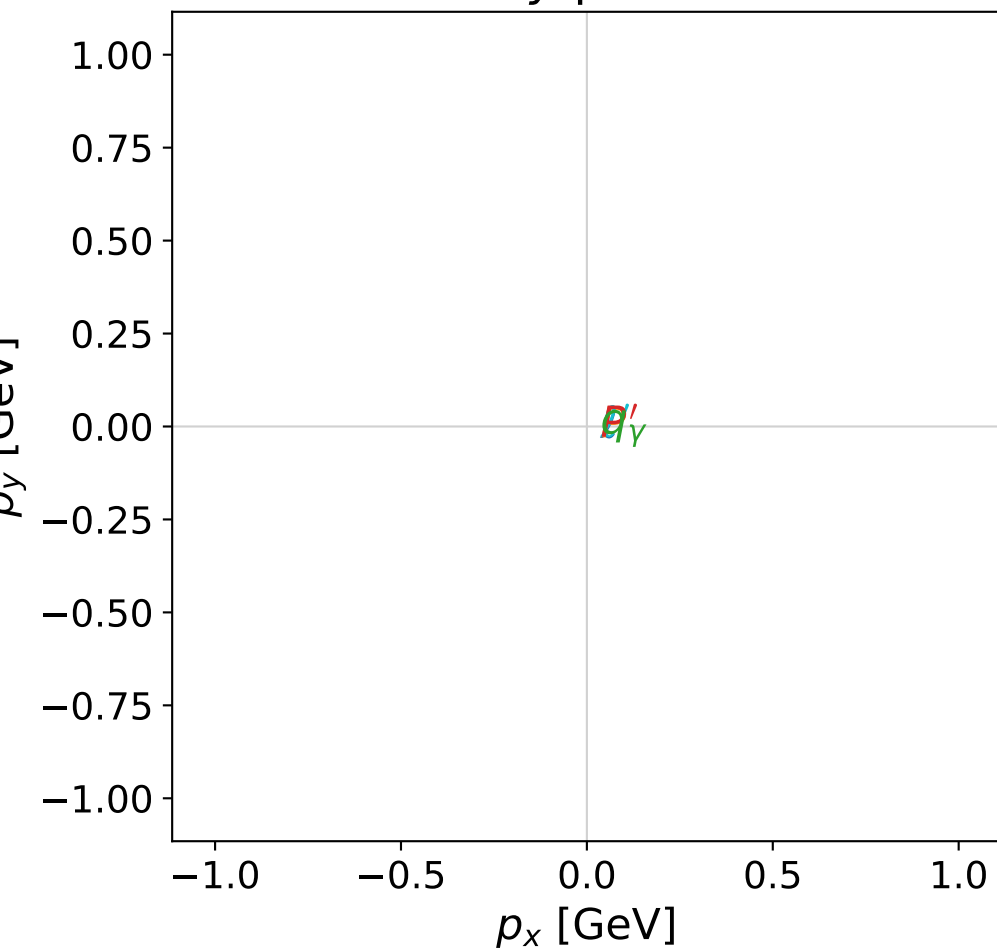


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



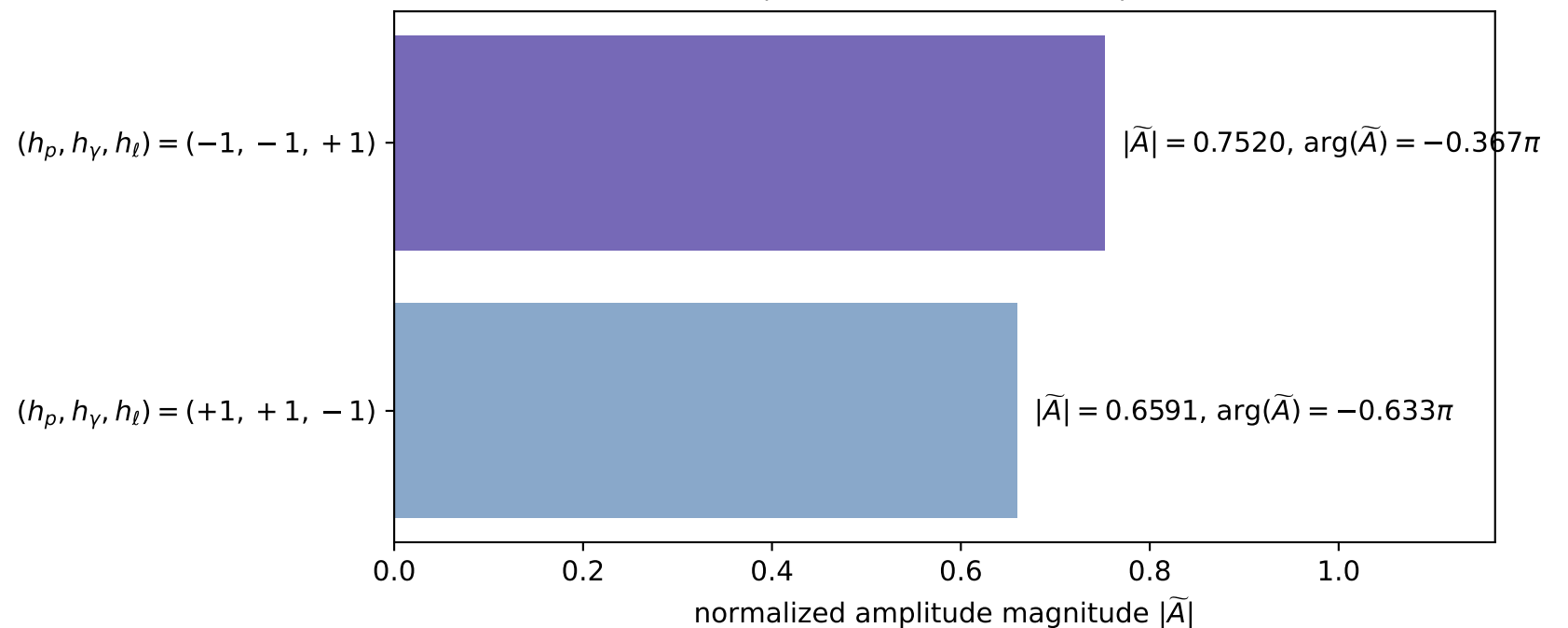
dghz_mixing_angles_polarization_01_local_minimum_367
 $d_{\{\rm GHZ\}}=0.0171798$
 region: polarization_cluster_01_local_minimum_367
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0367
 lepton mixing angle: $\alpha_e=6.1169479\text{e-}06$ rad
 incoming lepton: $1|+\rangle +6.11695\text{e-}06|-\rangle$
 proton mixing angle: $\alpha_p=0$ rad
 incoming proton: $1|+\rangle +0|-\rangle$

$s=5.06282$, $\sqrt{s}=2.25007$, $|\vec{P}|=0.929521$, $|\vec{P}'|=0.929521$
 $\theta_{p'}=0$, $\theta_\gamma=3.14159$, $\phi_{p'}=2.9999$, $\phi_\gamma=5.13056$
 $E_\gamma=0.493368$, $\theta_{\ell'}=3.14159$, $\phi_{\ell'}=1.98897$
 $Q^2=1.56777\text{e-}07$, $x_B=7.06132\text{e-}08$, $t=-4.3871\text{e-}15$, $W^2=3.10007$, $y=0.530776$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.9295$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.9295$
 P $E= 1.3206$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.9295$
 ℓ' $E= 0.4362$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.4362$
 P' $E= 1.3206$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.9295$
 q_γ $E= 0.4934$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.4934$

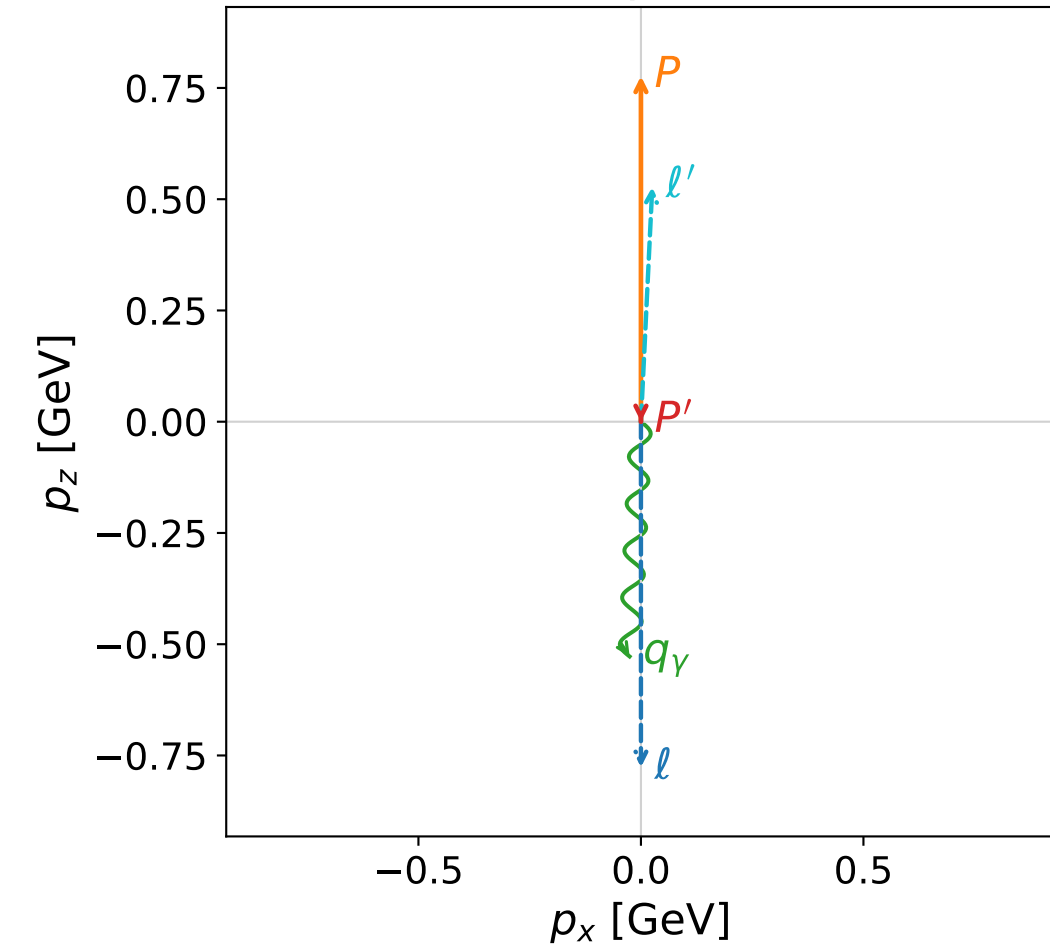
GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 43.446%, conserve 56.554%
 $R_{\text{flip/conserve}}=7.682\text{e-}01$; $I_h=+7.080\text{e-}19$
 virtual-photon powers: T 56.554%, L 43.446%
 $R_L/T=7.682\text{e-}01$; $I_{TL}=+0.000\text{e+}00$

Leading normalized final-state amplitudes (N=2, retained=100.0%)

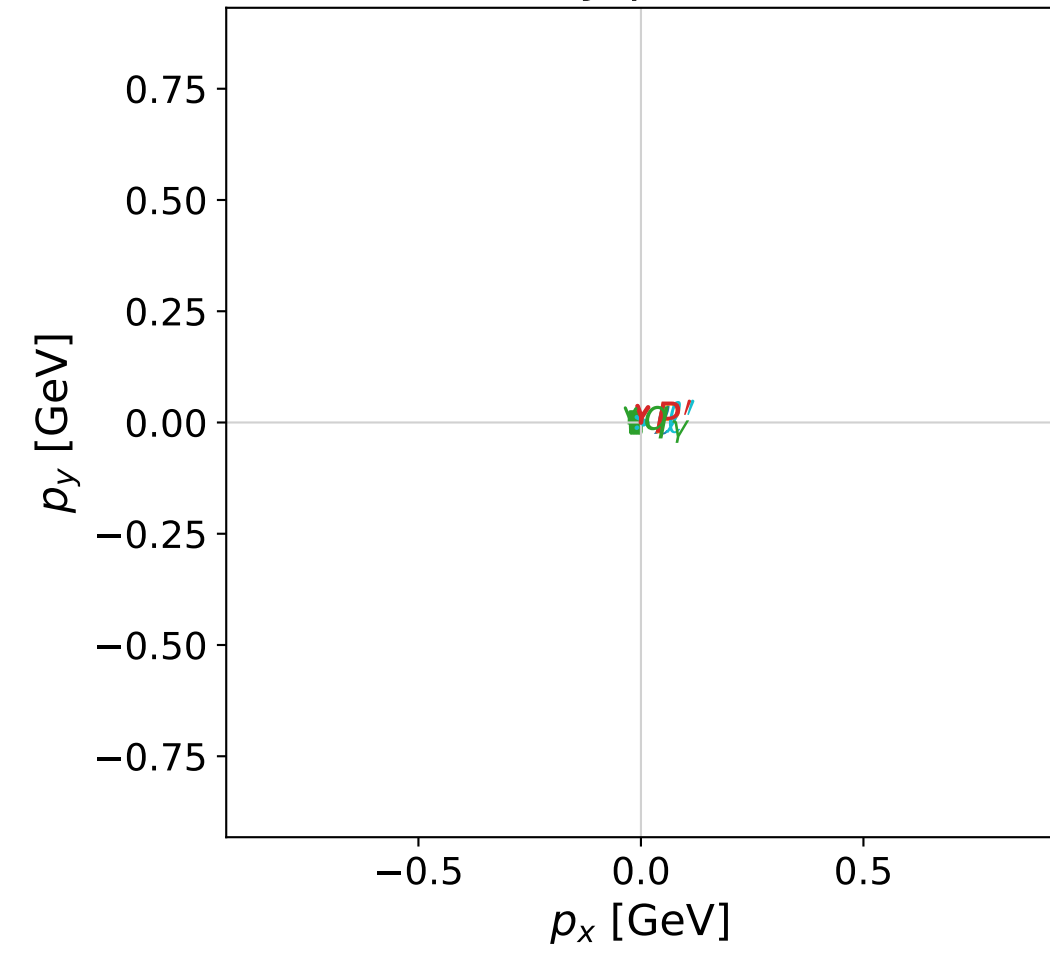


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



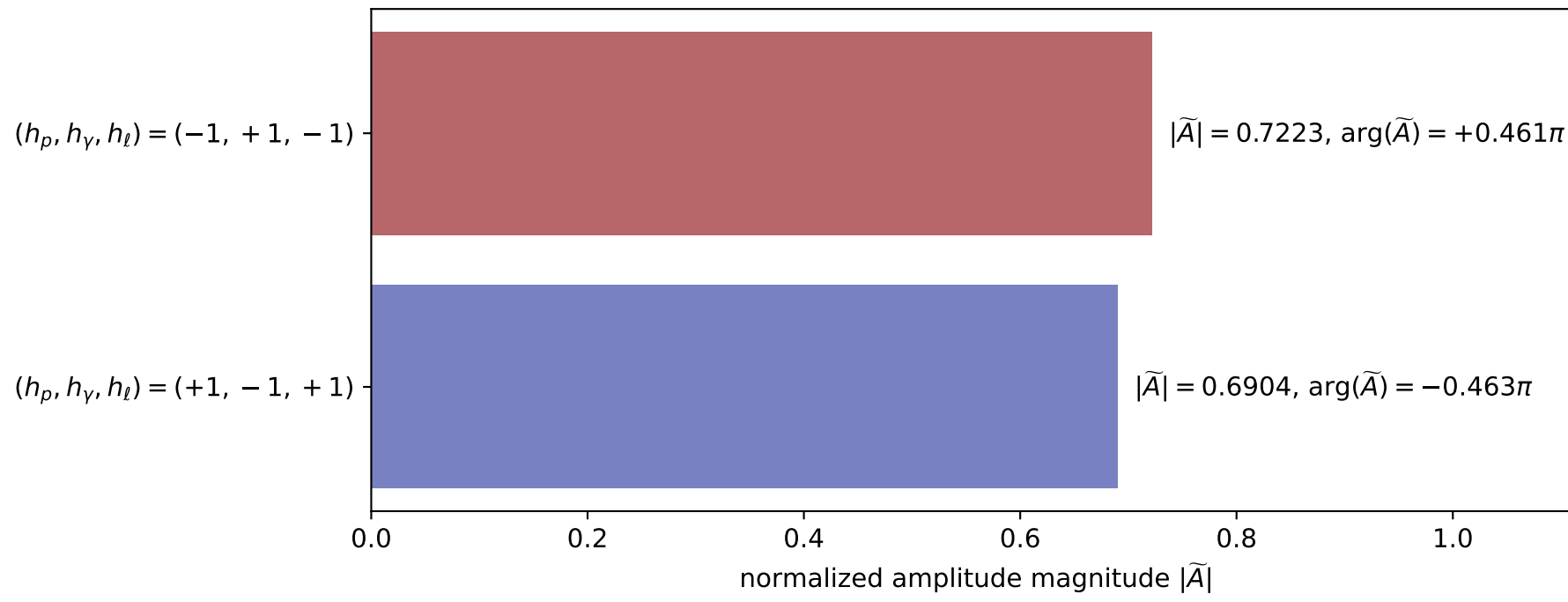
dghz_mixing_angles_polarization_01_local_minimum_369
 $d_{\text{GHZ}}=0.0208267$
 region: polarization_cluster_01_local_minimum_369
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0369
 lepton mixing angle: $\alpha_e=1.5707901$ rad
 incoming lepton: $6.21409\text{e-}06|+\rangle +1|-\rangle$
 proton mixing angle: $\alpha_p=3.1415775$ rad
 incoming proton: $-1|+\rangle +1.51327\text{e-}05|-\rangle$

$s=3.97747$, $\sqrt{s}=1.99436$, $|\vec{P}|=0.776596$, $|\vec{P}'|=4.18979\text{e-}06$
 $\theta_{P'}=3.14157$, $\theta_\gamma=3.09338$, $\phi_{P'}=4.59353$, $\phi_\gamma=3.13875$
 $E_\gamma=0.528177$, $\theta_{\ell'}=0.0482106$, $\phi_{\ell'}=6.28034$
 $Q^2=1.63978$, $x_B=0.62334$, $t=-0.524841$, $W^2=1.8707$, $y=0.849244$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.7766$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.7766$
 P $E= 1.2178$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.7766$
 ℓ' $E= 0.5282$ $p_x= 0.0255$ $p_y= -0.0001$ $p_z= 0.5276$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0000$
 q_γ $E= 0.5282$ $p_x= -0.0255$ $p_y= 0.0001$ $p_z= -0.5276$

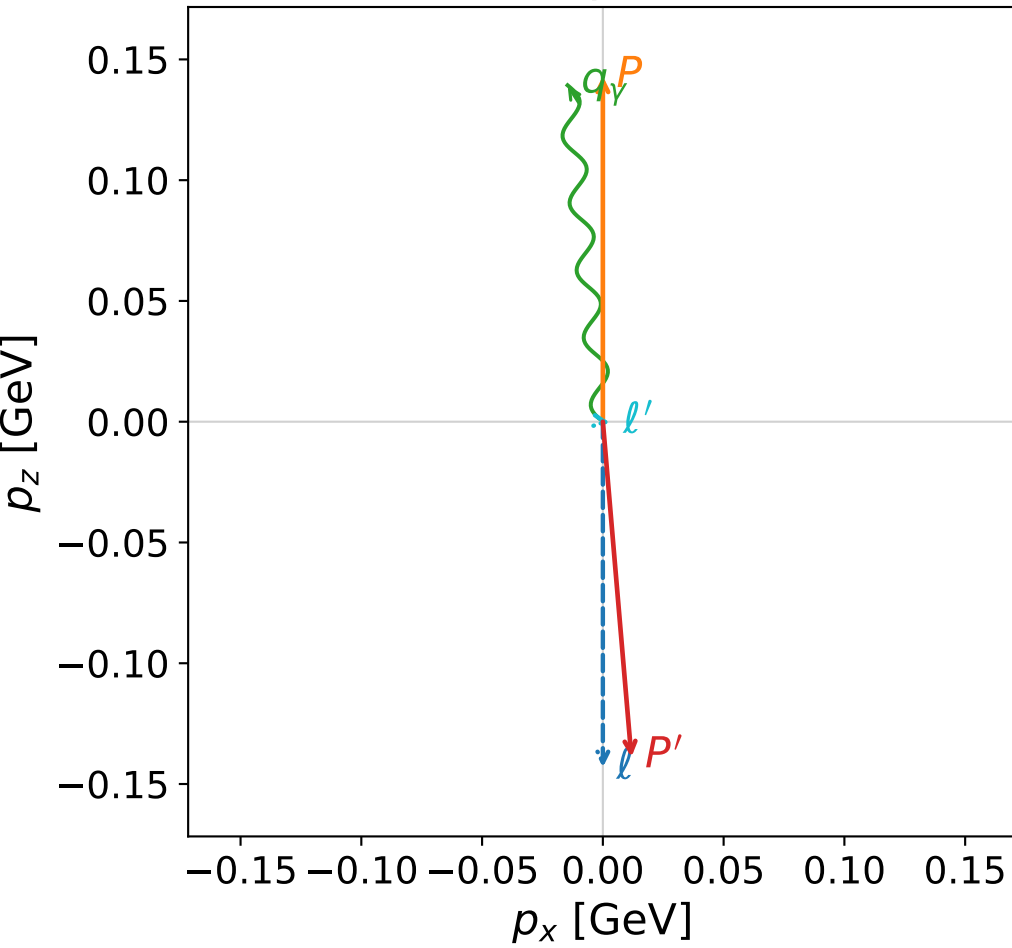
GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 47.737%, conserve 52.263%
 $R_{\text{flip/conserve}}=9.134\text{e-}01$; $I_h=+2.036\text{e-}04$
 virtual-photon powers: T 47.730%, L 52.270%
 $R_L/T=1.095\text{e+}00$; $I_{TL}=-5.199\text{e-}04$

Leading normalized final-state amplitudes
 (N=2, retained=99.8%)

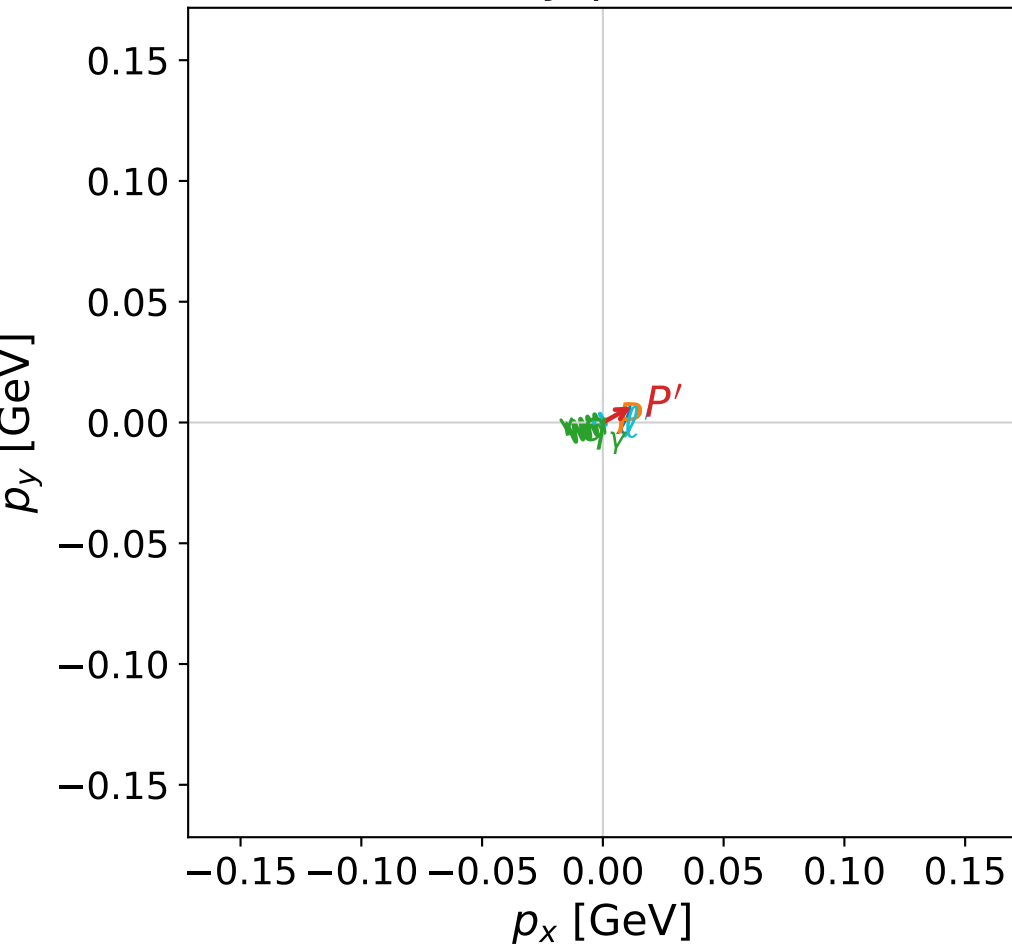


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_01_local_minimum_373

$d_{\{\rm GHZ\}}=0.0300609$

region: polarization_cluster_01_local_minimum_373

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0373

lepton mixing angle: $\alpha_e=0.74214409$ rad

incoming lepton: $0.737021|+\rangle +0.67587|-\rangle$

proton mixing angle: $\alpha_p=0.90015049$ rad

incoming proton: $0.621492|+\rangle +0.78342|-\rangle$

$s=1.19232$, $\sqrt{s}=1.09193$, $|\vec{P}|=0.143083$, $|\vec{P}'|=0.139544$

$\theta_{P'}=3.04448$, $\theta_Y=0.109503$, $\phi_{P'}=0.509518$, $\phi_Y=3.44722$

$E_Y=0.140121$, $\theta_{l'}=1.68565$, $\phi_{l'}=5.66363$

$Q^2=0.000885013$, $x_B=0.00289466$, $t=-0.0796895$, $W^2=1.1847$, $y=0.978442$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E=0.1431$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1431$

P $E=0.9489$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1431$

l' $E=0.0035$ $p_x=0.0028$ $p_y=-0.0020$ $p_z=-0.0004$

P' $E=0.9483$ $p_x=0.0118$ $p_y=0.0066$ $p_z=-0.1389$

q_Y $E=0.1401$ $p_x=-0.0146$ $p_y=-0.0046$ $p_z=0.1393$

GHZ mechanism: helicity conserving (full H G C)

helicity powers: flip 0.357%, conserve 99.643%

$R_{\text{flip/conserve}}=3.586\text{e-}03$; $I_h=+2.070\text{e-}04$

virtual-photon powers: T 99.187%, L 0.813%

$R_L/T=8.194\text{e-}03$; $I_{TL}=+1.078\text{e-}02$

Leading normalized final-state amplitudes
(N=2, retained=99.6%)

